

SEC P vs NP log 2026-05-19

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-21 10:32 UTC

SEC P vs NP — daily log 2026-05-19

112 cycles recorded

Decision Tree Depth and Communication Complexity via Lifting

- **Verdict:** INCONCLUSIVE
- **Bridge:** `DECISION_TREE_DEPTH` $\times$ `COMMUNICATION_COMPLEXITY`
- **Recorded:** 2026-05-19 00:11 UTC
- **Entry ID:** eeec3f948dcf

Statement

For any Boolean function f on n variables, if the decision tree depth of f is d , then the deterministic communication complexity $R_{\{1/3\}}(f)$ is $\Omega(d / \log n)$.

Rationale

This conjecture explores the relationship between the depth of a decision tree and the communication complexity of the function, leveraging lifting techniques to establish a lower bound. If true, it would imply that functions with deeper decision trees require more communication, which is a key insight in understanding the interplay between these complexity measures.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [2512.11833v1] Soft Decision Tree classifier: explainable and extendable PyTorch implementation - [1302.4207v1] A composition theorem for decision tree complexity - [2105.01963v3] One-way communication complexity and non-adaptive decision trees - [0911.3482v5] Complexity of Networks (reprise) - [0101006v1] On Complexity and Emergence

Empirical Test

- exit code: 124, elapsed: 240.11s

TIMEOUT after 240s

Judge reasoning

The test timed out after 240s (returncode 124) without producing any RESULT line or data, so the pre-registered support condition cannot be evaluated. | next: Reduce the parameter grid (e.g., restrict to $n \in \{8, 16, 24\}$ and sample fewer depths) and/or optimize the deterministic CC computation to fit within the time budget.

Laplacian Apolar Rank Caps Delorme-Poljak Max-CUT SoS-2 Gap

- **Verdict:** INCONCLUSIVE
- **Bridge:** Curto-Fialkow real moment problem / apolarity & Hankel flat-extension theory (real algebraic geometry). The Hankel rank of the Laplacian moment sequence $m_k(G) := \text{tr}(L_G^k)$ is by the Hamburger truncated moment theorem the smallest support size of an atomic measure on $\mathbb{R}_{\geq 0}$ reproducing the moments — equivalently the number of distinct Laplacian eigenvalues (Curto-Fialkow 1996; Lasserre 2009 ‘Moments, Positive Polynomials and their Applications’; Blekherman-Parrilo-Thomas 2012 ‘Semidefinite Optimization and Convex Algebraic Geometry’; Reznick 1992). The invariant uses real-eigenvalue clustering and rank-of-Hankel — non-ring (no $\mathbb{F}_q$ polynomial extension preserves cluster count of real eigenvalues under polynomial ring lifts), Aaronson-Wigderson algebrization-safe. An arXiv search ‘Curto-Fialkow flat extension’ AND (‘Max-Cut’ OR ‘Delorme-Poljak’ OR ‘SoS integrality gap’) returns 0 direct hits with <5 adjacent papers (Lasserre hierarchy convergence proofs, Helton-Nie semidefinite representability), none using Hankel-apolar rank of the Laplacian-moment sequence as an SoS-2 Max-CUT tightness invariant. Distinct from blacklisted Lee-Yang zeros of the cut polynomial (complex roots, not real moment-sequence), Fekete-capacity (transfinite diameter of spectrum), free 4-cumulant excess (Voiculescu NC partition statistics), Kashin L^1/L^2 flatness (single-eigenvector L^{1/L^2} ratio), and Halász L^2 discrepancy (CDF deviation). $\times$ Degree-2 Sum-of-Squares / Delorme-Poljak integrality gap $\rho(G) := n \cdot \lambda_{\max}(L_G) / (4 \cdot \text{MC}(G)) - 1$ for the Max-CUT problem on connected k -regular graphs (Goemans-Williamson 1995; Delorme-Poljak 1993; Schoenebeck 2008; Barak-Steurer 2014). The structural form of the Hankel-rank invariant (a count of distinct Laplacian eigenvalues, a property of the GRAPH WIRING — two non-isomorphic graphs with identical Max-CUT value can have very different $h(G)$) shields it from the Razborov-Rudich natural-proofs barrier; using only real-eigenvalue grouping and integer counting it lies outside the algebrization closure of Aaronson-Wigderson.
- **Recorded:** 2026-05-19 00:24 UTC
- **Entry ID:** 378d9a4dead5

Statement

For every connected k -regular simple graph G on $n \in [8, 20]$ vertices with $k \in \{3, 4\}$, let $h(G)$ be the number of distinct nonzero Laplacian eigenvalues of L_G (equivalently the rank of the truncated Hankel matrix $H_d(G)_{\{i,j\}} = \text{tr}(L_G^{\{i+j+1\}})$ for $d \geq n$, the Curto-Fialkow apolar rank of the Laplacian moment sequence). Let $\text{MC}(G)$ be the exact Max-CUT value and $\rho(G) := n \cdot \lambda_{\max}(L_G) / (4 \cdot \text{MC}(G)) - 1$ be the Delorme-Poljak SoS-2 integrality gap. Then $n \cdot \rho(G) \leq h(G) \cdot \log_2(n+1)$; a single instance with $n \cdot \rho(G) > 2 \cdot h(G) \cdot \log_2(n+1)$ refutes the conjecture.

Rationale

Apolarity / Curto-Fialkow theory identifies the Hankel rank of a moment sequence with the minimum number of atoms in a representing measure — a canonical real-algebraic-geometry obstruction. Graphs whose Laplacian spectra collapse onto few atoms (distance-regular, SRG, complete bipartite) tend to admit tight Delorme-Poljak SoS-2 certificates because the symmetry forces the optimal pseudoexpectation onto a low-dimensional invariant subspace. The conjecture promotes the folk ‘symmetric spectra = tight SoS-2’ heuristic to a quantitative apolar bound, suggesting Hankel rank as the moment-matrix property P sought by the SOS_DEGREE focus.

Novelty

- Judge: NOVEL over 7 arXiv hits

Top hits consulted: - [1107.1630v3] On the Integrality Gap of the Subtour LP for the 1,2-TSP - [2404.18142v1] Revisiting Majumdar-Ghosh spin chain model and Max-cut problem using variational quantum algorithms - [2411.11721v4] Eigenvalues of the Neumann magnetic Laplacian in

the unit disk - [0812.4567v1] On degenerate Hamburger moment problem and extensions of positive semidefinite Hankel block matrices - [1508.00617v2] Hankel determinants of random moment sequences

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_06da5d96.py", line 123, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_06da5d96.py", line 96, in run_trial
    eigenvalues = laplacian_eigenvalues(graph)
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_06da5d96.py", line 58, in laplacian_eig
    eigenvalues = gaussian_elimination(L)
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_06da5d96.py", line 29, in gaussian_elim
    factor = A[j][i] / A[i][i]
              ~~~~~^~~~~~
```

ZeroDivisionError: float division by zero

Judge reasoning

The test crashed with a ZeroDivisionError in the custom gaussian_elimination routine before any trials completed, so no r-values were produced and neither the support nor falsification condition can be evaluated. | next: Replace the hand-rolled Gaussian elimination with numpy.linalg.eigvalsh (or use pivoting) to robustly compute Laplacian eigenvalues, then rerun the 240-seed sweep.

Cross-Read Kronecker Sum Rank Lower-Bounds Read-Twice BP for IP₂

- **Verdict:** INCONCLUSIVE
- **Bridge:** GCT-style plethystic Kronecker-product orbit-tangent obstructions via the bit-flip operators $D_p := T_p^1 - T_p^0$ in $Z^{\{w \times w\}}$ (infinitesimal $GL(w)$ -tangent directions to the BP's orbit-closure in the Mulmuley-Sohoni 2001 / Burgisser-Ikenmeyer 2013 sense), aggregated as the cross-read Kronecker sum $K(P) := \sum_v D_{\{p_1(v)\}}(x) D_{\{p_2(v)\}}$ in $Z^{\{w^2 \times w^2\}}$. The invariant is STRUCTURAL on the BP wiring (two read-twice BPs computing the same function can have very different K), shielding from Razborov-Rudich natural-proofs; rank over Z has no canonical F_q polynomial lift preserving the integer rank value across the Kronecker sum (the rank of a Kronecker SUM is not preserved by polynomial ring lifts, unlike the rank of a single Kronecker product), so the bridge is Aaronson-Wigderson algebrization-safe. An arXiv search 'Kronecker sum bit-flip' AND ('read-twice branching program' OR 'Borodin Razborov Smolensky' OR 'GCT plethysm') returns 0 direct hits with <5 adjacent papers (all on Kronecker-coefficient plethysm computations or layer-product BP lower bounds, never on the cross-read SUM of bit-flip Kronecker products). Distinct from blacklisted Generated-Algebra-Dimension (Wedderburn span via Gauss-Jordan closure), Schatten Stable-Rank of Layer-Difference Stack (Schatten $2/\inf$ ratio on a vertically stacked layer-difference matrix, NOT a Kronecker sum), Cycle-Type Mixing (Plancherel S_5 distance), and Coxeter-Inversion Spread (S_5 reflection-length word statistic) - none uses the cross-read Kronecker SUM of bit-flip operators. × Read-twice oblivious branching program size $s(P)$ for $IP_2(x,y) = XOR_k (x_k \text{ AND } y_k)$ on $2n$ inputs under the adversarial variable order $x_1, \dots, x_n, y_1, \dots, y_n, x_1, \dots, x_n, y_1, \dots, y_n$ (Borodin-Razborov-Smolensky 1993 regime), instantiating the focus's 'distinguishing tensor' formalism with $\rho(P) = \log_2 \text{rank } K(P) = O(\log s(P))$ but $\rho(\text{any RT-BP for } IP_2) = \Omega(n)$.

- **Recorded:** 2026-05-19 00:32 UTC
- **Entry ID:** fd57783d2ed4

Statement

Let P be a read-twice oblivious BP with width w and 0/1-valued layer transition matrices T_p^b in $\{0,1\}^{w \times w}$ for layers p in $[4n]$ and bits b in $\{0,1\}$, with each variable v read at exactly two positions $p_1(v) < p_2(v)$. Define the bit-flip operator $D_p := T_p^1 - T_p^0$ in $Z^{w \times w}$ and the cross-read Kronecker sum $K(P) := \sum \text{over all } 2n \text{ variables } v \text{ of } D_{p_1(v)} \otimes D_{p_2(v)}$ in $Z^{w^2 \times w^2}$, and let $\rho(P) := \text{rank}_Q(K(P))$. Conjecture: (a) $\rho(P) \leq w(P)^2 \leq s(P)^2$ for every read-twice oblivious BP P (trivial dimension cap), and (b) for every read-twice oblivious BP P computing IP_2 under the adversarial order above, $\rho(P) \geq 2^n / 4$. Equivalently, any read-twice oblivious BP for IP_2 has size at least $2^{\{n/2\}/2}$; a single (P, IP_2) pair with $\rho(P) < 2^{n/4}$ falsifies (b).

Rationale

In Mulmuley-Sohoni GCT the orbit-closure of a polynomial is probed by $GL(w)$ -equivariant tangent directions; the bit-flip operators D_p are exactly the layer-wise infinitesimal generators of the BP's representation variety, and their cross-read Kronecker product encodes the plethystic coupling that READ-ONCE BPs cannot create. For IP_2 under adversarial order, each variable v forces the two reads at $p_1(v), p_2(v)$ to jointly implement an AND-XOR coupling: when bits are flipped the resulting $D_{p_1(v)} \otimes D_{p_2(v)}$ term must contribute a fresh rank-1 block to $K(P)$, and the n distinct variables produce $2n$ rank-1 blocks whose supports cannot algebraically cancel without enlarging the width. The fooling-set / matrix-rank machinery is non-ring and structural on the BP wiring (not the truth table), shielding the bridge from both Razborov-Rudich natural proofs and Aaronson-Wigderson algebrization.

Novelty

- Judge: NOVEL over 8 arXiv hits

Top hits consulted: - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [1601.02745v1] Basic Reasoning with Tensor Product Representations - [1511.07136v1] Identity Testing and Lower Bounds for Read- k Oblivious Algebraic Branching Programs - [1806.02734v3] Spectral lower bounds for the orthogonal and projective ranks of a graph - [2107.09834v4] Communication lower bounds for nested bilinear algorithms via rank expansion of Kronecker products

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f8c519ad.py", line 91, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f8c519ad.py", line 61, in run_trial
    rho_ip2 = compute_rho(n)
              ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f8c519ad.py", line 28, in compute_rho
    P_IP2 = build_read_twice_bp(n)
            ^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f8c519ad.py", line 23, in build_read_twice_bp
    D_p.append([[T_p_1[i][j] - T_p_0[i][j] for j in range(w)] for i in range(w)])
                ^^^^^^^^^^^^^
```

TypeError: 'int' object is not subscriptable

Judge reasoning

The test crashed with a `TypeError` in `build_read_twice_bp` before producing any rho data, so neither the support nor falsification condition could be evaluated. | next: Fix the BP construction bug (ensure T_{p_0} and T_{p_1} are 2D $w \times w$ matrices, not ints) and rerun the pre-registered protocol for n in $\{2,3,4,5\}$.

SoS Cone-Gram Stable Rank Caps AC^0 Depth-d Size for PARITY

- **Verdict:** INCONCLUSIVE
- **Bridge:** Sum-of-Squares hierarchy / Lasserre stable-rank geometry of structural Gram matrices on cone-indicator features (Lasserre 2001; Parrilo 2003; Barak-Steurer 2014; Schur 1923 Hadamard-product PSD theory; Rudelson-Vershynin 2007 stable rank). The invariant $\psi(C) := \sum_i \lambda_i(K(C)) / \lambda_{\max}(K(C))$ — the Schatten-1/Schatten- ∞ stable rank of the gate-cone-indicator Gram matrix $K(C)[g,h] = |\text{cone}(g) \cap \text{cone}(h)|/n$ — is the SoS-degree-2 second-moment / pseudoexpectation functional of the structural feature ensemble $\{1_{\{\text{cone}(g)\}}/\sqrt{n} \mid g \in V(C)\}$ viewed as a Lasserre relaxation of the gate-incidence polytope. It is a non-ring, real-spectral Schatten-ratio (no F_q polynomial extension preserves $\sum \lambda / \lambda_{\max}$ under polynomial ring lifts), Aaronson-Wigderson algebrization-safe. STRUCTURAL on the circuit DAG (two AC^0 circuits with the same Boolean function can have very different cone Gram spectra), so it sidesteps the Razborov-Rudich barrier that killed the prior ‘Lasserre Moment-Rank Gap of Accept Set’ attempt (which operated on truth-table accept sets). $\text{Sym}(n)$ -equivariant under input permutations ($K(C)$ transforms by simultaneous permutation, preserving spectrum), fulfilling the focus’s natural-group-action requirement. An arXiv search ‘Lasserre stable rank’ AND (‘ AC^0 ’ OR ‘PARITY lower bound’ OR ‘switching lemma’) returns 0 direct hits with <5 adjacent papers (all on SoS-Max-Cut / SoS-CSP integrality gaps, never on gate-cone Gram spectra of AC^0 circuits). $\times AC^0$ circuit size $s(C)$ and depth d for the explicit n -bit PARITY function, in the Håstad 1986 switching-lemma / Razborov-Smolensky / Beame 1994 regime, accessed through the depth- d $2^{\Theta(n \{1/(d-1)\})}$ lower bound (open: tight constants and a non-restriction-based proof). In the Cook-Reckhow-Krajíček bounded-arithmetic framework, an AC^0 PARITY size lower bound is the canonical $V^0 \sqsubset V^0(\oplus)$ separation stepping stone.
- **Recorded:** 2026-05-19 00:39 UTC
- **Entry ID:** 63e0e3f43696

Statement

For every AC^0 circuit C of depth $d \geq 2$ on n input variables that computes PARITY_n exactly, the gate-cone-indicator Gram matrix $K(C) \in \mathbb{R}^{|V| \times |V|}$ with $K[g,h] := |\text{cone}(g) \cap \text{cone}(h)|/n$ satisfies $\psi(C) := (\sum_i \lambda_i(K(C))) / \lambda_{\max}(K(C)) \leq C_0 \cdot (\log_2 s(C))^{d-1}$, equivalently $\log_2 s(C) \geq \psi(C)^{1/(d-1)/C_0}$, for an absolute constant $C_0 \leq 4$. The bound is non-vacuous because PARITY requires every input in $\text{cone}(\text{output})$, forcing $\psi(C)$ to scale at least as $n^{(d-2)/(d-1)}$ on any depth- d AC^0 realization, matching Håstad’s lower-bound shape without invoking random restrictions. A single AC^0 circuit C computing PARITY with $\psi(C)^{1/(d-1)} > 4 \cdot \log_2 s(C)$ falsifies the conjecture.

Rationale

The cone Gram matrix $K(C)$ is the natural SoS-degree-2 Lasserre moment-matrix of the structural feature ensemble $\{1_{\{\text{cone}(g)\}}\}_g$, and its stable rank $\psi(C)$ is a Schur-Horn-type combinatorial second-moment measuring how many ‘effective independent input directions’ the wiring of C must explore — exactly the resource that Håstad’s switching lemma shows PARITY forces large via random restrictions, but here computed deterministically from the DAG alone. Because $\text{cone}(g)$ depends only on the wiring (not the gate functions or truth table), two AC^0 circuits with identical Boolean function can have very different $\psi(C)$, giving a Razborov-Rudich-safe structural functional within the SoS hierarchy. The bound’s exponent $(d-1)$ mirrors Håstad’s tight $2^{\Theta(n \{1/(d-1)\})}$

shape, so if it holds even loosely it yields a restriction-free route to AC^0 PARITY lower bounds and a candidate ψ for the focus's polynomial-time natural-invariant target.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7c7fd2fe.py", line 151, in <module>
    trial = run_trial(seed)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7c7fd2fe.py", line 108, in run_trial
    cones = [reverse_bfs(g, i) for i in range(n)]
              ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7c7fd2fe.py", line 28, in reverse_bfs
    if not visited[neighbor]:
        ~~~~~^~~~~~
```

TypeError: list indices must be integers or slices, not tuple

Judge reasoning

The test crashed with a `TypeError` in `reverse_bfs` (list indexed by a tuple) before producing any $r(C)$ values, so neither the support nor falsification condition could be evaluated. | next: Fix the cone-construction bug in `reverse_bfs` (ensure neighbor is an integer node id, not a (node, edge) tuple) and rerun the 540-instance sweep.

Talagrand L1 Influence Spread of Clause-Falsification Polynomial Lower-Bounds Tree-Resolution

- **Verdict:** INCONCLUSIVE
- **Bridge:** Talagrand L1-influence-vector / KKL-Friedgut Boolean Fourier analysis — the Talagrand 1994 functional $T(p) := \sum_i \sqrt{\text{Inf}_i(p)}$, with $\text{Inf}_i(p) = \sum_{S \ni i} \hat{p}(S)^2$, applied as a SYN-TACTIC Fourier-analytic invariant of the clause-falsification pseudo-Boolean polynomial $p_F(x) := \#\{C \in F : x \text{ falsifies } C\}$ of a CNF (a property of the clause LIST — multiplicities and polarities — not of the truth table; two unsat 3-CNFs with identical satisfiability function $\equiv \text{FALSE}$ can have very different $T(F)$). Distinct from blacklisted Mansour L1 mass $\sum_S |\hat{p}(S)|$ (a coefficient-indexed absolute-value sum), Bonami 4/2 ratio $\|p\|_4/\|p\|_2$ (a hypercontractive norm ratio), Bourgain NSM $\sum_S (1 - \rho^{|S|}) \hat{p}(S)^2$ (a level-weighted L^2 mass), hypercontractive term-pair noise log-ratio (a Rényi-2-to-4 kernel statistic), and cross-side Fourier symmetric-difference mass (a truth-table communication-matrix Fourier object that hit NATURAL_PROOFS) — Talagrand L1 is the unique coordinate-indexed $\sum_i \sqrt{(\sum_{S \ni i} \hat{p}(S)^2)}$ functional with an OUTER $\sqrt{}$ over per-variable L^2 influence buckets. The functional uses $\sqrt{}$ on integer-valued partial sums and a per-coordinate sum that does NOT extend to a polynomial F_q lift (square-root of an L^2 influence has no polynomial-ring surrogate), Aaronson-Wigderson algebrization-safe; structural shielding from Razborov-Rudich since $T(F)$ is not a poly-time-computable property of the truth table of F (which is $\equiv \text{FALSE}$). ArXiv ‘Talagrand L1 influence sum’ AND (‘DPLL’ OR ‘tree-Resolution’ OR ‘CNF refutation’ OR ‘proof complexity’) returns 0 direct hits and <5 adjacent papers (KKL/Friedgut juntas, never on syntactic clause-falsification polynomials). × Tree-like Resolution / DPLL refutation size $t(F)$ for unsatisfiable 3-CNFs F in the Ben-Sasson-Wigderson 2001 / Beame-Karp-Pitassi-Saks 2002 / Krajíček 1995 bounded-arithmetic regime where $\neg F$ is

Σ^b_0 and $\log_2 t(F)$ lower-bounds V^0 -witnessing complexity; the canonical Frege/Extended-Frege depth stepping stone via Pudlák-Buss feasible interpolation.

- **Recorded:** 2026-05-19 00:46 UTC
- **Entry ID:** 9e12285a05e1

Statement

For every unsatisfiable 3-CNF F on $n \leq 40$ variables with m clauses, let $p_F : \{0,1\}^n \rightarrow \mathbb{Z}_{\geq 0}$ count falsified clauses, expand p_F in the Walsh basis (only subsets S with $|S| \leq 3$ are non-zero), and define $T(F) := \sum_{i=1}^n \sqrt{(\sum_{S \ni i, 1 \leq |S| \leq 3} \hat{p}_F(S)^2)}$. Then $\log_2(1 + t(F)) \geq 0.10 \cdot T(F) / \sqrt{m}$, where $t(F)$ is the DPLL refutation size (backtrack count) under the unit-propagation + first-unassigned variable order. A single F violating this inequality (i.e. $\log_2(1+t^*(F)) < 0.10 \cdot T(F)/\sqrt{m}$) refutes the conjecture.

Rationale

Clause-falsification influence $\text{Inf}_i(p_F)$ measures how strongly variable i ‘pivots’ the count of broken clauses, and Talagrand’s L^1 -spread functional $\sum_i \sqrt{\text{Inf}_i}$ interpolates between the variance $\sum_i \text{Inf}_i (= I(p_F))$, the total clause-sensitivity) and the maximum influence KKL bound, capturing per-variable contention without collapsing to a coefficient sum (Mansour) or a hypercontractive norm ratio (Bonami). Resolution / DPLL must spend $\geq \Omega(\text{width})$ steps to refute, and Ben-Sasson-Wigderson’s width-size tradeoff together with Talagrand’s variance-spread inequality suggest the clause-list-side $T(F)$ functional should track refutation width, hence $\log t^*$. The bridge is structural (not a truth-table property), uses non-polynomial operations ($\sqrt{\cdot}, |\cdot|$), and lies outside both the Razborov-Rudich constructivity regime and the Aaronson-Wigderson algebrization closure.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_972a393d.py", line 175, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_972a393d.py", line 132, in run_trial
    if not is_unsatisfiable(cnf):
            ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_972a393d.py", line 33, in is_unsatisfiable
    n = max(int(literal[2:]) if literal.startswith('x') else int(literal[3:]) for literal in cnf.replac
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_972a393d.py", line 33, in <genexpr>
    n = max(int(literal[2:]) if literal.startswith('x') else int(literal[3:]) for literal in cnf.replac
            ^^^^^^^^^^^^^^^^^
```

ValueError: invalid literal for int() with base 10: ''

Judge reasoning

The test crashed with a ValueError during CNF parsing (is_unsatisfiable) before any instances were evaluated, so no violation count or Pearson correlation was produced. The pre-registered support condition cannot be assessed. | next: Fix the CNF literal parser in is_unsatisfiable (handle the negation prefix and empty tokens correctly) and rerun the pre-registered sweep over (n, α) with 30 seeds.

Border Rank Lower Bound for Communication Complexity of Disjointness

- **Verdict:** INCONCLUSIVE
- **Bridge:** ALGEBRAIC_GEOMETRY (border rank of tensors) $\times$ COMMUNICATION_COMPLEXITY
- **Recorded:** 2026-05-19 01:09 UTC
- **Entry ID:** d76acad80bde

Statement

The randomized communication complexity of the DISJ_n matrix is $\Omega(\log(\text{border_rank}(M)))$, where $\text{border_rank}(M)$ is the border rank of the communication matrix over the real numbers.

Rationale

The border rank of a tensor captures its asymptotic complexity, and higher border ranks imply greater information-theoretic requirements for communication. This conjecture links algebraic geometry's tensor rank framework to communication complexity, leveraging the known $\Omega(n)$ lower bound for DISJ_n as a baseline.

Novelty

- Judge: NOVEL over 2 arXiv hits

Top hits consulted: - [s2:1608.07486] A $2n^2 - \log(n) - 1$ lower bound for the border rank of matrix multiplication - [RX025] A $2n^2 - \log_2(n) - 1$ lower bound for the border rank of matrix multiplication

Empirical Test

- exit code: 0, elapsed: 0.20s

```
_value': 24, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 24, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 18, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 15, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 12, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 4, 'instances_tested': 1, 'conjec
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 22, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 20, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 6, 'instances_tested': 1, 'conjec
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 35, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 13, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 38, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 16, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 14, 'instances_tested': 1, 'conje
RESULT: SUPPORTED mean=19.5 std=10.465021102065045 support_fraction=1.0
```

Judge reasoning

The test output only reports a single 'communication_complexity' integer per trial with no border_rank values shown, so the pre-registered criterion (comparing $R(\text{DISJ}_n)$ to $\log_2(\text{border_rank_mean})$) cannot be verified; even if computed, the bound $\log(\text{border_rank}) \leq n$ is trivially satisfied by $R(\text{DISJ}_n) = \Theta(n)$, making the 'SUPPORTED' result vacuous. | next: Re-run with explicit border_rank values logged per n, and compare against a nontrivial benchmark function where border rank is provably super-pol

Communication Complexity Lower Bound for Disjointness Implies Monotone Circuit Size Lower Bound

- **Verdict:** INCONCLUSIVE
- **Bridge:** COMMUNICATION_COMPLEXITY × MONOTONE_CIRCUIT_LOWER_BOUNDS
- **Recorded:** 2026-05-19 01:23 UTC
- **Entry ID:** bf58b88b218f

Statement

For the disjointness function on n -bit inputs, if the deterministic communication complexity is $\Omega(n)$, then every monotone circuit computing it requires size $\Omega(2^{\{n/2\}})$

Rationale

The Karchmer-Wigderson game links communication protocols to circuit depth, but monotone circuits require stronger lower bounds. Disjointness has known $\Omega(n)$ communication complexity, and its monotone circuit size lower bound remains open. A direct connection would unify these complexity measures via combinatorial geometry of protocols.

Novelty

- Judge: NOVEL over 2 arXiv hits

Top hits consulted: - [s2:2506.12420] Deterministic Lifting Theorems for One-Way Number-on-Forehead Communication - [3406325.3451069] Lower bounds for monotone arithmetic circuits via communication complexity

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 89, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 67, in run_trial
    D = compute_communication_complexity()
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 46, in compute_communication_complexity
    protocol_tree_search(inputs, 0)
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 43, in protocol_tree_search
    protocol_tree_search(left_partition, depth + 1)
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 43, in protocol_tree_search
    protocol_tree_search(left_partition, depth + 1)
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 43, in protocol_tree_search
    protocol_tree_search(left_partition, depth + 1)
```

[Previous line repeated 1 more time]

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 44, in protocol_tree_search
    protocol_tree_search(right_partition, depth + 1)
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 43, in protocol_tree_search
    protocol_tree_search(left_partition, depth + 1)
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 43, in protocol_tree_search
    protocol_tree_search(left_partition, depth + 1)
```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 43, in protocol_tree_search(left_partition, depth + 1)
[Previous line repeated 989 more times]
RecursionError: maximum recursion depth exceeded

Judge reasoning

The test crashed with a RecursionError in protocol_tree_search before producing any data, so neither the communication complexity D nor the monotone circuit size S was measured. With no metrics computed and the critic challenging, the pre-registered support condition cannot be evaluated. | next: Rewrite protocol_tree_search iteratively (or cap depth at $2n$ and prune by rectangle monochromaticity) so $D(\text{DISJ}_4)$ and $S(\text{DISJ}_4)$ can actually be computed before re-running the 10-seed trial.

Cubical First-Betti Linearly Lower-Bounds DNF-MCSP

- **Verdict:** INCONCLUSIVE
- **Bridge:** Cubical combinatorial topology — the cubical subcube complex $K_f := \{\sigma \subseteq \{0,1\}^n : \sigma \text{ is a subcube and } \sigma \subseteq f^{-1}(1)\}$ with its boundary maps over F_2 , $\beta_1(K_f)$ computed as $\text{nullity}(\partial_1) - \text{rank}(\partial_2)$ (Kaczynski-Mischaikow-Mrozek 2004 ‘Computational Homology’; Edelsbrunner-Letscher-Zomorodian 2002; Carlsson 2009 ‘Topology and data’). A non-ring, F_2 -rank-of-cubical-boundary functional: ranks of integer cellular boundaries do NOT extend to a polynomial F_q lift of the Boolean ring (the cubical boundary involves signed face inclusion not preserved by polynomial ring lifts), so the bridge is Aaronson-Wigderson algebrization-safe. The invariant is RELATIONAL (a categorical functor $(\text{Bool}^n \rightarrow \text{Top} \rightarrow \text{Vect}\{F_2\})$ applied to the subcube cover poset), not a single truth-table number — sidestepping trap (j). An arXiv search ‘cubical homology’ AND (‘MCSP’ OR ‘minimum DNF’ OR ‘meta-complexity’) returns 0 direct hits with <5 adjacent papers (all on TDA of biological data or persistent-homology classifiers, never on DNF-MCSP). Distinct from blacklisted Mahler-measure (single complex-analytic truth-table number), FCA antichain width (Galois-lattice width on assignments $\times$ literals), treewidth of PI compatibility graph (graph-minor invariant), and independence-complex Euler $\tilde{\chi}$ (Stanley-Reisner alternating sum). $\times$ DNF-MCSP / minimum DNF size $\sigma_{\text{DNF}}(f)$ for $f: \{0,1\}^n \rightarrow \{0,1\}$ in the Quine-McCluskey / Masek 1979 / Umans 1999 / Allender-Hellerstein-McCabe-Pitassi-Saks / Hirahara-Oliveira meta-complexity regime, where σ_{DNF} is the optimum of a prime-implicant set-cover and the canonical NP-hard meta-complexity object whose hardness fuels Hirahara’s worst-case-to-average-case program. Sparsity escape from Razborov-Rudich: ‘ $\sigma_{\text{DNF}}(f) \leq s$ ’ is rare among random f (only $\sim 2^{\{O(s \log n)\}}$ functions admit it, of $2^{\{2n\}}$ total), so any lower bound $\sigma_{\text{DNF}} \geq \varphi(f)$ inherits sparsity from the cap side and does not violate Razborov-Rudich largeness.
- **Recorded:** 2026-05-19 01:33 UTC
- **Entry ID:** 9020e29f840c

Statement

Let $K_f = \{\sigma \subseteq \{0,1\}^n : \sigma \text{ is a subcube and } \sigma \subseteq f^{-1}(1)\}$ be the cubical subcube complex of $f^{-1}(1)$, and $\beta_1 := \beta_1(K_f, F_2)$ its first F_2 -Betti number. Then for every Boolean $f: \{0,1\}^n \rightarrow \{0,1\}$ satisfying $\beta_1(K_f) \geq 1$, the minimum DNF size obeys $\sigma_{\text{DNF}}(f) \geq 3 \cdot \beta_1(K_f)$. One counterexample (a single f with $\sigma_{\text{DNF}}(f) < 3 \cdot \beta_1(K_f)$ and $\beta_1(K_f) \geq 1$) refutes the conjecture.

Rationale

By the nerve theorem applied to any DNF cover $F = T_1 \vee \dots \vee T_s$ of f (each T_i a subcube $\subseteq f^{-1}(1)$), the nerve N_F is homotopy equivalent to K_f , giving $\beta_1(N_F) = \beta_1$. Trivially N_F has s vertices and any 1-cycle uses ≥ 3 of them, yielding only $\sigma_{\text{DNF}} \geq \sqrt{(2\beta_1)}$; the conjecture asserts the much stronger LINEAR bound — each independent topological hole costs 3 fresh terms in any

optimal cover because subcubes encircling a hole are ‘rigid’ (cannot be shared across distinct holes without collapsing the cycle). This is a clean nerve-rigidity strengthening pioneered by no prior MCSP work.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_80da2c5c.py", line 105, in <module>
    trial = run_trial(seed)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_80da2c5c.py", line 84, in run_trial
    beta_1, sigma_DNF = dnf_size(f, n)
                        ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_80da2c5c.py", line 25, in dnf_size
    beta_1 = compute_beta_1(K_f)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_80da2c5c.py", line 30, in compute_beta_1
    n = len(K_f[0])
        ~~~~
```

IndexError: list index out of range

Judge reasoning

The test crashed with an IndexError in compute_beta_1 (empty K_f list) before any instances were evaluated, so no data was produced to assess the pre-registered support or falsification condition.
| next: Fix compute_beta_1 to handle the empty subcube complex case ($\beta_1 = 0$ when K_f is empty) and rerun the 270-triple sweep.

Fourier Spectral Entropy of Clause-Falsification Polynomial Bounds Tree-Resolution

- **Verdict:** INCONCLUSIVE
- **Bridge:** Fourier-analytic information theory — Shannon entropy of the normalized Walsh-Hadamard spectrum of a multilinear pseudo-Boolean polynomial (Friedgut-Kalai 1996 FEI conjecture; O’Donnell 2014 ‘Analysis of Boolean Functions’ Ch. 9; Wan-Wang-Yu 2014 on FEI for low-degree functions; Chakraborty-Kulkarni-Lokam-Saurabh 2016 ‘Upper bounds on Fourier entropy’). The functional $H_F(F) := -\sum_S q_F(S) \cdot \log_2 q_F(S)$, where $q_F(S) := \hat{p}_F(S)^2 / \|p_F\|^2$, is the Shannon entropy of the SPECTRAL probability distribution viewed as a measure on the Boolean character lattice 2^1 . It is distinct from Mansour $L^1 \sum |\hat{p}(S)|$ (linear in $|\hat{p}|$, no log), Bourgain NSM $\sum_S (1 - \rho^{|S|}) \hat{p}(S)^2$ (level-indexed L^2 weight, not entropy), Bonami $4/2$ ratio $\|p\|_4 / \|p\|_2$ (norm ratio), Talagrand $\sum_i \sqrt{\text{Inf}_i}$ (per-coordinate L^2 -root), term-pair noise log-ratio (Rényi-2-to-4 hypercontractive kernel), and cross-side symdiff mass (truth-table CC-matrix Fourier). It uses log of a normalized ratio of squares, so it admits no polynomial $F \rightarrow q$ extension (Shannon entropy of a probability distribution is not preserved under polynomial-ring lifts of the Boolean ring), making the bridge Aaronson-Wigderson algebrization-safe. Razborov-Rudich-safe because $H_F(F)$ is a property of the CLAUSE LIST (multiplicities,

¹_n

polarities), so two unsat 3-CNFs F_1, F_2 with identical satisfiability function (both $\equiv \text{FALSE}$) can have very different H_F . An arXiv search ‘Fourier entropy’ AND (‘DPLL’ OR ‘tree-Resolution’ OR ‘CNF refutation’ OR ‘proof complexity’) returns 0 direct hits and <5 adjacent papers (all on FEI for ± 1 Boolean f or random-DNF entropy, never on the clause-falsification polynomial of an unsat CNF). $\times$ Tree-like Resolution / DPLL refutation size $t(F)$ for unsatisfiable 3-CNFs F in the Chvátal-Szemerédi 1988 / Ben-Sasson-Wigderson 2001 / Beame-Karp-Pitassi-Saks 2002 / Krajíček 1995 bounded-arithmetic regime, where $\neg F$ is Σ^b_0 and $\log_2 t(F)$ lower-bounds V^0 -witnessing complexity; the canonical Frege/Extended-Frege depth stepping stone via Pudlák-Buss feasible interpolation.

- **Recorded:** 2026-05-19 01:40 UTC
- **Entry ID:** 363fda00c17f

Statement

Let F be an unsatisfiable 3-CNF on n variables with m clauses. Define $p_F : \{-1, +1\}^n \rightarrow \mathbb{R}$ by $p_F(x) := \sum_{C \in F} \mathbb{1}[x \text{ falsifies } C]$; its Walsh-Hadamard spectrum $\hat{p}_F(S)$ is supported on $|S| \leq 3$ and admits the closed form $\hat{p}_F(\emptyset) = m/8$, $\hat{p}_F(\{i\}) = -\sum_{C \ni i} \varepsilon_i^C / 8$, $\hat{p}_F(\{i, j\}) = \sum_{C \supseteq \{i, j\}} \varepsilon_i^C \varepsilon_j^C / 8$, $\hat{p}_F(\{i, j, k\}) = -\sum_{C = \{i, j, k\}} \varepsilon_i^C \varepsilon_j^C \varepsilon_k^C / 8$ where $\varepsilon_i^C \in \{\pm 1\}$ is the polarity of variable i in clause C . Let $q_F(S) := \hat{p}_F(S)^2 / \|p_F\|^2$ and $H_F(F) := -\sum_{\{S : |S| \leq 3\}} q_F(S) \cdot \log_2 q_F(S)$. Conjecture: there exists a universal constant $c > 0$ such that for every unsat 3-CNF F at clause-to-variable ratio $\alpha \in [4.2, 5.0]$, $\log_2 t(F) \geq c \cdot n \cdot H_F(F) / \log_2(1 + n + n^2 + n^3)$. One counterexample (an unsat 3-CNF whose ratio $\log_2 t(F) \cdot \log_2(1 + n + n^2 + n^3) / (n \cdot H_F(F))$ is arbitrarily small) refutes it.

Rationale

The spectral entropy H_F measures how DELOCALIZED the Walsh-Hadamard energy of the clause-falsification polynomial is across degree- ≤ 3 character modes; intuitively, a CNF whose Fourier mass concentrates on a few low-degree characters admits a short tree-Resolution refutation (a few decisions kill the dominant modes), while a CNF whose Fourier mass spreads broadly forces DPLL to branch in many directions. The Friedgut-Kalai FEI heuristic — entropy is bounded by total influence — suggests an inverse linear relationship between spectral concentration and combinatorial complexity, and here we are normalizing entropy by the log of the number of degree- ≤ 3 characters to get an n -scaled functional. The bridge is genuinely under-explored: spectral entropy of the syntactic clause polynomial has never been used as a DPLL lower-bound certificate.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6e131d60.py", line 139, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6e131d60.py", line 111, in run_trial
    spectrum = walsh_hadamard_transform(cnf)
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6e131d60.py", line 59, in walsh_hadamard_transform
    spectrum = [0] * (2 ** n_vars)
                ^~~~~~
```

OverflowError: cannot fit 'int' into an index-sized integer

Judge reasoning

The test crashed with an OverflowError when attempting to allocate a 2^n -sized array for the Walsh-Hadamard transform, so no $R(F)$ values or Pearson correlations were produced. Without data, neither the support nor the falsification condition can be evaluated. | next: Rewrite walsh_hadamard_transform to exploit the closed-form spectrum (support only on $|S| \leq 3$, $O(n^3)$ coefficients) rather than allocating a 2^n array, then rerun the pre-registered protocol.

2-Adic Cokernel Type of Sign Matrix Caps Randomized CC

- **Verdict:** INCONCLUSIVE
- **Bridge:** Hall algebra of finite abelian 2-groups / Cohen-Lenstra heuristics on integer-matrix cokernels — the partition $\lambda_2(M) \vdash v_2(|\det M|)$ encoding the 2-Sylow structure of $\text{cok}(M) = \mathbb{Z}^n / M \cdot \mathbb{Z}^n$ as a finite abelian 2-group (Hall 1959; Macdonald 1995 ‘Symmetric Functions and Hall Polynomials’ Ch. II ‘The Hall algebra’; Cohen-Lenstra 1984 ‘Heuristics on class groups’; Friedman-Washington 1989 random matrix cokernels; Wood 2017 ‘Random integral matrices’). The invariant is computed via Smith normal form mod 2^N in $O(n^3 \cdot \log n)$ pure-Python big-int Gaussian elimination, sidestepping any Coxeter/Hecke machinery. An arXiv search ‘Cohen-Lenstra’ AND (‘communication complexity’ OR ‘sign rank’ OR ‘discrepancy’) returns 0 direct hits with <5 adjacent papers (all on lattice-based cryptography or chip-firing critical groups, never on CC matrices). Distinct from blacklisted 2-Sylow rank of Tseitin critical group $K(G)$ (an invariant of a charged GRAPH, not a sign matrix) and from Specht-isotype / cycle-type / RSK / hook-length attempts (which projected via S_n -characters or partition statistics, never via the Hall-algebra 2-adic cokernel type of an integer matrix). × Two-party randomized communication complexity $R_{1/3}(M)$ for $\{\pm 1\}$ -sign matrices $M \in \{\pm 1\}^{n \times n}$, in the Yao 1979 / Razborov 1992 DISJ regime, accessed via the Linial-Shraibman γ_2 / Klartag-Regev corruption lower bound and the Forster sign-rank route to UPP^{cc} . The invariant $\lambda_2(M)$ is STRUCTURAL on the sign assignment: it is preserved under the full CC symmetry group (row/column permutation + signed-diagonal pre/post conjugation $D_1 \cdot M \cdot D_2$ with $D_i \in \{\pm 1\}^n$ diagonal), since these act as $\mathbb{Z}$ -module automorphisms of $\mathbb{Z}^n$ and leave the cokernel isomorphism class fixed — but it is NOT a poly-time-computable property of the underlying Boolean function (two CC matrices with identical function up to relabeling can have differing λ_2 after lift to $\mathbb{Z}$), shielding from Razborov-Rudich.
- **Recorded:** 2026-05-19 01:51 UTC
- **Entry ID:** 7227efbf367

Statement

For every sign matrix $M \in \{\pm 1\}^{n \times n}$, let $v_2(|\det M|)$ be the 2-adic valuation of the determinant computed over $\mathbb{Z}$, and define $\rho(M) := v_2(|\det M|) / n \in [0, (\log_2 n)/2]$. We conjecture there exists an absolute constant $c \in (0, 4)$ such that for all $n \geq 4$ and all $M \in \{\pm 1\}^{n \times n}$ with $\gamma_2(M) \leq g$ (the Linial-Shraibman γ_2 -norm, equivalent to $R_{1/3}(M)$ up to log factors): $\rho(M) \leq c \cdot (\log_2 g) / (\log_2 n)$. One sign matrix M with $\rho(M) \cdot \log_2 n > c \cdot \log_2 \gamma_2(M)$ for $c = 4$ falsifies the conjecture.

Rationale

Sign matrices with arithmetic structure that forces $|\det M|$ to absorb a large 2-power (Sylvester-Hadamard, IP_k , Paley) are precisely those where the $\mathbb{Z}$ -module $\text{cok}(M)$ is far from the Cohen-Lenstra random model — and these are also the canonical hard CC functions (IP_k achieves $R = \Omega(k)$, Hadamard achieves $\gamma_2 = \Theta(\sqrt{n})$). Conversely, low-CC matrices admit small monochromatic-rectangle covers, which when lifted to $\mathbb{Z}$ force $\text{cok}(M)$ to have small 2-Sylow rank because the cover provides a near-rank-1 integer decomposition. The Hall-algebra 2-Sylow type encodes finer structure than rank alone (it tracks the elementary-divisor partition), so it can detect structured matrices that fool spectral / Frobenius-norm CC bounds.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1703.01572v1] The Smith normal form of a specialized Giambelli-type matrix
- [2211.09736v2] Equidistributions of Sign Patterns of the Liouville Function and Normal Numbers
- [1602.00166v2] Smith Normal Form in Combinatorics - [1211.0398v1] Extending valuations to formal completions - [1503.07648v2] Sign rank versus VC dimension

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cbb8938a.py", line 151, in <module>
    trial = run_trial(seed)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cbb8938a.py", line 122, in run_trial
    det = abs(det_mod_2k(M, n * math.ceil(math.log2(n))))
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cbb8938a.py", line 94, in det_mod_2k
    factor = (A[j][i] * pow(A[i][i], -1, 2**k)) % (2**k)
              ~~~~~
```

ValueError: base is not invertible for the given modulus

Judge reasoning

The test crashed with a ValueError in det_mod_2k (non-invertible base modulo 2^k) before any of the 630 instances produced data, so neither the support nor falsification condition can be evaluated. The critic also challenged due to the crash. | next: Replace the modular inverse-based Gaussian elimination with a direct integer determinant (e.g., sympy.Matrix.det or fraction-free Bareiss algorithm) and re-run the 630-instance sweep.

vdW 3-AP Density of MAJ-Lifted NW Designs Tracks Overlap k

- **Verdict:** INCONCLUSIVE
- **Bridge:** Arithmetic Ramsey theory — quantitative van der Waerden / Behrend AP-density bounds (Behrend 1946; Szemerédi 1975; Bloom-Sisask 2020 ‘Breaking the logarithmic barrier in Roth’s theorem’; Kelley-Meka 2023). The 3-AP density functional ρ_3 on Boolean sequences is a comparison/counting statistic (no F_q polynomial extension preserves monochromatic-AP counts under polynomial ring lifts), Aaronson-Wigderson algebrization-safe. Distinct from Chang spectrum / Plünnecke / additive-energy / Sidon attempts: those measure Fourier-spectrum or sumset-cardinality on truth-table or wire-label objects, while ρ_3 counts monochromatic 3-APs in the OUTPUT POSITION sequence of a MAJ-lifted design. ArXiv ‘van der Waerden’ AND (‘Nisan-Wigderson’ OR ‘NW design’ OR ‘combinatorial design pseudorandom generator’) returns 0 direct hits and <5 adjacent papers (Behrend-style cap-set constructions; never on NW designs). $\times$ Nisan-Wigderson PRG combinatorial-design overlap parameter k controlling seed-blow-up 2^k (Nisan-Wigderson 1994; Impagliazzo-Wigderson 1997; Klivans-van Melkebeek 2002; Kabanets-Impagliazzo 2004), in the BPP=P-under-hardness regime. The invariant is STRUCTURAL on $D = (S_1, \dots, S_m)$ (two designs supporting the same Boolean PRG function can yield very different ρ_3), shielding from Razborov-Rudich natural proofs; the MAJ-lift breaks F_2 ring structure since MAJ has no $o(\sqrt{l})$ -degree polynomial approximation (Razborov-Smolensky), shielding from algebrization.
- **Recorded:** 2026-05-19 02:06 UTC
- **Entry ID:** 4bcf619e2ecb

Statement

Let $D = (S_1, \dots, S_m)$ be an (l, k) -NW design on $[n]$ with $|S_i| = l$ and $\max_{\{i < j\}} |S_i \cap S_j| \leq k$ ($k \geq 1, l \geq 4$). Define the MAJ-lift $\chi_D : \{0, 1\}^n \rightarrow \{0, 1\}^m$ by $\chi_D(z)_i = \text{MAJ}(z|_{S_i})$ (lexicographic tie-break = 0). Let $\rho_3(D) := \mathbb{E}_{\{z \sim U(\{0, 1\}^n)\}} [\#\{(a, d) : 1 \leq a, a+2d \leq m, d \geq 1, \chi_D(z)_a = \chi_D(z)_{a+d} = \chi_D(z)_{a+2d}\}] / N_3(m)$ where $N_3(m)$ is the total number of 3-APs in $[m]$. Conjecture: there exist universal constants $0 < c \leq C < \infty$ (independent of n, l, k, m) such that for every such NW design D , $c \cdot (k/l)^{1/2} \leq \rho_3(D) - 1/4 \leq C \cdot (k/l)^{1/2}$. A single design with $\rho_3(D)$ outside this band falsifies it.

Rationale

NW PRG fooling power degrades like 2^k in the overlap parameter, but k is a global combinatorial property of the design that should manifest in second-order output statistics. Monochromatic 3-AP density is the simplest non-trivial AP statistic and probes 3-wise correlation across position triples $(i, i+d, i+2d)$ — exactly the structure that Behrend/Bloom-Sisask measure. If the conjecture holds, ρ_3 gives a black-box, sub-quadratic-time structural certificate of design overlap, which feeds the Kabanets-Impagliazzo hardness-vs-randomness program by lower-bounding the seed-length penalty of any design substrate from output statistics alone.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c15756de.py", line 79, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c15756de.py", line 54, in run_trial
    chi_D = [MAJ(z, s) for z in range(2**n) for s in S]
            ^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c15756de.py", line 30, in MAJ
    count = sum(1 for x in z if x in S)
            ^^^^^^^^^^^^^^^^^^^^^^^^^
```

TypeError: 'int' object is not iterable

Judge reasoning

The test crashed with a `TypeError` in the MAJ function (iterating over an int instead of a bit vector) before producing any data, so neither the Pearson correlation nor the R-ratio criterion could be evaluated. | next: Fix the MAJ implementation to iterate over the bits of z restricted to S (e.g., represent z as a tuple/list of bits or use bitmask operations) and rerun the pre-registered 30-seed sweep.

Lorentzian Defect of Cut Polynomial Lower-Bounds Max-CUT SoS-2 Gap

- **Verdict:** INCONCLUSIVE
- **Bridge:** Lorentzian polynomials / Brändén-Huh log-concavity and the matroid Hodge-Riemann relations of Adiprasito-Huh-Katz — the normalized log-concavity condition $\tilde{a}_{j-1} \cdot \tilde{a}_{j+1} \leq \tilde{a}_j^2$ with $\tilde{a}_j := a_j / C(m, j)$ characterizing when a homogenized univariate coefficient vector lies in the closure of the Lorentzian cone; the defect $LD := \max_j$

$\max(0, \log(\tilde{a}_{j-1} \cdot \tilde{a}_{j+1} / \tilde{a}_j^2))$ is a sup-of-log-ratios functional that uses only integer counts, binomial normalisation, and log of ratios, with no F_q polynomial extension (binomial normalisation and log are not preserved under polynomial ring lifts), so the bridge is Aaronson–Wigderson algebrization-safe. Distinct from blacklisted Lee–Yang zero cluster angle (complex roots of the same polynomial), Kashin L^1 -flatness (single-eigenvector L^1/L^2 ratio), Curto–Fialkow Hankel-apolar rank (real moment problem on L_G spectrum), free 4-cumulant excess (Voiculescu NC partitions on squared singular values), Fekete capacity (transfinite diameter of Laplacian spectrum), hypercontractive 4/2 ratio (norm ratio of the truth table), and Schur–Horn majorization gap (NATURAL_PROOFS-rejected). An arXiv search ‘Lorentzian polynomial’ AND (‘Max-Cut’ OR ‘Delorme–Poljak’ OR ‘SoS integrality gap’) returns 0 direct hits with <5 adjacent papers (Anari–Liu–Oveis–Gharan–Vinzant on log-concavity for matroid-base sampling, never on SoS-2 integrality gaps). × Degree-2 Sum-of-Squares / Delorme–Poljak Max-CUT integrality gap $\rho(G) := n \cdot \lambda_{\max}(L_G) / (4 \cdot \text{MC}(G)) - 1$ for connected 3-regular graphs G , in the Goemans–Williamson 1995 / Delorme–Poljak 1993 / Schoenebeck 2008 / Barak–Steurer 2014 SOS_HIERARCHY regime; the canonical stepping stone toward unconditional SoS-degree- d Max-CUT lower bounds via matroidal obstructions on degree-2 pseudoexpectation moment matrices.

- **Recorded:** 2026-05-19 02:13 UTC
- **Entry ID:** dad52f056827

Statement

Let G be a connected 3-regular graph on n vertices with $m = 3n/2$ edges and cut-size distribution $a_j(G) := \#\{S \subseteq V(G) : |\delta_G(S)| = j\}$, and let $\text{LD}(G) := \max\{0, \max_{\{1 \leq j \leq m-1, a_{j-1}a_{j+1} > 0\}} \log((\tilde{a}_{j-1} \cdot \tilde{a}_{j+1}) / \tilde{a}_j^2)\}$ with $\tilde{a}_j := a_j / C(m, j)$ be the normalized Lorentzian defect of the cut-generating polynomial $P_G(z) = \sum_j a_j z^j$. Then for every such G , $\text{LD}(G) \geq 0.05 \cdot \rho(G)$, where $\rho(G) = n \cdot \lambda_{\max}(L_G) / (4 \cdot \text{MC}(G)) - 1$ is the Delorme–Poljak SoS-2 integrality gap. Equivalently, any connected 3-regular G whose cut-polynomial coefficient sequence is normalized log-concave (Lorentzian, $\text{LD}=0$) must have $\rho(G) = 0$ (SoS-2 tight).

Rationale

The Brändén–Huh Lorentzian property of a homogeneous polynomial is exactly the matroidal Hodge–Riemann signature condition; if $P_G(z, w)$ (homogenized) were Lorentzian, the symmetric bilinear form on $(R^{\wedge} V, J_G)$ induced by the cut-direction Gram would inherit a one-positive-eigenvalue signature compatible with the Laplacian’s spectral gap, forcing the SoS-2 pseudoexpectation cone to align with the integer-cut cone. Conversely the Delorme–Poljak gap measures the angle between the leading Laplacian eigenvector and the cut polytope’s max-cut face, an angle that combinatorially obstructs the Newton-inequality regularity of (a_j) . Bridging matroid log-concavity to the SoS-2 obstruction is essentially unexplored: it bypasses the NATURAL_PROOFS trap that killed Schur–Horn (the invariant is structural on the graph, not on a truth table) and the ALGEBRIZATION trap that closes ring-only constructions (binomial normalisation and log of ratios admit no F_q polynomial surrogate).

Novelty

- Judge: NOVEL over 2 arXiv hits

Top hits consulted: - [s2:2407.19905] The Bidirected Cut Relaxation for Steiner Tree has Integrality Gap Smaller Than 2 - [LIPIcs.SWAT.2024.23] A Logarithmic Integrality Gap for Generalizations of Quasi-Bipartite Instances of Directed Steiner Tree

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b1bc8fd9.py", line 103, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b1bc8fd9.py", line 71, in run_trial
    lambda_max = laplacian_eigenvalues(n, m)[0]
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b1bc8fd9.py", line 42, in laplacian_eig
    return [eigenvalue for eigenvalue in numpy.linalg.eigvalsh(L) if eigenvalue >= 0]
           ^^^^^^
```

NameError: name 'numpy' is not defined

Judge reasoning

The test crashed with a NameError ('numpy' not defined) before any of the 210 instances were evaluated, so no data on the $LD(G) \geq 0.05 \cdot p(G)$ inequality was produced. Per Rule 2, a crashed test yields INCONCLUSIVE. | next: Fix the import (add 'import numpy' or use the already-imported alias) in laplacian_eigenvalues and rerun the full 210-instance sweep.

Level-1 Fourier Mass Fraction of Term Family Bounds Monotone CLIQUE DNF

- **Verdict:** INCONCLUSIVE
- **Bridge:** Fourier-analytic hypercontractivity / KKL low-level concentration on syntactic term-incidence empirical measures $\times$ Monotone DNF leaf size $L(F)$ for the k -CLIQUE indicator F^*_v on K_v with $k = \lfloor \sqrt{v} \rfloor$, $n = v(v-1)/2$ edge variables (Razborov 1985 / Andreev / Alon-Boppana monotone regime)
- **Recorded:** 2026-05-19 02:22 UTC
- **Entry ID:** 9ec4f1b18412

Statement

For a monotone DNF $F = \bigvee_{i=1}^s T_i$ over n variables, view the term-support indicators $\{1\{T_i\}\}_{i=1}^s \subset \{0,1\}^n$ as an empirical sample and let $\hat{\mu}_F(S) := (1/s) \sum_i (-1)^{|S \cap T_i|}$ be the Walsh-Fourier coefficient of the empirical measure at $S \subseteq [n]$. Define the level-1 low-frequency fraction $\lambda(F) := (\sum_{|S|=1} \hat{\mu}_F(S)^2) / (\sum_{1 \leq |S| \leq 2} \hat{\mu}_F(S)^2)$ and the complexity measure $\mu(F) := -\log_2 \max(\lambda(F), 1/n)$. Conjecture: (a) for every monotone DNF F of leaf size $L(F) = \sum_i |T_i|$, $\mu(F) \leq 2 \cdot \log_2(L(F)+2) + 4$; (b) for the k -CLIQUE indicator F^*_v on K_v with $k = \lfloor \sqrt{v} \rfloor$, $\mu(F^*_v) \geq (1/8) \cdot v$ for all $v \geq 4$. Equivalently, $\log_2 L(F^*_v) \geq \Omega(v) = \Omega(\sqrt{v})$, a monotone CLIQUE leaf-size lower bound for $k = \sqrt{v}$.

Rationale

By KKL/Friedgut, a Boolean-valued empirical measure with level-1 mass dominating its low-level spectrum behaves like a junta of few variables — exactly the regime of small DNFs whose terms touch few variables and whose Fourier signal sits on singletons. The k -CLIQUE indicator's term family is the $(v \text{ choose } k)$ edge-sets of k -cliques, an S_v -equivariant design that uniformizes level-1 character sums and pushes ≤ 2 -level mass onto edge-pair correlations, suppressing λ to $O(1/v)$. The measure is SYNTACTIC on the term list (two monotone DNFs computing the same Boolean function can have very different μ), so it is shielded from Razborov-Rudich, and uses $\log_2$ of an $L^{2/L-2}$ fraction over level-1 and level-2 characters — a non-ring functional with no F_q polynomial lift (Aaronson-Wigderson algebrization-safe).

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 196.97s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b489d7d.py", line 94, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b489d7d.py", line 50, in run_trial
    mu_F = complexity_measure(level_1_low_frequency_fraction(F))
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b489d7d.py", line 28, in level_1_low_f
    sum_level_1 = sum(walsh_fourier_coefficient(F, {j+1 for j in range(n) if (i >> j) & 1})**2 for i in r
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b489d7d.py", line 28, in <genexpr>
    sum_level_1 = sum(walsh_fourier_coefficient(F, {j+1 for j in range(n) if (i >> j) & 1})**2 for i in r
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b489d7d.py", line 23, in walsh_fourier
    sum_term += F[i] * term
                ^^^
```

IndexError: list index out of range

Judge reasoning

The test crashed with an IndexError in walsh_fourier_coefficient before producing any RESULT line, so neither ensemble (A) nor (B) was evaluated. Per the rules, a crashed test yields INCONCLUSIVE. | next: Fix the indexing bug (likely a 0-based vs 1-based variable indexing mismatch between F and the subset construction {j+1 ...}) and re-run all 30 seeds plus $v \in \{4..8\}$.

Dilworth Width of Gate-Cone Poset Lower-Bounds AC^0 PARITY Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Order theory — Dilworth's theorem and König-matching computation of poset width on Boolean-lattice containment orders (Dilworth 1950 'A decomposition theorem for partially ordered sets'; Mirsky 1971; Fulkerson 1956 'Note on Dilworth's decomposition theorem'; Trotter 1992 'Combinatorics and Partially Ordered Sets'). A non-ring set-inclusion functional: width via Hopcroft-Karp matching on the strict-comparability bipartite graph uses only Boolean subset tests and integer matching, with NO polynomial F_q extension (containment of supports in 2^2 is not preserved under polynomial-ring lifts of the Boolean ring), so the bridge is Aaronson-Wigderson algebrization-safe. $\times AC^0$ depth-d circuit size $s(C)$ for the explicit n-bit PARITY function, in the Håstad 1986 switching-lemma / Razborov-Smolensky / Beame 1994 regime; the canonical $2^{\Theta(n \{1/(d-1)\})}$ target and the $V^0 \sqcap V^0(\oplus)$ bounded-arithmetic stepping stone.
- **Recorded:** 2026-05-19 02:31 UTC
- **Entry ID:** f2533eda1d01

²_n

Statement

Let C be an AC^0 circuit of depth d and size $s = |V(C)|$ with $C \equiv \text{PARITY}_n$, and define the gate-cone poset $P(C)$ on $V(C)$ by $g \sqsubseteq h$ iff $\text{support}(g) \subseteq \text{support}(h)$ where $\text{support}(g) := \{x \in \{0,1\}^n : g(x)=1\}$; let $w(C) := \text{width}(P(C))$ be Dilworth's maximum-antichain size, computed by König matching as $|V(C)| - \mu(B(C))$ on the strict-comparability bipartite graph $B(C)$. Conjecture: there exists an absolute constant $c > 0$ such that $w(C) \geq c \cdot \log_2(s)$ for every AC^0 circuit C computing PARITY_n ; equivalently $s \leq 2^{\{w(C)/c\}}$. A single AC^0 instance $C \equiv \text{PARITY}_n$ with $w(C)/\log_2(s) < c$ for every fixed c refutes it.

Rationale

A pairwise-incomparable family of gate cones is a 'structural fooling set' in the containment lattice — antichains here play the role of rectangle-discrepancy fooling sets that LIFTING transfers from query (decision-tree certificates) to communication, but for SIZE rather than depth, since min chain cover (Dilworth dual) equals the minimum number of monotone monocoloured chains needed to schedule the gate evaluation. The invariant is structural on the circuit wiring — two AC^0 circuits with identical truth table (e.g. one with redundant subsumed gates) can have very different widths, shielding from Razborov-Rudich natural proofs — and the Boolean-lattice subset relation does not algebrize.

Novelty

- Judge: NOVEL over 8 arXiv hits

Top hits consulted: - [2304.02770v2] Tight Correlation Bounds for Circuits Between AC^0 and TC^0 - [2511.07739v3] A Lower Bound for the Fourier Entropy of Boolean Functions on the Biased Hypercube - [2504.19966v3] Quantum circuit lower bounds in the magic hierarchy - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1505.02372v3] On Asymptotic Gate Complexity and Depth of Reversible Circuits With Additional Memory

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_00eea011.py", line 133, in <module>
    trial = run_trial(seed)
            ~~~~~^~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_00eea011.py", line 110, in run_trial
    w_C = hopcroft_karp(build_bipartite_graph(C, assignments))
            ~~~~~^~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_00eea011.py", line 42, in build_bipartite
    G.extend([(g, h) for h in C if set(support(g, assignments)) <= set(support(h, assignments))])
            ~~~~~^~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_00eea011.py", line 37, in support
    return [i for i, bit in enumerate(assignments) if gate[i] == 1]
            ~~~~~^~~~~
```

TypeError: 'int' object is not subscriptable

Judge reasoning

The test crashed with a `TypeError` in the `support()` function ('int' object is not subscriptable), indicating a bug in how gates are represented or indexed; no instances were evaluated, so neither the support nor falsification condition can be assessed. | next: Fix the gate representation in `support()`: ensure each gate is stored as a truth-table vector (list/array indexable by assignment index) rather than a scalar, then re-run the 360-instance sweep.

Mean Log-Bias of Internal Gates Lower-Bounds AC^0 PARITY Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Information geometry — mean Kullback-Leibler divergence from Bernoulli(1/2) of the internal-gate bias empirical measure (Csiszár 1975 ‘I-divergence geometry of probability distributions’; Amari-Nagaoka 2000 ‘Methods of Information Geometry’; Cover-Thomas 2006). The functional uses log of empirical probabilities and per-gate $\min(p, 1-p)$, with NO polynomial F_q extension (log and min are not preserved under polynomial-ring lifts of the Boolean ring), so the bridge is Aaronson-Wigderson algebrization-safe. An arXiv search ‘mean gate-bias KL divergence’ AND (‘ AC^0 ’ OR ‘switching lemma’ OR ‘PARITY lower bound’) returns 0 direct hits and <5 adjacent papers (all on noise-sensitivity of the OUTPUT function in the Kahn-Kalai-Linial / Bourgain / O’Donnell tradition, never on the empirical measure of INTERNAL gate biases as a structural circuit invariant). Distinct from blacklisted Lasserre Moment-Rank Gap on accept set (truth-table side; NATURAL_PROOFS-rejected), SoS Cone-Gram Stable Rank (Schatten-1/ ∞ spectrum on a Gram matrix), Talagrand L^1 influence spread / Fourier spectral entropy / Mansour L^1 / Bonami 4/2 / Bourgain NSM (all on the OUTPUT polynomial’s Fourier transform, not internal gate biases), and Dilworth width / Gromov hyperbolicity / Mostar / Forman-Ricci (all combinatorial-graph functionals on the DAG, never an information-geometric KL-from-uniform on Bernoulli pseudoexpectations). $\times AC^0$ depth-d circuit size $s(C)$ for the n-bit PARITY function in the Håstad 1986 switching-lemma / Razborov-Smolensky / Beame 1994 regime, targeting the canonical $2^{\Theta(n)}\{1/(d-1)\}$ lower bound and the $V^0 \sqsubseteq V^0(\oplus)$ bounded-arithmetic separation. Gate biases p_g are exactly the SoS-degree-2 pseudoexpectations $\mathbb{E}[1_{\{g(x)=1\}}]$ under the uniform pseudo-distribution, so the invariant is a per-gate average of an SoS-2 functional — a Lasserre-pseudomoment-based circuit invariant in the Barak-Steurer 2014 / Grigoriev SoS hierarchy lineage.
- **Recorded:** 2026-05-19 02:40 UTC
- **Entry ID:** 8f8f4d4c4a54

Statement

Let C be an AC^0 depth-d circuit on n inputs with m internal gates $g_1, \dots, g_m$ (including the output, excluding input literals). Let $p_g := \Pr_{x \sim U_n}[g(x)=1]$ (estimated by $N=\min(2^n, 2^{14})$ random samples) and define $\psi(C) := (1/m) \cdot \sum_g [-\log_2(2 \cdot \max(\min(p_g, 1-p_g), 1/(2m)))]$. CONJECTURE: (a) $\psi(C) \leq \log_2 m + 1$ for every AC^0 circuit C [universal cap]; (b) for every AC^0 depth-d circuit C exactly computing PARITY_n, $\psi(C) \geq 0.2 \cdot n^{\{1/(d-1)\}}$ [PARITY witness]. Combining (a)+(b) yields $\log_2 \text{size}(C) \geq 0.2 \cdot n^{\{1/(d-1)\}} - 1$.

Rationale

Håstad’s switching lemma forces internal AC^0 gates whose downstream cone reaches PARITY’s perfectly balanced output to each carry exponentially small $\min(p_g, 1-p_g) \approx 2^{\{-n\{1/(d-1)\}\}}$; the mean-log-bias condenses this ‘bias bookkeeping’ into a single SoS-pseudoexpectation invariant. The invariant is RELATIONAL on the gate-bias multiset of C (two circuits with identical Boolean function can have very different gate-bias distributions because they wire intermediate gates differently), shielding from Razborov-Rudich natural-proofs and sidestepping the truth-table trap (j) that killed prior Lasserre Moment-Rank attempts. The natural S_n action on input bits permutes inputs but leaves the multiset $\{p_g\}$ fixed (uniform measure is S_n -invariant), giving the required non-trivial group equivariance.

Novelty

- Judge: NOVEL over 10 arXiv hits

Top hits consulted: - [2304.02770v2] Tight Correlation Bounds for Circuits Between AC^0 and TC^0
- [2604.11744v1] KL Divergence Between Gaussians: A Step-by-Step Derivation for the Variational

Autoencoder Objective - [2504.19966v3] Quantum circuit lower bounds in the magic hierarchy - [1205.6363v1] What Should Developers Be Aware Of? An Empirical Study on the Directives of API Documentation - [2103.17141v3] Yoneda's lemma for internal higher categories

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_916b5b32.py", line 161, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_916b5b32.py", line 105, in run_trial
    circuit = generate_hastad_circuit(n, d)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_916b5b32.py", line 63, in generate_hastad
    term &= ('AND', j)
```

TypeError: unsupported operand type(s) for &=: 'bool' and 'tuple'

Judge reasoning

The test crashed with a `TypeError` in `generate_hastad_circuit` (`term &= ('AND', j)` misuses `&` on `bool` and `tuple`) before producing any data, so neither the universal cap nor the PARITY lower bound was empirically evaluated. | next: Fix the circuit construction bug (initialize `term` as a tuple/AST node rather than a `bool`, e.g., `term = ('AND',)` and append children) and re-run the 30-seed sweep over (n,d) cells for E1/E2/E3.

Sandpile Group Exponent Lower-Bounds Tseitin Tree-Resolution Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Sandpile (critical) group exponent / largest invariant factor of the reduced graph Laplacian — the integer $d_{\{n-1\}}(\tilde{L}G) := \text{top invariant factor in the Smith normal form of } \tilde{L}G$ (Lorenzini 1991 'A finite group attached to the Laplacian of a graph'; Bacher-de la Harpe-Nagnibeda 1997 'The lattice of integer flows and the lattice of integer cuts of a finite graph'; Biggs 1999 'Chip-firing and the critical group of a graph'; Wood 2017 'The distribution of sandpile groups of random graphs', Cohen-Lenstra heuristics). For 3-regular expanders Wood's distribution predicts $K(G)$ is concentrated on a single $\approx$ cyclic factor of order $\approx$ #spanning trees (so $\log_2 d_{\{n-1\}} = \Theta(n)$), while for non-expanders (cycle-augmented trees, prism $C_n \square K_2$, bottleneck-glued cubic graphs) $K(G)$ splits into many small factors so $\log_2 d_{\{n-1\}} = O(\log n)$. The functional uses integer Smith-form (row/column operations + integer gcd) on a $(n-1) \times (n-1)$ integer matrix of entries ≤ 3 — $O(n^3)$ bigint ops, no F_q polynomial extension preserves the LARGEST invariant factor under polynomial ring lifts (Smith form is module-theoretic, not ring-polynomial), so the bridge is Aaronson-Wigderson algebrization-safe. Distinct from the blacklisted 2-Sylow RANK of $K(G)$ ($\dim_{F_2} K(G)/2K(G)$, an integer in $\{0,1,\dots,O(\log n)\}$ that does NOT separate expander from non-expander 3-regular graphs — for both, the 2-Sylow rank is $\approx O(1)$ by Cohen-Lenstra; only the EXPONENT scales with the spectral gap), distinct from Baker-Norine ω -gonality (a divisor-rank with Dhar-burning chip-firing predicate, computes $r_{BN}(D)$ for a (G,ω) -specific divisor, NOT a global invariant of $\tilde{L}$), and distinct from all spectral/metric/topological Tseitin attempts (effective resistance, $p=3$ modulus, Fiedler IPR, signed-Laplacian holonomy, Szegedy phase gap, Viennot heap, hook-length $\dim$, dismantlability core, cubical β_1). An arXiv search for 'sandpile group exponent' AND ('Tseitin' OR 'resolution refutation' OR 'tree-Resolution' OR 'proof complexity') returns 0 direct hits and <5 adjacent papers (all on Cohen-Lenstra distribution of $K(G)$ for random graphs or chip-firing dynamics, never on CNF refutation). $\times$ Tree-like Resolution / DPLL refutation size

$t(T(G,\omega))$ for Tseitin XOR formulas $T(G,\omega)$ on connected 3-regular graphs G with odd charge $\omega : V(G) \rightarrow F_2$ (Urquhart 1987 ‘Hard examples for resolution’; Ben-Sasson-Wigderson 2001 ‘Short proofs are narrow — resolution made simple’; Alekhnovich-Razborov 2001 expansion regime; Galesi-Talebanfard-Torán 2020 branchwidth bounds), in the Cook-Reckhow-Krajíček bounded-arithmetic framework where $\neg T(G,\omega)$ is Σ^b_0 and $\log_2 t(T(G,\omega))$ lower-bounds V^0 -witnessing complexity; the canonical Frege-depth stepping stone via Pudlák-Buss feasible interpolation toward the open Tseitin Extended-Frege lower bound.

- **Recorded:** 2026-05-19 02:48 UTC
- **Entry ID:** b1e77f376f5e

Statement

For every connected 3-regular graph G on n vertices and every odd Tseitin charge $\omega : V(G) \rightarrow F_2$ ($\sum_v \omega(v) = 1$), the tree-Resolution / DPLL refutation size satisfies $\log_2 t^*(T(G,\omega)) \geq c \cdot \log_2 d_{\{n-1\}}(\tilde{L}G) - C$ for absolute constants $c > 0$, $C \geq 0$, where $d_{\{n-1\}}(\tilde{L}G)$ is the largest invariant factor of the reduced combinatorial Laplacian $\tilde{L}G$ (any $(n-1) \times (n-1)$ principal submatrix of L_G in its Smith normal form). Equivalently, $\nu(G) := \log_2 d_{\{n-1\}}(\tilde{L}G)$ is a poly-time computable graph invariant with $\nu(G) = O(1)$ for graphs of bounded carving-width / bandwidth (non-expanders) and $\nu(G) = \Omega(n)$ for spectral expanders, and Tseitin tree-Resolution length is $\geq 2^{\Omega(\nu(G))}$. A single counterexample (G,ω) with $\nu(G) \geq \alpha \cdot n$ and $\log_2 t^*(T(G,\omega)) \leq \beta \cdot n$ for $\beta < c \cdot \alpha$ refutes the conjecture.

Rationale

Sandpile-group exponent is the integer-arithmetic shadow of the spectral gap: by Kirchhoff’s theorem $\prod d_i = \#\text{spanning trees}$, and for 3-regular expanders Wood’s Cohen-Lenstra distribution concentrates $K(G)$ on one $\approx$ cyclic factor, so $d_{\{n-1\}} \approx \prod d_i \approx c^n$; for non-expanders the invariant factors split (high 2-Sylow rank or many small primes), forcing $d_{\{n-1\}}$ to drop to $O(\text{poly } n)$. Tseitin lower bounds via Ben-Sasson-Wigderson width require a ‘global cycle-space mixing’ obstruction, exactly what a large exponent encodes — a large cyclic factor of $K(G)$ means the cycle space $Z^E \supset \ker \tilde{L}$ is densely packed by short integer relations only realisable along long expanding chains, which is precisely what blocks short Resolution refutations of $T(G,\omega)$. Unlike the 2-Sylow rank (which is $O(1)$ for both expanders and non-expanders by Cohen-Lenstra), the exponent scales with the spectral gap and thus is the right integer-arithmetic surrogate for Urquhart’s expansion lower bound.

Novelty

- Judge: NOVEL over 8 arXiv hits

Top hits consulted: - [1904.12209v2] Sandpile monomorphisms and limits - [0610012v1] Tevatron-for-LHC Report of the QCD Working Group - [2008.13672v1] Computing sandpile configurations using integer linear programming - [2309.16111v2] The relational complexity of linear groups acting on subspaces - [1908.06409v1] Schur multipliers of special p -groups of rank 2

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_814a9fa0.py", line 127, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_814a9fa0.py", line 106, in run_trial
    M.append([L_tilde_G[i][j] for j in range(i + 1, n - 1)])
            ^^^^^^^^^^^^^^^
```

IndexError: list index out of range

Judge reasoning

The test crashed with an `IndexError` while constructing the reduced Laplacian submatrix, so no data was produced and neither the support nor falsification conditions could be evaluated. | next: Fix the off-by-one in building L_tilde_G (use $\text{range}(i+1, n-1)$ on a correctly sized $(n-1) \times (n-1)$ matrix, or build the full upper triangle then take the principal submatrix) and rerun the 30-seed sweep across $n \in \{8, \dots, 20\}$ for ensembles E and N .

SAW Connective Growth of Gate DAG Lower-Bounds $\text{ACC}^0[2]$ for MOD_3

- **Verdict:** INCONCLUSIVE
- **Bridge:** Statistical mechanics of self-avoiding walks — Hammersley’s connective constant $\mu(G) = \lim_L c_L(G)^{1/L}$ on finite graphs (Hammersley 1957; Madras-Slade 1993 ‘The Self-Avoiding Walk’; Duminil-Copin-Smirnov 2012 honeycomb constant $\sqrt{(2+\sqrt{2})}$; Grimmett-Li 2017 ‘Connective constants and height functions for Cayley graphs’). A combinatorial-physics functional based on the set-predicate ‘walk is self-avoiding’ — counts can be enumerated by DFS with a visited-set, an $O(L \cdot \deg^L)$ routine; no F_q polynomial extension exists since ‘no vertex repeats’ is a Boolean lattice predicate on power-set membership, not a polynomial identity, so the bridge is Aaronson-Wigderson algebrization-safe. The invariant is STRUCTURAL on the gate-incidence multigraph G_C (two $\text{ACC}^0[2]$ circuits computing the same Boolean function can have different connective growth), shielding from Razborov-Rudich natural proofs. Distinct from blacklisted ACC^0 attempts: Plünnecke doubling (sumset on top-Fourier supports, NATURAL PROOFS-rejected), Sidon r^* (max representation in F_2^N on MOD-gate outputs), Chang spectrum (Fourier η -cardinality on wire labels), additive energy (L^2 on out-degree sequence), Gromov 4-point δ (max over distance differences), Forman-Ricci (Bochner edge-vertex sum), Mostar (edge distance asymmetry), Dilworth width (König matching on poset), SoS cone-gram stable rank (Schatten $1/\infty$ on Gram), Mean log-bias (KL on internal gate biases) — none uses SAW enumeration on the gate-incidence multigraph. An arXiv search ‘self-avoiding walk’ AND (‘ ACC^0 ’ OR ‘Williams algorithmic method’ OR ‘NEXP circuit lower bound’) returns 0 direct hits and <5 adjacent papers (all on SAWs on Cayley graphs in mathematical physics, never on ACC circuits). $\times \text{ACC}^0[2]$ circuit size $s(C)$ and depth $d(C)$ for the explicit function MOD_3 on n inputs ($q=3$ coprime to $m=2$), in the Williams 2011 / Williams 2014 ‘Nonuniform ACC circuit lower bounds’ / Murray-Williams 2018 algorithmic-method regime targeting $\text{NEXP} \not\subseteq \text{ACC}^0$ and explicit $\text{ACC}^0[2]$ lower bounds for MOD_3 . The canonical Razborov-Smolensky stepping stone: MOD_3 has no $o(\sqrt{n})$ -degree F_2 polynomial approximation, so any $\text{ACC}^0[2]$ biased toward MOD_3 must wire its DAG to accommodate this — predicting a structural lower bound on a non-spectral combinatorial invariant of the gate graph.
- **Recorded:** 2026-05-19 02:53 UTC
- **Entry ID:** 988e93f4afe3

Statement

Let C be an $\text{ACC}^0[2]$ circuit on n inputs with $s = |\text{gates}(C)|$, treated as an undirected multigraph G_C on gate vertices with an edge for every wire (parallel edges retained). Fix walk-length $L := \lfloor 2 \cdot \log_2(s+2) \rfloor$ and let $c_L(G_C)$ be the number of self-avoiding walks of length L starting from the output gate. Define the SAW connective growth $\rho(C) := \log_2(c_L(G_C)+1)/L$. The conjecture: for every $\text{ACC}^0[2]$ circuit C whose Boolean function f_C agrees with MOD_3 on at least $0.45 \cdot 2^n$ uniform inputs (bias $|\Pr[f_C = \text{MOD}_3] - 1/3| \geq 1/9$), the growth satisfies $\rho(C) \geq (1/8) \cdot \log_2(s) - 1$; equivalently, $c_L(G_C) \geq s^{\lfloor L/8 \rfloor / 2} L$. One counterexample (a biased circuit with low ρ) refutes it.

Rationale

MOD_3 is Razborov-Smolensky-hard to F_2 -approximate by low-degree polynomials, so any $\text{ACC}^0[2]$ circuit that biases toward MOD_3 must repeatedly funnel many F_2 -linear gate outputs through interleaved AND/OR layers; this wiring creates the local branching and reconvergence

that drive SAW count growth on G_C . SAW growth is a non-spectral, non-Fourier coarse-geometric obstruction that survives where ring-level invariants algebrize, and it is structural rather than functional, so it sidesteps Razborov–Rudich largeness while remaining sensitive to the underlying $ACC^0[2]$ resource. This puts the SAW connective constant in the same coarse niche as Forman–Ricci or Gromov hyperbolicity but with a fundamentally different (set-predicate, lattice-walk) computation.

Novelty

- Judge: NOVEL over 7 arXiv hits

Top hits consulted: - [1302.2796v2] Endless self-avoiding walks - [1307.6288v2] Self-avoiding quantum walks: realisations in subspaces and limit theorems - [9302003v1] New Lower Bounds on the Self-Avoiding-Walk Connective Constant - [1009.3029v1] Invariant Spectral Hashing of Image Saliency Graph - [2206.07869v1] Let Invariant Rationale Discovery Inspire Graph Contrastive Learning

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c639def0.py", line 131, in <module>
    trial_result = run_trial(seed)
                    ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c639def0.py", line 66, in run_trial
    bias = estimate_bias(circuit, n)
           ^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c639def0.py", line 49, in estimate_bias
    result = simulate_circuit(circuit, n)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c639def0.py", line 38, in simulate_circuit
    b, a = stack.pop(), stack.pop()
           ^^^^^^^^^^^
```

IndexError: pop from empty list

Judge reasoning

The test crashed with an `IndexError` (pop from empty stack) in `simulate_circuit` before any retained instances could be evaluated, so no data was produced to assess the pre-registered support or falsification condition. | next: Fix the circuit simulator's stack-handling bug (ensure each gate has sufficient operands on the stack, likely by validating circuit generation or adjusting evaluation order) and rerun the 30-seed sweep.

Frobenius Fourier Defect of Barrington AND_n PBP Trajectory

- **Verdict:** INCONCLUSIVE
- **Bridge:** Non-abelian harmonic analysis / Diaconis–Shahshahani Fourier theory on the symmetric group S_5 (Diaconis 1988 ‘Group Representations in Probability and Statistics’; Diaconis–Shahshahani 1981; Saloff-Coste 1997) $\times$ Width-5 Barrington permutation branching programs (Barrington 1989) realizing NC^1 / Ben-Or–Cleve 1992 formula-to-ABP simulation
- **Recorded:** 2026-05-19 03:06 UTC
- **Entry ID:** 51aafa940751

Statement

Let P_n be Barrington's canonical width-5 PBP for AND_n on $n = 2^k$ inputs ($k \geq 2$), built recursively by the depth-1 commutator program $[a(x_1), b(x_2)]$ for AND_2 using fixed non-commuting 5-cycles $a=(1\ 2\ 3\ 4\ 5)$, $b=(1\ 3\ 5\ 2\ 4)$, nested via the standard Barrington substitution scheme to length $L(n)=4^{k/2}$. For each input $x \in \{0,1\}^n$ let $\pi_0(x)=e$, $\pi_i(x)=g_1(x)\dots g_i(x) \in S_5$ be the prefix products and let $\bar{M}(P_n, x) := (1/L(n)) \cdot \sum_{i=1}^{L(n)} \text{Perm}(\pi_i(x)) \in \mathbb{R}^{5 \times 5}$ be the empirical average of the 5×5 permutation-representation matrices along the trajectory. Define the Frobenius Fourier defect $\bar{D}(P_n) := \mathbb{E}_x \|\bar{M}(P_n, x) - (1/5) \cdot J\|_F^2$ averaged over 30 uniform-random inputs $x \in \{0,1\}^n$. Then for all $n \in \{4, 8, 16, 32\}$ the sequence $\bar{D}(P_n)$ is strictly monotone non-increasing in n , and every consecutive log-log decay slope $\alpha_n := \log_2(\bar{D}(P_{n/2})/\bar{D}(P_n))$ lies in the open interval $(0.5, 3.0)$. A single n where monotonicity or the slope band fails falsifies the conjecture.

Rationale

Barrington's PBP for AND_n is a rigid nested-commutator scaffold in S_5 , structurally different from an i.i.d. random walk; Diaconis-Shahshahani would predict Frobenius defect $\Theta(1/L) = \Theta(1/n^2)$ for a random walk of length $L=n^2$, so any observed slope strictly below 2 detects the commutator's structural memory and gives a non-trivial structural fingerprint of formula depth $d=\log_2 n$. The Frobenius L^2 ratio uses real Schatten-2 norm with no F_q polynomial extension (ring lifts of permutation matrices do not preserve $\|\cdot\|_F^2/\|\cdot\|_F^2$ ratios), so the bridge is Aaronson-Wigderson algebrization-safe; $\bar{D}$ depends on the PBP wiring (two PBPs computing AND_n can have very different $\bar{D}$), shielding from Razborov-Rudich.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5a7bd9ab.py", line 117, in <module>
    trial_result = run_trial(seed)
                   ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5a7bd9ab.py", line 63, in run_trial
    if x[literal_index - 1] == 0:
       ~~~~~
```

IndexError: tuple index out of range

Judge reasoning

Test crashed with IndexError before producing data, preventing verification of conjecture | next:
Fix index out-of-range error in test code by validating tuple bounds

Eigenvalue Gap in SOS Moment Matrices for Max-CUT Approximation

- **Verdict:** INCONCLUSIVE
- **Bridge:** REAL_ALGEBRAIC_GEOMETRY $\times$ SUM_OF_SQUARES_HIERARCHY
- **Recorded:** 2026-05-19 03:17 UTC
- **Entry ID:** b99dacf3b7fe

Statement

For any max-CUT instance on n vertices, if there exists a 0.879-approximator, then the corresponding degree- d SOS moment matrix M must satisfy that all eigenvalues of M lie within $[-1, 1]$. If M has an eigenvalue outside this interval, then no such approximator exists. This property $P(M)$ is preserved under polynomial reductions.

Rationale

The SOS hierarchy's moment matrices encode correlations between variables in the relaxation. A 0.879-approximator for max-CUT requires the matrix to capture the graph's structure tightly, restricting eigenvalues to $[-1, 1]$. Violating this bound would imply the relaxation cannot achieve the desired approximation, aligning with Bouland-Kothari's SOS lower bounds for planted clique. The property is reduction-invariant because reductions preserve the problem's algebraic structure.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_27a96c15.py", line 104, in <module>
    result = run_trial(seed)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_27a96c15.py", line 84, in run_trial
    eigenvalues = compute_eigenvalues(M)
                  ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_27a96c15.py", line 66, in compute_eigen
    y = matrix_multiplication(matrix, x)
      ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_27a96c15.py", line 35, in matrix_multip
    result[i][j] += A[i][k] * B[k][j]
                  ~~~~^
```

TypeError: 'float' object is not subscriptable

Judge reasoning

Test crashed due to a type error in matrix operations, preventing data collection. The pre-registered support condition cannot be evaluated without successful runs. | next: Debug the matrix multiplication code to handle float values properly and rerun tests with multiple seeds

Matroid Rank Complexity of Monotone DNF for k-CLIQUE

- **Verdict:** INCONCLUSIVE
- **Bridge:** Matroid Theory $\times$ Monotone Circuit Complexity
- **Recorded:** 2026-05-19 03:33 UTC
- **Entry ID:** 1c08aaba24ea

Statement

There exists a matroid M on the vertex set of the complete graph K_n such that (i) the rank function $r(M)$ is submodular under the conjunction of DNF terms, (ii) for any monotone DNF formula of size $\text{poly}(n)$, $r(M)$ is $O(\log n)$, and (iii) for the k -CLIQUE indicator function, $r(M)$ is $\Omega(n)$.

Rationale

The rank function of a matroid is inherently submodular, which aligns with requirement (i). By associating each term in the DNF with a subset in the matroid, the rank could measure the complexity of the DNF. For functions with small DNF size, the rank would be logarithmic, but for k -CLIQUE, which requires exponentially many terms, the rank would scale linearly with n . This connects matroid theory to circuit complexity, offering a new angle for lower bounds.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.23s

```
c_name': 'matroid_rank', 'metric_value': 25.0, 'instances_tested': 30, 'conjecture_holds': False, 'c
TRIAL: {'metric_name': 'matroid_rank', 'metric_value': 23.0, 'instances_tested': 30, 'conjecture_hol
TRIAL: {'metric_name': 'matroid_rank', 'metric_value': 9.0, 'instances_tested': 30, 'conjecture_hol
TRIAL: {'metric_name': 'matroid_rank', 'metric_value': 38.0, 'instances_tested': 30, 'conjecture_hol
TRIAL: {'metric_name': 'matroid_rank', 'metric_value': 16.0, 'instances_tested': 30, 'conjecture_hol
TRIAL: {'metric_name': 'matroid_rank', 'metric_value': 36.0, 'instances_tested': 30, 'conjecture_hol
TRIAL: {'metric_name': 'matroid_rank', 'metric_value': 19.0, 'instances_tested': 30, 'conjecture_hol
TRIAL: {'metric_name': 'matroid_rank', 'metric_value': 17.0, 'instances_tested': 30, 'conjecture_hol
```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7eea0a2f.py", line 93, in <module>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                        ~~~~~^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7eea0a2f.py", line 93, in <genexpr>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                        ~~~~~^
KeyError: 'seed'
```

Judge reasoning

Test crashed with KeyError 'seed' before producing valid results, preventing reliable evaluation of conjecture support/falsification. | next: Fix test code to handle seed tracking and re-run experiments with proper error handling

Communication Complexity Lower Bound for AC^0 PARITY via Real Algebraic Geometry

- **Verdict:** INCONCLUSIVE
- **Bridge:** COMMUNICATION_COMPLEXITY $\times$ REAL_ALGEBRAIC_GEOMETRY
- **Recorded:** 2026-05-19 03:40 UTC
- **Entry ID:** b77a7e4e1eda

Statement

For any AC^0 circuit C computing PARITY on n inputs, the communication complexity of the Karchmer-Wigderson game for C is $\Omega(\log n)$. This invariant $\psi(C)$ = communication complexity of C 's function is polynomial-time computable and invariant under input permutations.

Rationale

The Karchmer-Wigderson game links circuit size to communication complexity. By framing PARITY's complexity in real algebraic geometry (via polynomial representations of communication protocols), we isolate a structural invariant. This avoids natural proofs by focusing on non-trivial geometric obstructions to AC^0 computation.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.03s

```
'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 27, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 21, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 18, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 15, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 7, 'instances_tested': 1, 'conjec
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 25, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 23, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 9, 'instances_tested': 1, 'conjec
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 38, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 16, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 36, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 19, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 17, 'instances_tested': 1, 'conje
RESULT: SUPPORTED mean=20.766666666666666 std=9.03579302305866 support_fraction=1.0
```

Judge reasoning

Single-instance testing cannot meet the 80% support threshold for statistical validity. | next: Test ≥ 1000 instances to validate support fraction and ensure statistical significance

Log-Size Submodularity in Monotone DNF for k-CLIQUE

- **Verdict:** INCONCLUSIVE
- **Bridge:** BOOLEAN_FUNCTION_APPROXIMATION $\times$ MONOTONE_CIRCUIT_COMPLEXITY
- **Recorded:** 2026-05-19 03:56 UTC
- **Entry ID:** 2222bcac7069

Statement

For any monotone DNF formulas A and B , the logarithm of the number of terms in $A \wedge B$ is at most the sum of the logarithms of the number of terms in A and B . Furthermore, any monotone DNF formula representing the k -CLIQUE function has $\Omega(n)$ log-size, while poly-sized DNF formulas have $O(\log n)$ log-size.

Rationale

The submodularity of log-size under conjunction implies that combining DNFs doesn't exponentially increase the term count, which could be leveraged to show lower bounds. The $\Omega(n)$ bound for k-CLIQUE suggests that efficient DNF representations are impossible, aligning with known lower bounds.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.13s

```
ed': 30, 'conjecture_holds': False, 'counterexample': 'mean_dnf_size is too small'}
TRIAL: {'metric_name': 'mean_dnf_size', 'metric_value': 19.2, 'instances_tested': 30, 'conjecture_ho
TRIAL: {'metric_name': 'mean_dnf_size', 'metric_value': 19.433333333333334, 'instances_tested': 30,
TRIAL: {'metric_name': 'mean_dnf_size', 'metric_value': 18.9, 'instances_tested': 30, 'conjecture_ho
TRIAL: {'metric_name': 'mean_dnf_size', 'metric_value': 19.366666666666667, 'instances_tested': 30,
TRIAL: {'metric_name': 'mean_dnf_size', 'metric_value': 19.166666666666668, 'instances_tested': 30,
TRIAL: {'metric_name': 'mean_dnf_size', 'metric_value': 18.966666666666665, 'instances_tested': 30,
TRIAL: {'metric_name': 'mean_dnf_size', 'metric_value': 19.3, 'instances_tested': 30, 'conjecture_ho
TRIAL: {'metric_name': 'mean_dnf_size', 'metric_value': 19.133333333333333, 'instances_tested': 30,
```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_43ebdb7e.py", line 85, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_43ebdb7e.py", line 85, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~^
```

KeyError: 'seed'

Judge reasoning

Test crashed before completing validation; counterexample metrics may be unreliable | next: Re-run test with seed tracking enabled and error handling for metric collection

SOS Moment Matrix Eigenvalue Threshold for ACC⁰ Lower Bounds

- **Verdict:** INCONCLUSIVE
- **Bridge:** SUM_OF_SQUARES_HIERARCHY $\times$ ACC_ZERO_CIRCUIT_COMPLEXITY
- **Recorded:** 2026-05-19 04:03 UTC
- **Entry ID:** 3654b18c7301

Statement

For an explicit function $f: \{0,1\}^n \rightarrow \{0,1\}$ in P , if the minimal eigenvalue $\lambda_{\min}$ of its degree- d SOS moment matrix satisfies $\lambda_{\min} \geq \Omega(1/\sqrt{n})$, then f cannot be computed by ACC⁰ circuits of size $o(n^{\frac{1}{4}})$ for any constant $d \geq 2$.

Rationale

The SOS moment matrix eigenvalue captures the ‘hardness’ of refuting f via low-degree polynomial approximations. If $\lambda_{\min}$ is sufficiently large, it implies the function resists ACC^0 computation, aligning with Williams’ $\text{NEXP} \not\subseteq \text{ACC}^0$ separation. This property is computable in subexponential time for $n \leq 40$ via numerical linear algebra, avoiding natural proofs barriers by focusing on structural eigenvalue thresholds rather than direct circuit analysis.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 34.49s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_40a05773.py", line 73, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_40a05773.py", line 57, in run_trial
    conjecture_holds = lambda_min >= Fraction(1, math.sqrt(n))
                        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/usr/lib/python3.12/fractions.py", line 277, in __new__
    raise TypeError("both arguments should be ")
```

TypeError: both arguments should be Rational instances

Judge reasoning

Test crashed due to type error in comparison; support condition not met | next: Fix type error in $\lambda_{\min}$ comparison by converting $\text{math.sqrt}(n)$ to Fraction or rational type

Fourier Coefficient Concentration in ACC^0 Circuits

- **Verdict:** INCONCLUSIVE
- **Bridge:** $\text{FOURIER_ANALYSIS} \times \text{ACC_ZERO_CIRCUIT_COMPLEXITY}$
- **Recorded:** 2026-05-19 04:17 UTC
- **Entry ID:** 971d5ab3c3ef

Statement

A Boolean function f on n variables cannot be computed by an ACC^0 circuit of size $o(2^{\{n/2\}})$ if its Fourier transform has a coefficient $|F(S)| \geq 1/\sqrt{n}$ for some subset S of size $\Omega(n)$.

Rationale

This conjecture connects ACC^0 circuit complexity to the structure of Fourier coefficients, leveraging the known lower bounds for functions with large Fourier coefficients. If true, it would provide a natural property (Fourier concentration) that excludes ACC^0 circuits while being computable in subexponential time via Fourier transform analysis on small n .

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.11s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_aele000f.py", line 52, in <module>
    trial_result = run_trial(seed)
                    ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_aele000f.py", line 22, in run_trial
    ft = fourier_transform(f, n)
        ^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_aele000f.py", line 41, in fourier_trans
    term = f(x) * math.exp(-2j * math.pi * S * x / (1 << n))
           ^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_aele000f.py", line 21, in <lambda>
    f = lambda x: sum(x[i] for i in range(n)) % 2 # Example function
                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_aele000f.py", line 21, in <genexpr>
    f = lambda x: sum(x[i] for i in range(n)) % 2 # Example function
                ^^^
```

TypeError: 'int' object is not subscriptable

Judge reasoning

Test crashed due to TypeError in Fourier transform computation, preventing data collection. The crash indicates a code error rather than a theoretical failure. | next: Fix the test code to handle x as a list/array instead of integer, then re-run with proper input types

Schur-Weyl Multiplicity Gap in Symmetric vs Antisymmetric Decompositions

- **Verdict:** INCONCLUSIVE
- **Bridge:** SCHUR_WEYL_DUALITY × GEOMETRIC_COMPLEXITY_THEORY
- **Recorded:** 2026-05-19 04:24 UTC
- **Entry ID:** 38826d199a13

Statement

For all integers $n \geq 2$, the multiplicity of the irreducible representation indexed by the partition $\lambda = (n-1, 1)$ in the decomposition of the symmetric square of the standard representation of $GL(n)$ is strictly greater than its multiplicity in the decomposition of the antisymmetric square of the standard representation.

Rationale

The symmetric square decomposition (related to the permanent) and antisymmetric square decomposition (related to the determinant) exhibit distinct representation-theoretic structures. By comparing multiplicities of key irreducible components (e.g., $\lambda = (n-1, 1)$), we isolate a quantitative invariant preserved under linear substitutions. A gap in these multiplicities could imply structural differences in their orbit closures, advancing the GCT program's goal to separate det and perm.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_115dcd49.py", line 72, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^^^
```

[illegible]

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_115dcd49.py", line 29, in multiplicity
    sym_multiplicity = hook_length_formula(n, partition[:]) / math.factorial(n)
                        ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_115dcd49.py", line 24, in hook_length_f
    total *= (n - i + j) / (partition[j] * (i + j))
               ^~~~~~
```

```
ZeroDivisionError: division by zero
```

Judge reasoning

```
parse fail | next: NONE
```

Communication Complexity Lower Bound for AC^0 Circuits via Real Polynomial Approximation

- **Verdict:** INCONCLUSIVE
- **Bridge:** SUM_OF_SQUARES_HIERARCHY $\times$ COMMUNICATION_COMPLEXITY
- **Recorded:** 2026-05-19 04:45 UTC
- **Entry ID:** bea3150f135a

Statement

For any Boolean function $f: \{0,1\}^n \rightarrow \{0,1\}$ that cannot be approximated by a real polynomial of degree d in the L_1 norm, its deterministic communication complexity is $\Omega(n^{1/2} / d)$. This implies that functions requiring high-degree polynomials for approximation also have high communication complexity, establishing a lifting connection between algebraic approximation and communication complexity.

Rationale

This conjecture ties the algebraic property of polynomial approximation (from SUM_OF_SQUARES_HIERARCHY to communication complexity). If a function needs a high-degree polynomial to approximate it, it suggests that the function has complex structure, which could translate to high communication complexity. This aligns with lifting theorems that connect different complexity measures.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 240.32s

TIMEOUT after 240s

Judge reasoning

Test timed out without producing results, preventing evaluation of support fraction or counterexamples. | next: Run test with extended timeout and debug crash logs

Super-Linear Lower Bound on Extended Frege Proofs for Monotone Tautologies

- **Verdict:** INCONCLUSIVE
- **Bridge:** PROOF_COMPLEXITY $\times$ MONOTONE_CIRCUIT_COMPLEXITY
- **Recorded:** 2026-05-19 04:56 UTC
- **Entry ID:** 0ea34fe2ce09

Statement

Every monotone tautology that requires exponential size in monotone circuits also requires super-linear size in Extended Frege proofs.

Rationale

Monotone tautologies are known to have high circuit complexity, and if their proof complexity is also high, this could provide a separation between circuit complexity and proof complexity. This conjecture builds on known results about monotone circuits and their relation to proof systems.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.03s

Proof length 1 is not super-linear for n=18'

TRIAL: {'metric_name': 'Extended Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjunctive': 1, 'linear for n=15'}

TRIAL: {'metric_name': 'Extended Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjunctive': 1, 'linear for n=7'}

TRIAL: {'metric_name': 'Extended Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjunctive': 1, 'linear for n=25'}

TRIAL: {'metric_name': 'Extended Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjunctive': 1, 'linear for n=23'}

TRIAL: {'metric_name': 'Extended Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjunctive': 1, 'linear for n=9'}

TRIAL: {'metric_name': 'Extended Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjunctive': 1, 'linear for n=38'}

TRIAL: {'metric_name': 'Extended Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjunctive': 1, 'linear for n=16'}

TRIAL: {'metric_name': 'Extended Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjunctive': 1, 'linear for n=36'}

TRIAL: {'metric_name': 'Extended Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjunctive': 1, 'linear for n=19'}

TRIAL: {'metric_name': 'Extended Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjunctive': 1, 'linear for n=17'}

RESULT: FALSIFIED counterexample="First failing seed" first_failing_seed=11

Judge reasoning

Safety rail: critic_challenge_falsified | original: Multiple counterexamples show Extended Frege proofs of size 1 for monotone tautologies, directly contradicting the super-linear lower bound claim. | next: Analyze the specific $n=15$, $n=7$, and $n=25$ cases to understand why these tautologies admit such short proofs

Submodular Measure on Monotone DNF for k-CLIQUE Approximation

- **Verdict:** INCONCLUSIVE
- **Bridge:** MATROID THEORY $\times$ BOOLEAN_FUNCTION_APPROXIMATION
- **Recorded:** 2026-05-19 05:13 UTC
- **Entry ID:** c4e28a8fc7d7

Statement

There exists a submodular measure μ on monotone DNF formulas such that $\mu(f) \leq O(\log n)$ for any DNF formula f of size $\text{poly}(n)$, but $\mu(k\text{-CLIQUE}) \geq \Omega(n)$ for all $k \geq 2$. The measure μ is defined as the minimal number of terms in a DNF that approximates the $k\text{-CLIQUE}$ function with error $\leq 1/n^2$ under uniform distribution.

Rationale

This conjecture leverages matroid rank functions (submodularity) to quantify the structural complexity of DNF approximations. By tying μ to the approximation error of $k\text{-CLIQUE}$, it bridges matroid theory (field_A) with boolean function approximation (field_B). If true, it would provide a new combinatorial barrier for monotone circuits, as the submodular measure would grow linearly with n for $k\text{-CLIQUE}$ but remain logarithmic for other functions.

Novelty

- Judge: NOVEL over 2 arXiv hits

Top hits consulted: - [s2:1512.03726] Quantitative estimates in approximation by Bernstein-Durrmeyer-Choquet operators with respect to monotone and submodular - [s2:2308.05588] Banzhaf Values for Facts in Query Answering

Empirical Test

- exit code: 1, elapsed: 4.63s

```
'n=5, mu=4'}
TRIAL: {'metric_name': 'submodular_measure', 'metric_value': 3.2, 'instances_tested': 10, 'conjecture': 'true'}
TRIAL: {'metric_name': 'submodular_measure', 'metric_value': 3.2, 'instances_tested': 10, 'conjecture': 'true'}
TRIAL: {'metric_name': 'submodular_measure', 'metric_value': 3.2, 'instances_tested': 10, 'conjecture': 'true'}
TRIAL: {'metric_name': 'submodular_measure', 'metric_value': 3.1, 'instances_tested': 10, 'conjecture': 'true'}
TRIAL: {'metric_name': 'submodular_measure', 'metric_value': 3.5, 'instances_tested': 10, 'conjecture': 'true'}
TRIAL: {'metric_name': 'submodular_measure', 'metric_value': 3.2, 'instances_tested': 10, 'conjecture': 'true'}
TRIAL: {'metric_name': 'submodular_measure', 'metric_value': 3.1, 'instances_tested': 10, 'conjecture': 'true'}
TRIAL: {'metric_name': 'submodular_measure', 'metric_value': 3.5, 'instances_tested': 10, 'conjecture': 'true'}
TRIAL: {'metric_name': 'submodular_measure', 'metric_value': 3.4, 'instances_tested': 10, 'conjecture': 'true'}
TRIAL: {'metric_name': 'submodular_measure', 'metric_value': 3.1, 'instances_tested': 10, 'conjecture': 'true'}
TRIAL: {'metric_name': 'submodular_measure', 'metric_value': 3.3, 'instances_tested': 10, 'conjecture': 'true'}
```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5bfd6fd7.py", line 106, in <module>
    print(f"RESULT: FALSIFIED counterexample=\n{n={results[0]['n']}}, mu={results[0]['mu']}\n" first_fa
    ~~~~~~^~~~~~
```

KeyError: 'n'

Judge reasoning

Test crashed with KeyError: 'n', preventing reliable evaluation of counterexamples | next: Debug the test code's KeyError handling for 'n' key access

Eigenvalue Threshold in SOS Moment Matrices for ACC⁰ Lower Bounds

- **Verdict:** INCONCLUSIVE
- **Bridge:** SUM_OF_SQUARES_HIERARCHY × ACC_ZERO_CIRCUIT_COMPLEXITY
- **Recorded:** 2026-05-19 05:29 UTC
- **Entry ID:** 072fac5ab465

Statement

For any function $f \in \{0,1\}^n$ computable by an ACC⁰ circuit of size s and depth d , the minimum eigenvalue $\lambda_{\min}$ of the SOS moment matrix $M_k(f)$ satisfies $\lambda_{\min} \geq \Omega(s^{\{-1/2\}})$ for $k = \Omega(\log s)$. This implies that any SOS-based approximation of f with error $\varepsilon < \Omega(s^{\{-1/2\}})$ requires $\Omega(\log s)$ levels of the hierarchy, thereby bounding the depth of ACC⁰ circuits computing f .

Rationale

This conjecture leverages the connection between SOS hierarchies and ACC⁰ circuit complexity, building on Williams' 2011 framework. By analyzing eigenvalue thresholds, it avoids Fourier-based techniques that trigger natural proofs, instead focusing on algebraic geometry and real polynomial approximations. The testable hypothesis ties spectral properties of moment matrices to circuit depth, aligning with the approach's target separation of NEXP $\not\subseteq$ ACC⁰.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 30.62s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4e3a3179.py", line 92, in <module>
    trial_result = run_trial(seed)
    ~~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4e3a3179.py", line 72, in run_trial
    lambda_min = compute_eigenvalue(M)
    ~~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4e3a3179.py", line 52, in compute_eigen
    Av = matrix_multiplication(A, v)
    ~~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4e3a3179.py", line 25, in matrix_multip
    m, k = len(A), len(B[0])
    ~~~~~~^~~~~~
```

TypeError: object of type 'float' has no len()

Judge reasoning

Test crashed due to type error in matrix multiplication, preventing evaluation of eigenvalue threshold. | next: Fix matrix multiplication to handle numeric types, then re-run trials with diverse ACC⁰ circuit parameters

Positivstellensatz Rank Deficiency in Max-CUT SOS Relaxations

- **Verdict:** INCONCLUSIVE
- **Bridge:** REAL_ALGEBRAIC_GEOMETRY × SUM_OF_SQUARES_HIERARCHY
- **Recorded:** 2026-05-19 05:45 UTC
- **Entry ID:** a7db7b29ff30

Statement

For any degree- d SOS relaxation of Max-CUT on n variables, the moment matrix M_d must satisfy $\text{rank}(M_d) \geq \Omega(n^{\{1.5\}})$ to achieve approximation ratio better than 0.878. If a relaxation achieves 0.879-approximation with $\text{rank}(M_d) = o(n^{\{1.5\}})$, then its Gram matrix violates the Positivstellensatz condition for non-negative polynomials on the hypercube.

Rationale

The Positivstellensatz characterizes non-negative polynomials on the hypercube as sums of squares plus terms from the feasible region. A rank deficiency in the SOS moment matrix would imply the relaxation cannot capture the necessary algebraic structure to approximate Max-CUT beyond the Goemans-Williamson threshold. This connects spectral properties of the moment matrix to algebraic certification of positivity, avoiding natural proofs by focusing on structural rather than combinatorial barriers.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.10s

```
e': 'Positivstellensatz violation'}
TRIAL: {'metric_name': 'rank(M_2)', 'metric_value': 16, 'instances_tested': 1, 'conjecture_holds': F
TRIAL: {'metric_name': 'rank(M_2)', 'metric_value': 16, 'instances_tested': 1, 'conjecture_holds': F
TRIAL: {'metric_name': 'rank(M_2)', 'metric_value': 16, 'instances_tested': 1, 'conjecture_holds': F
TRIAL: {'metric_name': 'rank(M_2)', 'metric_value': 16, 'instances_tested': 1, 'conjecture_holds': F
TRIAL: {'metric_name': 'rank(M_2)', 'metric_value': 16, 'instances_tested': 1, 'conjecture_holds': F
TRIAL: {'metric_name': 'rank(M_2)', 'metric_value': 16, 'instances_tested': 1, 'conjecture_holds': F
TRIAL: {'metric_name': 'rank(M_2)', 'metric_value': 16, 'instances_tested': 1, 'conjecture_holds': F
TRIAL: {'metric_name': 'rank(M_2)', 'metric_value': 16, 'instances_tested': 1, 'conjecture_holds': F
TRIAL: {'metric_name': 'rank(M_2)', 'metric_value': 16, 'instances_tested': 1, 'conjecture_holds': F
TRIAL: {'metric_name': 'rank(M_2)', 'metric_value': 16, 'instances_tested': 1, 'conjecture_holds': F

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2a4619b3.py", line 117, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~^~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2a4619b3.py", line 117, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
```

~^~^~^~^~^~^

KeyError: 'seed'

Judge reasoning

Test crashed before completing evaluation; counterexamples may be invalid due to incomplete data.
| next: Debug seed handling in test code and re-run with stricter error checking

Width-5 ABPs for Parity Require Depth $\Omega(\log n)$

- **Verdict:** INCONCLUSIVE
- **Bridge:** ALGEBRAIC_BRANCHING_PROGRAMS (ABPs) $\times$ NC¹_CIRCUIT_COMPLEXITY
- **Recorded:** 2026-05-19 06:05 UTC
- **Entry ID:** aceb7cb67219

Statement

Any width-5 ABP computing the parity function on n bits must have depth $\Omega(\log n)$.

Rationale

Barrington's theorem shows NC¹ functions can be computed by width-5 ABPs, but parity (in $AC^0 \subseteq NC^1$) may require deeper ABPs. This conjecture explores depth-width trade-offs for parity, leveraging known AC^0 depth lower bounds to infer ABP depth constraints.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_67e018d3.py", line 81, in <module>
    trial_result = run_trial(seed)
                   ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_67e018d3.py", line 54, in run_trial
    abps = list(generate_abps())
             ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_67e018d3.py", line 25, in generate_abps
    for left in generate_abps():
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_67e018d3.py", line 25, in generate_abps
    for left in generate_abps():
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_67e018d3.py", line 25, in generate_abps
    for left in generate_abps():
```

```
[Previous line repeated 995 more times]
```

RecursionError: maximum recursion depth exceeded

Judge reasoning

Test crashed due to infinite recursion, preventing data collection | next: Fix generate_abps() recursion and re-run with seed sampling

Treewidth Exponential Lower Bound on Tseitin Resolution Length

- **Verdict:** INCONCLUSIVE
- **Bridge:** graph theory (treewidth) $\times$ resolution proof complexity
- **Recorded:** 2026-05-19 06:14 UTC
- **Entry ID:** 1c724d8f6762

Statement

For a graph G with treewidth $\nu(G)$, the Tseitin formula on G requires Resolution refutations of length $\geq 2^{\Omega(\nu(G))}$. $\nu(G)$ is poly-time computable and $O(1)$ for all non-expanders.

Rationale

High treewidth implies complex hierarchical structure, which may necessitate exponential-length refutations in resolution due to the need to propagate constraints across multiple layers of the tree decomposition. Non-expanders (e.g., trees) have constant treewidth and admit polynomial-size refutations.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7e16453d.py", line 92, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7e16453d.py", line 74, in run_trial
    tw = treewidth(G)
        ^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7e16453d.py", line 32, in treewidth
    tw = treewidth([G[i] for i in range(n) if i != u and i != v])
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7e16453d.py", line 31, in treewidth
    G[u].remove(v)
```

ValueError: list.remove(x): x not in list

Judge reasoning

Test crashed due to ValueError, preventing data collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Fix the graph node removal logic to handle missing elements, then re-run experiments with multiple seeds

Non-Commutative Rank Lower Bound for Disjointness Matrices

- **Verdict:** INCONCLUSIVE
- **Bridge:** COMMUNICATION_COMPLEXITY $\times$ NONCOMMUTATIVE_RANK
- **Recorded:** 2026-05-19 06:20 UTC
- **Entry ID:** f46ef08c667f

Statement

The non-commutative rank of the disjointness matrix M_n (where $M_n[x][y] = 1$ iff $x \cap y = \emptyset$) satisfies $\text{rank}_{\text{nc}}(M_n) = \Omega(n)$.

Rationale

Non-commutative rank connects to algebraic branching programs (ABPs) via Barrington's theorem, which equates NC^1 circuits to width-5 ABPs. The disjointness matrix's high non-commutative rank would imply exponential ABP size, translating to $\Omega(n)$ communication complexity via the log-rank conjecture. This avoids algebraic barriers by focusing on matrix structure rather than polynomial identities.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.13s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b5ffe8ca.py", line 68, in <module>
    trial_result = run_trial(seed)
    ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b5ffe8ca.py", line 53, in run_trial
    rank_nc = non_commutative_rank(gaussian_elimination(M))
    ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b5ffe8ca.py", line 40, in non_commutative_rank
    n = len(A)
    ~~~~~
```

TypeError: object of type 'NoneType' has no len()

Judge reasoning

Test crashed due to TypeError, preventing data collection. The pre-registered support condition cannot be evaluated without valid results. | next: Fix the non_commutative_rank function to handle NoneType inputs and re-run tests with multiple seeds

Seed Length Optimality for Nisan-Wigderson PRGs in Fooling AC^0 Circuits

- **Verdict:** INCONCLUSIVE
- **Bridge:** PSEUDORANDOMNESS $\times$ CIRCUIT_COMPLEXITY
- **Recorded:** 2026-05-19 06:26 UTC
- **Entry ID:** fe4202f3ac75

Statement

The Nisan-Wigderson PRG requires a seed length of $\Omega(\log n)$ to fool AC^0 circuits of size $\text{poly}(n)$, and this bound is tight under the assumption that $\text{EXP} \neq \text{P/poly}$.

Rationale

The Nisan-Wigderson PRG's seed length is critical for derandomization. If the seed length is optimal, it implies that hardness assumptions (like $\text{EXP} \neq \text{P/poly}$) are necessary for derandomization, linking PRG design to circuit complexity via the hardness vs randomness paradigm.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
6, 'conjecture_holds': False, 'counterexample': 'Failed to fool AC^0 circuit of size 5'}
TRIAL: {'metric_name': 'seed_length', 'metric_value': 3, 'instances_tested': 6, 'conjecture_holds': False}
TRIAL: {'metric_name': 'seed_length', 'metric_value': 3, 'instances_tested': 6, 'conjecture_holds': False}
TRIAL: {'metric_name': 'seed_length', 'metric_value': 3, 'instances_tested': 6, 'conjecture_holds': False}
TRIAL: {'metric_name': 'seed_length', 'metric_value': 3, 'instances_tested': 6, 'conjecture_holds': False}
TRIAL: {'metric_name': 'seed_length', 'metric_value': 3, 'instances_tested': 6, 'conjecture_holds': False}
TRIAL: {'metric_name': 'seed_length', 'metric_value': 3, 'instances_tested': 6, 'conjecture_holds': False}
TRIAL: {'metric_name': 'seed_length', 'metric_value': 3, 'instances_tested': 6, 'conjecture_holds': False}
TRIAL: {'metric_name': 'seed_length', 'metric_value': 3, 'instances_tested': 6, 'conjecture_holds': False}
```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_218e7362.py", line 84, in <module>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                        ~~~~~^~~~~~
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_218e7362.py", line 84, in <genexpr>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                        ~~~~~^~~~~~
```

KeyError: 'seed'

Judge reasoning

Test crashed due to missing 'seed' key, preventing reliable evaluation of conjecture status | next:
Fix test to handle missing keys and re-validate counterexample for AC^0 circuit size 5

Real Polynomial Approximation Degree Lower Bound for AC^0 Circuits Computing PARITY

- **Verdict:** INCONCLUSIVE
- **Bridge:** $\text{REAL_ALGEBRAIC_GEOMETRY} \times \text{CIRCUIT_COMPLEXITY}$
- **Recorded:** 2026-05-19 06:45 UTC
- **Entry ID:** 4d43beaa3b30

Statement

Any AC^0 circuit computing PARITY on n inputs requires depth $\Omega(\sqrt{n})$, and the minimum degree of a real polynomial approximating PARITY with error $\varepsilon = 1/n$ is $\Omega(\sqrt{n})$, implying that the circuit's size is exponential in depth.

Rationale

The degree of a real polynomial approximating PARITY is tied to the circuit's depth. Since AC^0 circuits cannot compute PARITY exactly, their approximation degree must be high, leading to a lower bound on circuit size. This replaces the switching lemma with a geometric invariant under input permutations (symmetry of PARITY).

Novelty

- Judge: NOVEL over 2 arXiv hits

Top hits consulted: - [s2:2502.15928] EHands: Quantum Protocol for Polynomial Computation on Real-Valued Encoded States - [s2:2411.00976] Low-degree approximation of QAC^0 circuits

Empirical Test

- exit code: 124, elapsed: 246.41s

TIMEOUT after 240s

Judge reasoning

Test timed out before producing results, preventing validation of support fraction or counterexample detection. | next: Retry test with extended timeout and debug resource constraints

Super-Exponential Lower Bound on Extended Frege Proofs for Ramsey-Type Tautologies

- **Verdict:** INCONCLUSIVE
- **Bridge:** PROOF_SYSTEMS $\times$ COMBINATORIAL_PRINCIPLES
- **Recorded:** 2026-05-19 06:53 UTC
- **Entry ID:** a8c40c3a34db

Statement

For any Ramsey-type tautology encoding the non-existence of monochromatic cliques in colored graphs, the size of Extended Frege proofs is super-exponential in the input size.

Rationale

Ramsey-type tautologies encode combinatorial principles with inherent structural complexity. By linking their proof size to the combinatorial hardness of Ramsey numbers, we hypothesize that Extended Frege cannot efficiently prove these tautologies, leveraging the interplay between proof complexity and combinatorial thresholds.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```

instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'mapping_undefined'}
TRIAL: {'metric_name': 'proof_size', 'metric_value': 64, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'proof_size', 'metric_value': 22801, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'proof_size', 'metric_value': 15876, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'proof_size', 'metric_value': 196, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'proof_size', 'metric_value': 148225, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'proof_size', 'metric_value': 3364, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'proof_size', 'metric_value': 96721, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'proof_size', 'metric_value': 7744, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'proof_size', 'metric_value': 3481, 'instances_tested': 1, 'conjecture_holds': 1}

```

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e0bd3894.py", line 63, in <module>
    first_failing_seed = next(res["seed"] for res in results if not res["conjecture_holds"])
                        ~~~~~^~~~~~
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e0bd3894.py", line 63, in <genexpr>
    first_failing_seed = next(res["seed"] for res in results if not res["conjecture_holds"])
                        ~~~~~^~~~~~

```

KeyError: 'seed'

Judge reasoning

Test crashed before producing reliable data, and the counterexample 'mapping_undefined' is ambiguous. | next: Fix the test to handle seed parameters properly and re-run with validated input configurations

Resolution Width Lower Bound for Tseitin Formulas on Expander Graphs

- **Verdict:** INCONCLUSIVE
- **Bridge:** GRAPH_EXPANSION $\times$ RESOLUTION_WIDTH
- **Recorded:** 2026-05-19 07:02 UTC
- **Entry ID:** 4c6e3be1d344

Statement

For any expander graph G with expansion Φ , the resolution width of the Tseitin formula $T(G, \omega)$ is $\Omega(1/\Phi)$, where ω is a balanced assignment.

Rationale

Expander graphs have strong connectivity properties, which may necessitate wider proofs in resolution systems. The conjecture links the expansion parameter to the required proof width, suggesting that graphs with higher expansion require wider proofs, thus increasing the complexity.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.09s

```

value': 4, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'resolution_width', 'metric_value': 4, 'instances_tested': 1, 'conjecture_holds': 1}

```

```

TRIAL: {'metric_name': 'resolution_width', 'metric_value': 4, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'resolution_width', 'metric_value': 4, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'resolution_width', 'metric_value': 4, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'resolution_width', 'metric_value': 4, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'resolution_width', 'metric_value': 4, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'resolution_width', 'metric_value': 4, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'resolution_width', 'metric_value': 4, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'resolution_width', 'metric_value': 4, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'resolution_width', 'metric_value': 4, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'resolution_width', 'metric_value': 4, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'resolution_width', 'metric_value': 4, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'resolution_width', 'metric_value': 4, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'resolution_width', 'metric_value': 4, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'resolution_width', 'metric_value': 4, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'resolution_width', 'metric_value': 4, 'instances_tested': 1, 'conjecture_hol
RESULT: SUPPORTED mean=4.0 std=0.0 support_fraction=1.0

```

Judge reasoning

Single test case meets support fraction but critic challenges validity due to insufficient diversity. Pre-registered criterion requires $\geq 80\%$ seeds, which is technically met but lacks statistical confidence. | next: Test with multiple expander graphs and varying expansion parameters to validate across diverse instances

Orbit Closure Dimension Separation for Determinant vs Permanent

- **Verdict:** INCONCLUSIVE
- **Bridge:** GEOMETRIC_COMPLEXITY_THEORY $\times$ ALGEBRAIC_GEOMETRY
- **Recorded:** 2026-05-19 07:08 UTC
- **Entry ID:** 14187b61dea5

Statement

The dimension of the orbit closure of the determinant polynomial under the action of GL_n is strictly less than the dimension of the orbit closure of the permanent polynomial for all $n \geq 2$.

Rationale

This conjecture aligns with GCT's goal of separating VP and VNP via orbit closure dimensions. The determinant's orbit closure (a GL_n -invariant) has a known dimension formula, while the permanent's orbit closure is conjectured to have higher dimension, reflecting its computational hardness. Avoids algebrization barriers by focusing on geometric invariants rather than polynomial identities.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2ebc93f8.py", line 85, in <module>
    result = run_trial(seed)
            ~~~~~

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2ebc93f8.py", line 59, in run_trial

```

```
det_dim = orbit_closure_dimension(n)
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2ebc93f8.py", line 53, in orbit_closure
    raise NotImplementedError("Mapping undefined for n > 3")
NotImplementedError: Mapping undefined for n > 3
```

Judge reasoning

Test crashed for $n > 3$ due to undefined mapping, preventing dimension calculations. Insufficient data to evaluate conjecture for $n \geq 4$. | next: Implement orbit closure dimension computation for $n > 3$ in the test code

Symmetric Group Multiplicity Gap in Permanent vs Determinant Decompositions

- **Verdict:** INCONCLUSIVE
- **Bridge:** Schur-Weyl duality $\times$ algebraic combinatorics
- **Recorded:** 2026-05-19 07:14 UTC
- **Entry ID:** 2ad894e61d43

Statement

For all integers $n \geq 3$, the multiplicity of the Young diagram $\lambda = (n-1, 1)$ in the decomposition of the permanent polynomial under the symmetric group S_n action is strictly greater than its multiplicity in the determinant polynomial's decomposition.

Rationale

The permanent and determinant have distinct symmetry properties under the symmetric group: the determinant decomposes with non-zero coefficients only for alternating partitions, while the permanent's decomposition includes symmetric partitions. The $(n-1, 1)$ hook shape represents a critical case where symmetric and alternating components intersect. A multiplicity gap here would reflect fundamental asymmetry in their representation-theoretic structures, providing a quantitative barrier for orbit closure containment.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.03s

```
nstances_tested": 0, "conjecture_holds": False, "counterexample": "mapping_undefined"}
TRIAL: {"seed": 463, "metric_name": "Multiplicity Gap", "metric_value": None, "instances_tested": 0,
TRIAL: {"seed": 503, "metric_name": "Multiplicity Gap", "metric_value": None, "instances_tested": 0,
TRIAL: {"seed": 547, "metric_name": "Multiplicity Gap", "metric_value": None, "instances_tested": 0,
TRIAL: {"seed": 593, "metric_name": "Multiplicity Gap", "metric_value": None, "instances_tested": 0,
TRIAL: {"seed": 631, "metric_name": "Multiplicity Gap", "metric_value": None, "instances_tested": 0,
TRIAL: {"seed": 677, "metric_name": "Multiplicity Gap", "metric_value": None, "instances_tested": 0,
TRIAL: {"seed": 727, "metric_name": "Multiplicity Gap", "metric_value": None, "instances_tested": 0,
TRIAL: {"seed": 773, "metric_name": "Multiplicity Gap", "metric_value": None, "instances_tested": 0,
TRIAL: {"seed": 821, "metric_name": "Multiplicity Gap", "metric_value": None, "instances_tested": 0,
TRIAL: {"seed": 877, "metric_name": "Multiplicity Gap", "metric_value": None, "instances_tested": 0,
TRIAL: {"seed": 929, "metric_name": "Multiplicity Gap", "metric_value": None, "instances_tested": 0,
```

RESULT: FALSIFIED counterexample="mapping_undefined" first_failing_seed=929

Judge reasoning

Test failed to execute for $n \geq 3$ with metric_value=None across all trials, invalidating the claimed counterexample. | next: Debug metric computation for Young diagram multiplicities in S_n representations

Eigenvalue Threshold in SOS Moment Matrices for Max-CUT Approximation

- **Verdict:** INCONCLUSIVE
- **Bridge:** REAL_ALGEBRAIC_GEOMETRY $\times$ COMMUNICATION_COMPLEXITY
- **Recorded:** 2026-05-19 07:24 UTC
- **Entry ID:** 03dda0d662e2

Statement

For any degree- d SOS relaxation of max-CUT on n -vertex graphs, if the moment matrix M has all eigenvalues λ_i satisfying $|\lambda_i| \leq C * \sqrt{n/d^3}$ for some universal constant C , then the relaxation cannot approximate max-CUT with ratio better than $0.878 - \varepsilon$ for any $\varepsilon > 0$. This eigenvalue bound is preserved under polynomial-time reductions between max-CUT instances.

Rationale

The conjecture ties SOS moment matrix spectral properties to approximation ratios, leveraging real algebraic geometry's Positivstellensatz and communication complexity's reduction invariance. If validated, it would establish a quantitative barrier for SOS-based max-CUT approximations, aligning with Bouland-Kothari's planted clique lower bounds while avoiding algebrization barriers.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.17s

```
tances_tested': 1, 'conjecture_holds': False, 'counterexample': 'mapping_undefined'}
TRIAL: {'metric_name': 'eigenvalue', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds':
TRIAL: {'metric_name': 'eigenvalue', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds':
TRIAL: {'metric_name': 'eigenvalue', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds':
TRIAL: {'metric_name': 'eigenvalue', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds':
TRIAL: {'metric_name': 'eigenvalue', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds':
TRIAL: {'metric_name': 'eigenvalue', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds':
TRIAL: {'metric_name': 'eigenvalue', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds':
TRIAL: {'metric_name': 'eigenvalue', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds':
TRIAL: {'metric_name': 'eigenvalue', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds':
TRIAL: {'metric_name': 'eigenvalue', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds':
```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_db715b38.py", line 98, in <module>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                        ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_db715b38.py", line 98, in <genexpr>
```

```
first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                        ~~~~~^
```

KeyError: 'seed'

Judge reasoning

Test crashed with KeyError: 'seed' before producing valid results | next: Re-run test with detailed logging to diagnose 'mapping_undefined' error

Multiplicity of Hook Representation in Symmetric Powers of Permanent vs Determinant

- **Verdict:** INCONCLUSIVE
- **Bridge:** Schur-Weyl duality $\times$ Algebraic combinatorics
- **Recorded:** 2026-05-19 07:30 UTC
- **Entry ID:** 920462144f4f

Statement

For all integers $n \geq 2$ and $m < n^{\{1.5\}}$, the multiplicity of the hook-shaped irreducible representation $\lambda = (n-1, 1)$ in the decomposition of the symmetric power S^m (standard representation of S_n) satisfies $\mu(\text{perm}_n, \lambda) > \mu(\det_m \circ O(1), \lambda)$, where $\mu(f, \lambda)$ denotes the multiplicity of λ in f 's decomposition under linear substitutions.

Rationale

Hook-shaped representations capture symmetry-breaking patterns critical for distinguishing permanent/determinant orbits. By comparing multiplicities in symmetric powers, we leverage Schur-Weyl duality to isolate algebraic invariants preserved under linear substitutions, aligning with Mulmuley-Sohoni's geometric complexity theory framework.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.03s

```
TRIAL: {'seed': 11, **result}
TRIAL: {'seed': 23, **result}
TRIAL: {'seed': 37, **result}
TRIAL: {'seed': 53, **result}
TRIAL: {'seed': 71, **result}
TRIAL: {'seed': 89, **result}
TRIAL: {'seed': 103, **result}
TRIAL: {'seed': 127, **result}
TRIAL: {'seed': 149, **result}
TRIAL: {'seed': 167, **result}
TRIAL: {'seed': 191, **result}
TRIAL: {'seed': 211, **result}
TRIAL: {'seed': 233, **result}
TRIAL: {'seed': 257, **result}
TRIAL: {'seed': 277, **result}
TRIAL: {'seed': 311, **result}
```

```

TRIAL: {'seed': 347, **result}
TRIAL: {'seed': 389, **result}
TRIAL: {'seed': 421, **result}
TRIAL: {'seed': 463, **result}
TRIAL: {'seed': 503, **result}
TRIAL: {'seed': 547, **result}
TRIAL: {'seed': 593, **result}
TRIAL: {'seed': 631, **result}
TRIAL: {'seed': 677, **result}
TRIAL: {'seed': 727, **result}
TRIAL: {'seed': 773, **result}
TRIAL: {'seed': 821, **result}
TRIAL: {'seed': 877, **result}
TRIAL: {'seed': 929, **result}
RESULT: SUPPORTED mean=41/2 std=0 support_fraction=1.0

```

Judge reasoning

Safety rail: critic_challenge | original judge: All 29 trials (100%) support the conjecture with mean ratio 41/2 and zero variance, meeting the 80% threshold unambiguously. | next: Theoretically prove the asymptotic dominance of hook multiplicities in symmetric powers of permanents over determinants

Expansion Parameter Determines Resolution Length for Tseitin Formulas

- **Verdict:** INCONCLUSIVE
- **Bridge:** spectral_graph_theory × resolution_proof_complexity
- **Recorded:** 2026-05-19 07:48 UTC
- **Entry ID:** 6653cb6e91d8

Statement

Let G be a graph with edge expansion $h(G) = \min_{S \subseteq V, |S| \leq n/2} (|E(S, V \setminus S)|)/|S|$. The Tseitin formula on G requires Resolution length $\geq 2^{\Omega(h(G))}$. For non-expanders ($h(G) = O(1)$), Resolution length is polynomial.

Rationale

High edge expansion forces long resolution proofs by creating bottlenecks that require repeated propagation of contradictions across cuts. This connects to the cumulative entropy framework $\Phi(\pi) = \sum H(C_t)$ where expansion increases the entropy of clause propagation steps.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.03s

```

}

```

```

TRIAL: {'metric_name': 'Resolution length', 'metric_value': 3.4822022531844965, 'instances_tested': 1}
TRIAL: {'metric_name': 'Resolution length', 'metric_value': 2.0, 'instances_tested': 1, 'conjecture': 'true'}
TRIAL: {'metric_name': 'Resolution length', 'metric_value': 2.2973967099940698, 'instances_tested': 1, 'conjecture': 'true'}
TRIAL: {'metric_name': 'Resolution length', 'metric_value': 3.4822022531844965, 'instances_tested': 1, 'conjecture': 'true'}

```

```

TRIAL: {'metric_name': 'Resolution length', 'metric_value': 2.6390158215457884, 'instances_tested': 1}
TRIAL: {'metric_name': 'Resolution length', 'metric_value': 6.062866266041593, 'instances_tested': 1}
TRIAL: {'metric_name': 'Resolution length', 'metric_value': 1.515716566510398, 'instances_tested': 1}
TRIAL: {'metric_name': 'Resolution length', 'metric_value': 3.4822022531844965, 'instances_tested': 1}
TRIAL: {'metric_name': 'Resolution length', 'metric_value': 6.062866266041593, 'instances_tested': 1}
TRIAL: {'metric_name': 'Resolution length', 'metric_value': 3.4822022531844965, 'instances_tested': 1}
TRIAL: {'metric_name': 'Resolution length', 'metric_value': 3.4822022531844965, 'instances_tested': 1}
TRIAL: {'metric_name': 'Resolution length', 'metric_value': 3.0314331330207964, 'instances_tested': 1}
TRIAL: {'metric_name': 'Resolution length', 'metric_value': 4.0, 'instances_tested': 1, 'conjecture_holds': False}
RESULT: SUPPORTED mean=3.821939387749721 std=0 support_fraction=1.0

```

Judge reasoning

parse_fail | next: NONE

Discrepancy Lower Bound for Inner Product in High Dimensions

- **Verdict:** INCONCLUSIVE
- **Bridge:** DISCREPANCY_METHODS $\times$ COMMUNICATION_COMPLEXITY
- **Recorded:** 2026-05-19 07:56 UTC
- **Entry ID:** 717e13caf4bb

Statement

For any n -dimensional vector space over the reals, the discrepancy of the inner product function between two vectors is at least $\Omega(\sqrt{\log n})$ when the vectors are chosen from a certain set of unit vectors. This discrepancy lower bound implies that the communication complexity of computing the inner product is $\Omega(\sqrt{\log n})$ bits.

Rationale

This conjecture connects geometric discrepancy with communication complexity, leveraging dispersion methods to establish a lower bound. The discrepancy here measures how evenly the inner product values are distributed over the input space, and a lower bound on discrepancy would translate to a lower bound on the communication required to compute the function accurately.

Novelty

- Judge: INCONCLUSIVE over 7 arXiv hits

Top hits consulted: - [2307.07605v1] First-order Methods for Affinely Constrained Composite Non-convex Non-smooth Problems: Lower Complexity Bound and Near-o - [0111062v2] One-way communication complexity and the Neciporuk lower bound on formula size - [9906008v2] A Lower Bound on the Average-Case Complexity of Shellsort - [1112.2000v3] A discrepancy lower bound for information complexity - [2101.04525v2] A comparison of optimisation algorithms for high-dimensional particle and astrophysics applications

Empirical Test

- exit code: 1, elapsed: 5.55s

```

sted': 5000, 'conjecture_holds': False, 'counterexample': 'mean_discrepancy=0.12789783157274617, low
TRIAL: {'metric_name': 'discrepancy', 'metric_value': 0.12793937463321467, 'instances_tested': 5000,
TRIAL: {'metric_name': 'discrepancy', 'metric_value': 0.1277042472587447, 'instances_tested': 5000,
TRIAL: {'metric_name': 'discrepancy', 'metric_value': 0.12827864896168484, 'instances_tested': 5000,
TRIAL: {'metric_name': 'discrepancy', 'metric_value': 0.12792222855940152, 'instances_tested': 5000,

```



```

TRIAL: {'metric_name': 'discrepancy', 'metric_value': 0.1261024102089387, 'instances_tested': 5000,
TRIAL: {'metric_name': 'discrepancy', 'metric_value': 0.12550336206180757, 'instances_tested': 5000,
TRIAL: {'metric_name': 'discrepancy', 'metric_value': 0.12699092074535773, 'instances_tested': 5000,
TRIAL: {'metric_name': 'discrepancy', 'metric_value': 0.12437400121686437, 'instances_tested': 5000,
TRIAL: {'metric_name': 'discrepancy', 'metric_value': 0.1262358236054102, 'instances_tested': 5000,

```

Judge reasoning

Test crashed before producing reliable results, preventing validation of conjecture status | next:
 Debug and rerun test with proper seed tracking implementation

Matrix Rank Limitation in ACC⁰ Circuits for Bounded Depth

- **Verdict:** INCONCLUSIVE
- **Bridge:** ACC_ZERO_CIRCUIT_COMPLEXITY × MATRIX_THEORY
- **Recorded:** 2026-05-19 08:05 UTC
- **Entry ID:** b78d3530defc

Statement

For any ACC⁰ circuit computing a matrix M of size $n \times n$ with entries in $\{0,1\}$, the rank of M over the reals is at most $O((\log n)^{d})$, where d is the depth of the circuit. This implies that depth- d ACC⁰ circuits cannot approximate high-rank matrices beyond this bound.

Rationale

This conjecture leverages linear algebra to establish a structural limitation on ACC⁰ circuits. By bounding the real rank of matrices computable by shallow ACC⁰ circuits, it provides an alternative pathway to NEXP vs ACC⁰ separation, avoiding Fourier-analytic and tropical methods used in prior work.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.18s

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0ccdfa34.py", line 79, in <module>
    trial_result = run_trial(seed)
                    ~~~~~^~~~~~

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0ccdfa34.py", line 63, in run_trial
    expected_rank_bound = Fraction((math.log2(n) ** depth), 1).limit_denominator()
                           ~~~~~^~~~~~

File "/usr/lib/python3.12/fractions.py", line 277, in __new__
    raise TypeError("both arguments should be "
TypeError: both arguments should be Rational instances

```

Judge reasoning

Test crashed due to `TypeError` in `Fraction` initialization, preventing data collection. The crash indicates a code error rather than a mathematical refutation. | next: Fix the `Fraction` argument type by converting $\log_2(n)^{\text{depth}}$ to a rational number representation

Kronecker Coefficient Asymmetry in Tensor Products of Permanent Representations

- **Verdict:** INCONCLUSIVE
- **Bridge:** Schur-Weyl duality $\times$ Representation-theoretic complexity
- **Recorded:** 2026-05-19 08:24 UTC
- **Entry ID:** 6594e3bef68d

Statement

For all integers $n \geq 2$ and $m < n^{\{1.5\}}$, the Kronecker coefficient $g(\lambda, \mu, \nu)$ for the symmetric group S_n in the decomposition of the tensor product of the permanent's representation (λ) and the determinant's representation (μ) satisfies $g(\lambda, \mu, \nu) > g(\lambda', \mu', \nu')$ for all partitions λ', μ' of m where ν' corresponds to the determinant's decomposition.

Rationale

This conjecture leverages Schur-Weyl duality to compare orbit closures of `det` and `perm` by analyzing their decomposition under symmetric group actions. The Kronecker coefficients, which encode multiplicity in tensor products, are preserved under linear substitutions. By focusing on asymmetric multiplicities in tensor products, we sidestep prior symmetric square comparisons and target a novel representation-theoretic invariant. Computational verification for $n \leq 40$ is feasible using existing algorithms for Kronecker coefficient computation.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 1.73s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c7ef62ea.py", line 79, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c7ef62ea.py", line 40, in run_trial
    ls = partitions(n)
        ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c7ef62ea.py", line 25, in partitions
    for p in partitions(n-1):
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c7ef62ea.py", line 25, in partitions
    for p in partitions(n-1):
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c7ef62ea.py", line 25, in partitions
    for p in partitions(n-1):
             ^^^^^^^^^^^^^
```

```
[Previous line repeated 26 more times]
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c7ef62ea.py", line 27, in partitions
    new_p = p[:i] + [p[i]+1] + p[i+1:]
              ~^^^
```

IndexError: list index out of range

Judge reasoning

Test crashed due to infinite recursion in partition generation, preventing data collection | next: Fix the partitions function to handle base cases and re-run experiments with proper termination conditions

Spectral Gap Determines Resolution Length for Tseitin Formulas

- **Verdict:** INCONCLUSIVE
- **Bridge:** Spectral Graph Theory × Resolution Proof Complexity
- **Recorded:** 2026-05-19 08:34 UTC
- **Entry ID:** 947a41db1384

Statement

For a connected graph G , let $\lambda_2(G)$ denote the second smallest eigenvalue of the Laplacian matrix. The Tseitin formula on G requires Resolution refutations of length $\geq 2^{\Omega(1/\lambda_2(G))}$.

Rationale

The spectral gap $\lambda_2(G)$ quantifies the graph's connectivity. Smaller $\lambda_2(G)$ (indicating stronger expansion properties) correlates with exponential Resolution lower bounds for Tseitin formulas, as expanders inherently resist efficient refutations. This conjecture bridges spectral properties to proof complexity, avoiding prior expansion-based approaches while staying within GCT's algebraic geometry framework.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e4626fdc.py", line 110, in <module>
    result = run_trial(seed)
             ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e4626fdc.py", line 93, in run_trial
    lambda_2 = second_smallest_eigenvalue(L)
               ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e4626fdc.py", line 75, in second_smallest_eigenvalue
    eigenvals = sorted(eigenvalues(L))
                   ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e4626fdc.py", line 53, in eigenvalues
    char_poly = [c * x - matrix[i][i] * c for c in char_poly]
                  ^
```

NameError: name 'x' is not defined

Judge reasoning

Test crashed due to undefined variable 'x' in eigenvalue calculation, preventing data collection. | next: Fix the eigenvalue computation code to handle polynomial expansion properly

Secant Variety Dimension Lower Bound for Disjointness Communication Complexity

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic geometry of secant varieties $\times$ Communication complexity of partial functions
- **Recorded:** 2026-05-19 08:51 UTC
- **Entry ID:** e6a528777329

Statement

For the disjointness matrix M on n elements, the dimension of the secant variety of its rank locus is $\Omega(n)$, providing a lower bound on the randomized communication complexity of M .

Rationale

The dimension of the secant variety reflects the complexity of representing the matrix as a combination of rank-1 tensors, which in turn could reflect the information-theoretic requirements for communication. Higher-dimensional secant varieties imply a greater number of rank-1 components, suggesting higher communication complexity.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.03s

```
ion', 'metric_value': 10, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'secant_variety_dimension', 'metric_value': 10, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'secant_variety_dimension', 'metric_value': 10, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'secant_variety_dimension', 'metric_value': 10, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'secant_variety_dimension', 'metric_value': 10, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'secant_variety_dimension', 'metric_value': 10, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'secant_variety_dimension', 'metric_value': 10, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'secant_variety_dimension', 'metric_value': 10, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'secant_variety_dimension', 'metric_value': 10, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'secant_variety_dimension', 'metric_value': 10, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'secant_variety_dimension', 'metric_value': 10, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'secant_variety_dimension', 'metric_value': 10, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'secant_variety_dimension', 'metric_value': 10, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'secant_variety_dimension', 'metric_value': 10, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'secant_variety_dimension', 'metric_value': 10, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'secant_variety_dimension', 'metric_value': 10, 'instances_tested': 1, 'conje
RESULT: SUPPORTED mean=10.0 std=0.0 support_fraction=1.0
```

Judge reasoning

All trials used $n=1$, preventing observation of $\Omega(n)$ growth. Metric saturation at $n=1$ invalidates the conjecture's dimensional scaling claim. | next: Test with $n \geq 2$ to observe scaling behavior and validate $\Omega(n)$ lower bound

Real Polynomial Approximation Degree Lower Bound for AC^0 Circuits Computing Majority

- **Verdict:** INCONCLUSIVE
- **Bridge:** REAL_ALGEBRAIC_GEOMETRY $\times$ ACC_ZERO_CIRCUIT_COMPLEXITY
- **Recorded:** 2026-05-19 08:57 UTC
- **Entry ID:** 4a0e438912c7

Statement

For any AC^0 circuit computing the majority function on n inputs, the minimal degree of a real polynomial approximating it within $\varepsilon = 1/n$ is $\Omega(\sqrt{n})$. This implies that AC^0 circuits require size $\Omega(2^{\Omega(\sqrt{n})})$ to compute majority, leveraging real polynomial approximation barriers.

Rationale

Williams’ methods connect real polynomial approximations to circuit lower bounds. Majority’s symmetry and sensitivity suggest its approximation degree is inherently high, creating a barrier for AC^0 circuits. This conjecture avoids natural proofs by focusing on algebraic geometry rather than combinatorial counting.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```

approximation_degree', 'metric_value': 5, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': None
TRIAL: {'metric_name': 'approximation_degree', 'metric_value': 4, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 1}
TRIAL: {'metric_name': 'approximation_degree', 'metric_value': 6, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 1}
TRIAL: {'metric_name': 'approximation_degree', 'metric_value': 6, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 1}
TRIAL: {'metric_name': 'approximation_degree', 'metric_value': 9, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 1}
TRIAL: {'metric_name': 'approximation_degree', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': True, 'counterexample': None}
TRIAL: {'metric_name': 'approximation_degree', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': True, 'counterexample': None}
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TRIAL: {'metric_name': 'approximation_degree', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': True, 'counterexample': None}

```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1e15c5c6.py", line 79, in <module>
    first_failing_seed = next((r['seed'] for r in results if not r['conjecture_holds']), None)
    ~~~~~
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1e15c5c6.py", line 79, in <genexpr>
    first_failing_seed = next((r['seed'] for r in results if not r['conjecture_holds']), None)
    ~~~~~
```

KeyError: 'seed'

Judge reasoning

Test crashed before completing evaluation; partial trials show conjecture holds but insufficient data for conclusion | next: Implement error handling for n_out_of_range cases and re-run with larger instance set

Separation of Bounded Arithmetic Theories via Proof System Strength

- **Verdict:** INCONCLUSIVE
- **Bridge:** BOUNDED_ARITHMETIC_THEORIES $\times$ PROOF_SYSTEMS
- **Recorded:** 2026-05-19 09:04 UTC
- **Entry ID:** d242f33fe08b

Statement

Let S12 be the bounded arithmetic theory corresponding to polynomial-size circuits with bounded-depth. There exists a proof system P such that S12 cannot prove the existence of an optimal proof for all tautologies in P, while a stronger theory T2 proves this property for all tautologies in P. This separation implies that P's strength exceeds S12's proof capacity.

Rationale

The conjecture leverages bounded arithmetic hierarchies to formalize proof system capabilities, avoiding relativization barriers by focusing on inherent complexity gaps. It connects S12's circuit-based limitations to proof system properties, aligning with Cook-Reckhow-Krajíček's framework for separating theories via proof complexity.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
conjecture_holds': False, 'counterexample': 'S12 cannot prove the existence of an optimal proof for al
TRIAL: {'metric_name': 'optimal_proof_exists', 'metric_value': False, 'instances_tested': 1, 'conjec
TRIAL: {'metric_name': 'optimal_proof_exists', 'metric_value': False, 'instances_tested': 1, 'conjec
TRIAL: {'metric_name': 'optimal_proof_exists', 'metric_value': False, 'instances_tested': 1, 'conjec
TRIAL: {'metric_name': 'optimal_proof_exists', 'metric_value': False, 'instances_tested': 1, 'conjec
TRIAL: {'metric_name': 'optimal_proof_exists', 'metric_value': False, 'instances_tested': 1, 'conjec
TRIAL: {'metric_name': 'optimal_proof_exists', 'metric_value': False, 'instances_tested': 1, 'conjec
TRIAL: {'metric_name': 'optimal_proof_exists', 'metric_value': False, 'instances_tested': 1, 'conjec
TRIAL: {'metric_name': 'optimal_proof_exists', 'metric_value': False, 'instances_tested': 1, 'conjec
TRIAL: {'metric_name': 'optimal_proof_exists', 'metric_value': False, 'instances_tested': 1, 'conjec
```

Judge reasoning

The test crashed due to a KeyError, preventing reliable evaluation of the conjecture's validity. | next: Fix the test to handle 'seed' data properly and re-run with sufficient instances to meet the support fraction criterion.

SOS Moment Matrix Eigenvalue Decay and Approximation Ratio Trade-off

- **Verdict:** BARRIER_HIT
- **Bridge:** REAL_ALGEBRAIC_GEOMETRY $\times$ SUM_OF_SQUARES_HIERARCHY
- **Recorded:** 2026-05-19 09:09 UTC
- **Entry ID:** f273e79d00f1

Statement

For any Max-CUT instance with a 0.879-approximator, the minimal eigenvalue $\lambda_{\min}(M_d)$ of the degree- d SOS moment matrix satisfies $\lambda_{\min}(M_d) \geq \Omega(1/\sqrt{n})$. Conversely, if $\lambda_{\min}(M_d) < 1/\sqrt{n}$, then no such approximator exists.

Rationale

This conjecture ties the spectral properties of the SOS moment matrix to approximation guarantees. By bounding the minimal eigenvalue, it creates a quantitative barrier for SOS relaxations to approximate Max-CUT beyond the Goemans-Williamson ratio. The preservation under reductions ensures the property is robust across problem instances.

Novelty

- Judge: “ over 0 arXiv hits

Empirical Test

- exit code: -1, elapsed: 0.00s

(no output)

Judge reasoning

Rejected by ALGEBRIZATION barrier

Matroid Connectivity Determines Pseudorandom Generator Efficiency for Nisan-Wigderson Designs

- **Verdict:** INCONCLUSIVE
- **Bridge:** MATROID_THEORY $\times$ PSEUDORANDOMNESS
- **Recorded:** 2026-05-19 09:19 UTC
- **Entry ID:** f4e06590d9e7

Statement

Let M be a matroid of rank r over a ground set of size n , and let G be a Nisan-Wigderson pseudorandom generator with seed length s . The connectivity $\kappa(M)$ of M is inversely proportional to the seed length s required for G to fool circuits of depth $d = O(\log n)$. Specifically, for matroids with $\kappa(M) \geq \log n$, the seed length s is $\Omega(\log n)$, while for matroids with $\kappa(M) < \log n$, s is $\Omega(n/\kappa(M))$.

Rationale

Matroid connectivity (κ) measures the minimal number of elements whose removal disconnects the matroid. This structural property directly influences the expansion properties of the matroid's incidence graph, which in turn determines the pseudorandom generator's ability to fool circuits. High connectivity implies better expansion, requiring shorter seeds, while low connectivity necessitates longer seeds to maintain pseudorandomness.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b61de3bf.py", line 143, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b61de3bf.py", line 107, in run_trial
    k_M = matroid_connectivity(M)
          ^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b61de3bf.py", line 91, in matroid_connectivity
    if find_set(parent, u) != find_set(parent, v):
       ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b61de3bf.py", line 76, in find_set
    if parent[u] != u:
       ~~~~~^^
```

IndexError: list index out of range

Judge reasoning

Test crashed with IndexError in matroid connectivity calculation, preventing data collection | next:
Fix the index out-of-range error in the find_set function and rerun tests

Real Radical Containment in SOS Moment Matrices for Max-CUT

- **Verdict:** INCONCLUSIVE
- **Bridge:** REAL_ALGEBRAIC_GEOMETRY $\times$ SUM_OF_SQUARES_HIERARCHY
- **Recorded:** 2026-05-19 09:32 UTC
- **Entry ID:** 3d31a9bd1776

Statement

For any Max-CUT instance with n variables, if the degree- d SOS moment matrix M has a real radical containing a polynomial $p(x)$ that vanishes on the feasible region of the cut, then the approximation ratio of the SOS relaxation is bounded by $0.878 - \varepsilon$ for some $\varepsilon > 0$. Conversely, if $p(x)$ is not in the real radical, the approximation ratio must exceed 0.878.

Rationale

The real radical captures the algebraic structure of the feasible region for Max-CUT. By linking its containment in the SOS moment matrix to approximation guarantees, we exploit real algebraic geometry's ability to characterize non-negativity conditions. This connects to the SOS hierarchy's role in approximating combinatorial optimization problems.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s


```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_89f0cc2f.py", line 42, in run_trial
    simplicial_complex = parse_simplicial_complex(instance) # Placeholder function
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

NameError: name 'parse_simplicial_complex' is not defined. Did you mean: 'simplicial_complex'?

Judge reasoning

Test crashed due to undefined function; cannot evaluate conjecture validity | next: Implement parse_simplicial_complex function and re-run experiments

Newton Polytope Vertex Count Bounds SOS Rank for CSP Refutations

- **Verdict:** INCONCLUSIVE
- **Bridge:** REAL_ALGEBRAIC_GEOMETRY $\times$ CSP_LOWER_BOUNDS
- **Recorded:** 2026-05-19 09:51 UTC
- **Entry ID:** 0ccae330ab62

Statement

For a system of polynomial equations defining a CSP, the SOS rank required to refute the CSP is at least the number of vertices of the Newton polytope of the system. Equivalently, the minimal SOS rank needed to certify infeasibility of a CSP instance is lower-bounded by the vertex count of its defining polynomial's Newton polytope.

Rationale

The Newton polytope encodes the monomial structure of the polynomial constraints, and its vertex count reflects the combinatorial complexity of the feasible region. This geometric invariant may directly influence the SOS hierarchy's ability to find refutations, as vertices correspond to extremal points that require higher rank certificates for non-negativity.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_daf81ef3.py", line 98, in <module>
    result = run_trial(seed)
    ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_daf81ef3.py", line 24, in run_trial
    equations = [sum(coeff * x**i for i, coeff in enumerate(row)) for row in coefficients]
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_daf81ef3.py", line 24, in <genexpr>
    equations = [sum(coeff * x**i for i, coeff in enumerate(row)) for row in coefficients]
    ^
```

NameError: name 'x' is not defined

Judge reasoning

Test crashed due to undefined variable 'x' before producing results | next: Fix test code to define variables properly and re-run experiments

Schur-Weyl Multiplicity Gap in Symmetric Powers of Permutation Matrices

- **Verdict:** INCONCLUSIVE
- **Bridge:** Schur-Weyl Duality $\times$ Monotone Circuit Complexity
- **Recorded:** 2026-05-19 10:01 UTC
- **Entry ID:** b0bc1c3eee2c

Statement

For $n \geq 2$, the multiplicity of the trivial representation in the decomposition of $\text{Sym}^k(\text{Perm}_n)$ exceeds that in $\text{Sym}^k(\text{Det}_n)$ by $\Omega(n^{\{1/2\}})$ for all $k \leq n^{\{1/2\}}$, where Perm_n and Det_n are the $n \times n$ permutation and determinant matrices. This gap persists under linear substitutions preserving the symmetric group action.

Rationale

Schur-Weyl duality decomposes tensor powers into irreducible representations, revealing structural asymmetry between determinant (alternating) and permanent (symmetric) under symmetric group actions. The multiplicity gap reflects differing symmetry constraints, potentially exposing algebraic barriers to monotone circuit simulation.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 240.05s

TIMEOUT after 240s

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Optimize test parameters or increase time limit to resolve timeout

Non-Abelian Fourier Analysis Determines Resolution Complexity for Tseitin Formulas

- **Verdict:** INCONCLUSIVE
- **Bridge:** Non-Abelian Fourier Analysis (Representation Theory of Symmetric Group) $\times$ Resolution Complexity of Tseitin Formulas
- **Recorded:** 2026-05-19 10:11 UTC
- **Entry ID:** 50287d06e16b

Statement

For any Tseitin formula on a graph G , the resolution complexity is exponential in the number of irreducible representations of the symmetric group that appear in the adjacency matrix of G .

Rationale

The number of irreducible representations of the symmetric group that appear in the adjacency matrix of G reflects the symmetry and structure of the graph, which directly influences the complexity of resolution proofs for Tseitin formulas. This connects representation-theoretic properties to combinatorial proof complexity.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.11s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1b1a0953.py", line 86, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1b1a0953.py", line 69, in run_trial
    irreps_count = irreducible_representations(G)
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1b1a0953.py", line 51, in irreducible_r
    char_table = gaussian_elimination(char_table)
                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1b1a0953.py", line 31, in gaussian_elim
    A[j][i] /= factor
File "/usr/lib/python3.12/fractions.py", line 630, in reverse
    return monomorphic_operator(Fraction(a), b)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/usr/lib/python3.12/fractions.py", line 763, in _div
    raise ZeroDivisionError('Fraction(%s, 0)' % db)
ZeroDivisionError: Fraction(1, 0)
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing data collection. Pre-registered support condition cannot be evaluated without completed trials. | next: Debug the Gaussian elimination implementation to handle division by zero cases in fraction arithmetic

Semialgebraic Dimension Bounds for SOS Approximation of Max-CUT

- **Verdict:** INCONCLUSIVE
- **Bridge:** Real Algebraic Geometry $\times$ Sum-of-Squares Hierarchy
- **Recorded:** 2026-05-19 10:27 UTC
- **Entry ID:** 7443ec5addcc

Statement

For a random Max-CUT instance on n vertices, the minimal degree d required to achieve an α -approximation satisfies $d \geq \lceil \log(\chi(G)) \rceil$, where $\chi(G)$ is the chromatic number of the graph. This bound is preserved under polynomial-time reductions between Max-CUT instances.

Rationale

Semialgebraic sets defined by polynomial inequalities constrain the SOS hierarchy's ability to approximate Max-CUT. The chromatic number, a geometric invariant of the graph's clique structure, dictates the minimal degree needed to capture the problem's combinatorial complexity through real algebraic geometry.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.13s

```
tio', 'metric_value': 1.0, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS_approximation_ratio', 'metric_value': 1.0, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'SOS_approximation_ratio', 'metric_value': 1.0, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'SOS_approximation_ratio', 'metric_value': 1.0, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'SOS_approximation_ratio', 'metric_value': 1.0, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'SOS_approximation_ratio', 'metric_value': 1.0, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'SOS_approximation_ratio', 'metric_value': 1.0, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'SOS_approximation_ratio', 'metric_value': 1.0, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'SOS_approximation_ratio', 'metric_value': 1.0, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'SOS_approximation_ratio', 'metric_value': 1.0, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'SOS_approximation_ratio', 'metric_value': 1.0, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'SOS_approximation_ratio', 'metric_value': 1.0, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'SOS_approximation_ratio', 'metric_value': 1.0, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'SOS_approximation_ratio', 'metric_value': 1.0, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'SOS_approximation_ratio', 'metric_value': 1.0, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'SOS_approximation_ratio', 'metric_value': 1.0, 'instances_tested': 1, 'conje
RESULT: SUPPORTED mean=1.0 std=0.0 support_fraction=1.0
```

Judge reasoning

Safety rail: critic_challenge | original judge: All 100% tested instances (n=1) satisfy the conjecture with perfect metric value 1.0, meeting the $\geq 80\%$ support threshold. | next: Test conjecture on larger graphs with varying chromatic numbers to validate generalization

Convex Hull Surface Area and Resolution Proof Length

- **Verdict:** INCONCLUSIVE
- **Bridge:** Integral Geometry $\times$ Resolution Proof Length
- **Recorded:** 2026-05-19 10:34 UTC
- **Entry ID:** 4f6ca05fef4b

Statement

For a CNF formula with n variables and m clauses, the surface area of the convex hull of its clauses (embedded in $\{0,1\}^n$) is inversely proportional to the minimal resolution proof length. Specifically, $\text{SurfaceArea} \leq C * (\log m) / (\text{ProofLength})$ for some universal constant C .

Rationale

Integral geometry's convex hull measures the geometric 'spread' of clauses. Resolution proofs may exploit clause clustering, so wider hulls (higher surface area) could imply more dispersed interactions, requiring longer proofs. This links geometric structure to proof complexity.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.05s

```
90000.0, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'mapping_undefined'}
TRIAL: {'metric_name': 'SurfaceArea * ProofLength', 'metric_value': 90000.0, 'instances_tested': 1, '
TRIAL: {'metric_name': 'SurfaceArea * ProofLength', 'metric_value': 90000.0, 'instances_tested': 1, '
TRIAL: {'metric_name': 'SurfaceArea * ProofLength', 'metric_value': 90000.0, 'instances_tested': 1, '
TRIAL: {'metric_name': 'SurfaceArea * ProofLength', 'metric_value': 90000.0, 'instances_tested': 1, '
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TRIAL: {'metric_name': 'SurfaceArea * ProofLength', 'metric_value': 90000.0, 'instances_tested': 1, '
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TRIAL: {'metric_name': 'SurfaceArea * ProofLength', 'metric_value': 90000.0, 'instances_tested': 1, '
TRIAL: {'metric_name': 'SurfaceArea * ProofLength', 'metric_value': 90000.0, 'instances_tested': 1, '
TRIAL: {'metric_name': 'SurfaceArea * ProofLength', 'metric_value': 90000.0, 'instances_tested': 1, '
RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed=11
```

Judge reasoning

Single instance tested with $n \leq 15$ shows metric saturation (90000.0) rather than correlation. Small n may plateau surface area metrics, making inverse proportionality unverifiable. | next: Test with larger n (≥ 20) and multiple instances to observe metric behavior

Noncommutative L^p Norm Lower Bound for Disjointness Communication Complexity

- **Verdict:** INCONCLUSIVE
- **Bridge:** Noncommutative L^p Geometry $\times$ Communication Complexity of Disjointness
- **Recorded:** 2026-05-19 10:44 UTC
- **Entry ID:** 5958cc0b1ee5

Statement

For a random $n \times n$ matrix M with entries 1 iff $i \cap j = \emptyset$, the noncommutative L^p norm $\|M\|_p \geq \Omega(n^{\{1/2\}})$ for $p \in (1, 2)$. This bound is tight up to $\text{polylog}(n)$ factors.

Rationale

Noncommutative L^p spaces capture the ‘geometric complexity’ of matrix entries under tensor products, which mirrors the entanglement structure in multi-party communication. The $p=2$ case corresponds to operator norms, while $p \rightarrow 1$ recovers the max-entropy of disjointness witnesses.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.12s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c9e52a4c.py", line 74, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c9e52a4c.py", line 56, in run_trial
    norm = noncommutative_lp_norm(singular_values(M), p)
                                                ^
NameError: name 'p' is not defined
```

Judge reasoning

Test crashed due to undefined variable 'p', preventing data collection. Pre-registered support condition cannot be evaluated without valid trial results. | next: Fix the NameError in test_c9e52a4c.py by ensuring 'p' is properly defined in the noncommutative_lp_norm function call

Additive Energy Inverse Proportionality to ACC⁰ Circuit Size for Explicit Functions

- **Verdict:** INCONCLUSIVE
- **Bridge:** Additive Combinatorics $\times$ ACC⁰ Circuit Size
- **Recorded:** 2026-05-19 11:00 UTC
- **Entry ID:** 249d7867db76

Statement

For an explicit function f in P with n variables, the additive energy $E(f)$ of the set derived from its truth table is inversely proportional to the minimum ACC⁰ circuit size $S(f)$, i.e., $E(f) * S(f)^\beta \geq C * n^\alpha$ for constants $\alpha, \beta, C > 0$.

Rationale

Additive energy captures structural regularity, which may conflict with the structured computations of ACC⁰ circuits. High energy implies low circuit size, but this inverse relationship is conjectured to hold for explicit functions in P .

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 8.19s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_edbab641.py", line 113, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_edbab641.py", line 70, in run_trial
    energy = additive_energy(truth_table)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_edbab641.py", line 54, in additive_ener
    if (truth_table[i][j] == truth_table[k][l]) != (truth_table[i][k] == truth_table[j][l]):
                                   ~~~~~^
```

IndexError: list index out of range

Judge reasoning

Test crashed with IndexError before producing results, preventing evaluation of support/falsification
| next: Fix the IndexError in additive_energy function by validating indices before access

Spectral Gap Determines Tseitin Resolution Length

- **Verdict:** INCONCLUSIVE
- **Bridge:** Spectral Graph Theory $\times$ Resolution Proof Length
- **Recorded:** 2026-05-19 11:20 UTC
- **Entry ID:** 715ab68a1aef

Statement

For any connected graph G with $n \geq 3$, the resolution length of its Tseitin formula is $\geq 2^{\Omega(\mu)}$, where μ is the algebraic connectivity (smallest non-zero Laplacian eigenvalue) of G . For non-expanders ($\mu \leq 1/n$), this bound is polynomial in n .

Rationale

Algebraic connectivity captures graph expansion properties critical for Tseitin formula hardness. Spectral methods provide a concrete, poly-time computable invariant linking graph structure to resolution complexity, avoiding algebraization barriers.

Novelty

- Judge: NOVEL over 10 arXiv hits

Top hits consulted: - [0701895v1] An algebraic proof of Deligne's regularity criterion for integrable connections - [2103.09609v1] Characterizing Tseitin-formulas with short regular resolution refutations - [2209.05839v3] On bounded depth proofs for Tseitin formulas on the grid; revisited - [2403.19550v2] Spectral gap for surfaces of infinite volume with negative curvature - [2508.16353v2] On the asymptotic behavior of the spectral gap for discrete Schrödinger operators

Empirical Test

- exit code: 1, elapsed: 0.68s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_84b6cba1.py", line 123, in <module>
    result = run_trial(seed)
             ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_84b6cba1.py", line 101, in run_trial
     $\mu$  = algebraic_connectivity(G, n)
             ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_84b6cba1.py", line 71, in algebraic_con
    return min(eigenvalue for eigenvalue in eigenvalues if eigenvalue > 0)
             ~~~~~
```

ValueError: min() iterable argument is empty

Judge reasoning

Test crashed during algebraic connectivity calculation, preventing data collection. The error suggests potential issues with graph generation or eigenvalue computation for connected graphs. | next: NONE

Free Entropy Gap in Read-Twice Branching Programs

- **Verdict:** INCONCLUSIVE
- **Bridge:** Free Probability $\times$ Read-Twice Branching Programs
- **Recorded:** 2026-05-19 11:34 UTC
- **Entry ID:** 4eca2ec83f86

Statement

Let P be a read-twice branching program over n variables. The free entropy $\phi(P)$ of its transition matrix satisfies $\phi(P) \geq \log^2 n$, while read-once BPs have $\phi(P) \leq \log n$. For all P with size $\geq 2^{n/2}$, $\phi(P) \geq \Omega(\log^2 n)$ holds.

Rationale

Free entropy captures non-commutative randomness in transition matrices. Read-twice BPs exhibit structured dependencies that amplify spectral complexity, creating a gap detectable via free probability invariants. This mirrors the separation between read-once and read-many BPs in algebraic complexity.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 240.10s

TIMEOUT after 240s

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run test with stricter resource limits

Matroid Rank Deficit in Monotone DNF for k-CLIQUE

- **Verdict:** INCONCLUSIVE
- **Bridge:** Matroid Theory $\times$ Monotone Circuit Complexity
- **Recorded:** 2026-05-19 11:53 UTC
- **Entry ID:** b0d510339340

Statement

For any monotone DNF formula computing k -CLIQUE on n vertices, the maximum number of pairwise disjoint clauses is $\Omega(n)$, while for any DNF of size $\text{poly}(n)$, this number is $O(\log n)$.

Rationale

Matroid rank functions are submodular and capture combinatorial independence, making them natural for analyzing DNF structure. The conjecture links the inherent complexity of k-CLIQUE to matroid-theoretic parameters, potentially revealing structural barriers for monotone circuits.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b5f6a9c.py", line 77, in <module>
    result = run_trial(seed)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b5f6a9c.py", line 54, in run_trial
    metric_value = max_pairwise_disjoint_clauses(DNF)
                  ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b5f6a9c.py", line 41, in max_pairwise_
    clauses = [c for c in clauses if not c.intersection(selected_clause)]
              ~~~~~
```

AttributeError: 'tuple' object has no attribute 'intersection'

Judge reasoning

Test crashed due to AttributeError, preventing data collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Fix the 'tuple' object 'intersection' error in the max_pairwise_disjoint_clauses function

Secant Rank Lower Bound for Disjointness Communication Matrices

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic geometry of secant varieties × Communication complexity of DISJOINTNESS
- **Recorded:** 2026-05-19 12:15 UTC
- **Entry ID:** 902c529a20f5

Statement

For the $n \times n$ DISJOINTNESS matrix M_n , the secant rank $\text{sr}(M_n)$ satisfies $\text{sr}(M_n) \geq \Omega(n)$. This bound is tight for all $n \geq 1$, with equality achieved when M_n is the incidence matrix of a projective plane of order n .

Rationale

Secant rank measures the minimal number of rank-1 tensors needed to span a matrix, reflecting its geometric complexity. For DISJOINTNESS, this invariant exposes the inherent structure of disjointness constraints, which are algebraically rigid due to their incidence geometry. The projective plane construction provides a tight example where the secant rank scales linearly with n .

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 240.10s

TIMEOUT after 240s

Judge reasoning

Test timed out without producing results; cannot evaluate support fraction or counterexamples. |
next: Increase timeout duration and re-run the test with identical parameters

Resolution Width Lower Bound via Graph Betti Numbers in Tseitin Formulas

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic Topology × Resolution Width of Tseitin Formulas
- **Recorded:** 2026-05-19 12:29 UTC
- **Entry ID:** 802deca2e3a4

Statement

For a Tseitin formula derived from a connected graph G with m edges and n vertices, the resolution width is at least $(m - n + 1)$. This bound holds for all graphs where the Tseitin formula is unsatisfiable and the graph has no multiple edges or loops.

Rationale

Betti numbers capture the cyclomatic complexity of graphs, which directly relates to the number of independent cycles in the constraint graph. These cycles create structural dependencies that may necessitate wider resolution proofs, as each cycle could require additional clauses to resolve.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
stances_tested": 1, "conjecture_holds": False, "counterexample": "graph_not_connected"}
TRIAL: {"seed": 593, "metric_name": "resolution_width", "metric_value": None, "instances_tested": 1,
TRIAL: {"seed": 631, "metric_name": "resolution_width", "metric_value": None, "instances_tested": 1,
TRIAL: {"seed": 677, "metric_name": "resolution_width", "metric_value": None, "instances_tested": 1,
TRIAL: {"seed": 727, "metric_name": "resolution_width", "metric_value": None, "instances_tested": 1,
TRIAL: {"seed": 773, "metric_name": "resolution_width", "metric_value": None, "instances_tested": 1,
TRIAL: {"seed": 821, "metric_name": "resolution_width", "metric_value": None, "instances_tested": 1,
TRIAL: {"seed": 877, "metric_name": "resolution_width", "metric_value": None, "instances_tested": 1,
TRIAL: {"seed": 929, "metric_name": "resolution_width", "metric_value": None, "instances_tested": 1,
```

--STDERR--

Traceback (most recent call last):

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b6c98454.py", line 143, in <module>

Statement

For a Tseitin formula derived from an expander graph G with n vertices, the number of irreducible components in the Schur-Weyl decomposition of the tensor product of G 's adjacency matrix with itself is $\Omega(2^n)$.

Rationale

Expander graphs' high connectivity induces complex tensor structures that require more irreducible components in their decomposition, potentially correlating with the proof complexity of the Tseitin formula via representation-theoretic invariants.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 1.48s

```
ample': 'mapping_undefined'}
```

```
TRIAL: {'metric_name': 'irreducible_components', 'metric_value': 0, 'instances_tested': 1, 'conjecture': True}
TRIAL: {'metric_name': 'irreducible_components', 'metric_value': 0, 'instances_tested': 1, 'conjecture': True}
TRIAL: {'metric_name': 'irreducible_components', 'metric_value': 0, 'instances_tested': 1, 'conjecture': True}
TRIAL: {'metric_name': 'irreducible_components', 'metric_value': 0, 'instances_tested': 1, 'conjecture': True}
TRIAL: {'metric_name': 'irreducible_components', 'metric_value': 0, 'instances_tested': 1, 'conjecture': True}
TRIAL: {'metric_name': 'irreducible_components', 'metric_value': 0, 'instances_tested': 1, 'conjecture': True}
TRIAL: {'metric_name': 'irreducible_components', 'metric_value': 0, 'instances_tested': 1, 'conjecture': True}
TRIAL: {'metric_name': 'irreducible_components', 'metric_value': 0, 'instances_tested': 1, 'conjecture': True}
TRIAL: {'metric_name': 'irreducible_components', 'metric_value': 0, 'instances_tested': 1, 'conjecture': True}
```

```
--STDERR--
```

```
Traceback (most recent call last):
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4eae02df.py", line 90, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4eae02df.py", line 90, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~^
```

```
KeyError: 'seed'
```

Judge reasoning

Test crashed before producing valid data; no reliable results to assess conjecture. | next: Implement robust error handling for metric collection and re-run with concrete expander instances

Free Cumulant Magnitude Separation in Read-Twice BPs

- **Verdict:** INCONCLUSIVE
- **Bridge:** Free Probability $\times$ Read-Twice Branching Programs
- **Recorded:** 2026-05-19 13:09 UTC
- **Entry ID:** 5eaea01f8dee

Statement

For a read-twice BP P over n variables, the magnitude of the free cumulant of its input distribution satisfies $\|\kappa(P)\| = \Omega(\log n)$. For read-once BPs, $\|\kappa(P)\| = O(1)$.

Rationale

Free cumulants capture non-commutative dependencies between variables, which are inherent in read-twice BPs but absent in read-once. This separation could expose structural differences between BP classes via operator-algebraic invariants.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.03s

```
holds": True, "counterexample": ""}
TRIAL: {"seed": 463, "metric_name": "free_cumulant_magnitude", "metric_value": 0.36888794541139386,
TRIAL: {"seed": 503, "metric_name": "free_cumulant_magnitude", "metric_value": 0.36888794541139386,
TRIAL: {"seed": 547, "metric_name": "free_cumulant_magnitude", "metric_value": 0.36888794541139386,
TRIAL: {"seed": 593, "metric_name": "free_cumulant_magnitude", "metric_value": 0.36888794541139386,
TRIAL: {"seed": 631, "metric_name": "free_cumulant_magnitude", "metric_value": 0.36888794541139386,
TRIAL: {"seed": 677, "metric_name": "free_cumulant_magnitude", "metric_value": 0.36888794541139386,
TRIAL: {"seed": 727, "metric_name": "free_cumulant_magnitude", "metric_value": 0.36888794541139386,
TRIAL: {"seed": 773, "metric_name": "free_cumulant_magnitude", "metric_value": 0.36888794541139386,
TRIAL: {"seed": 821, "metric_name": "free_cumulant_magnitude", "metric_value": 0.36888794541139386,
TRIAL: {"seed": 877, "metric_name": "free_cumulant_magnitude", "metric_value": 0.36888794541139386,
TRIAL: {"seed": 929, "metric_name": "free_cumulant_magnitude", "metric_value": 0.36888794541139386,
RESULT: SUPPORTED mean=0.36888794541139386 std=0.0 support_fraction=1.0
```

Judge reasoning

parse_fail | next: NONE

Plethysm Coefficient Ratio in SOS Refutations for Tseitin Formulas

- **Verdict:** INCONCLUSIVE
- **Bridge:** Schur-Weyl Duality $\times$ Sum-of-Squares Refutation Size
- **Recorded:** 2026-05-19 13:22 UTC
- **Entry ID:** e8c1cae80bbb

Statement

For a Tseitin formula over an n -vertex expander graph, the ratio of plethysm coefficients $\lambda[\mu]$ in the decomposition of $\text{Sym}^{k(\text{Sym}_m(\mathbb{C}^n))}$ satisfies $\lambda[\mu] \geq \Omega(n^{\lfloor k/2 \rfloor})$ for all $k \leq \log n$, while SOS refutation size is $\Theta(n^{\lfloor k/2 \rfloor})$

Rationale

Schur-Weyl duality provides a combinatorial framework to analyze tensor rank, which directly relates to SOS hierarchy's ability to refute CSPs. The plethysm coefficients capture symmetry-breaking patterns that could create inherent barriers for sos algorithms.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.03s

```
_coefficient < n^(k/2) for k=18, m=2"}

```

```
TRIAL: {"seed": 503, "metric_name": "plethysm_coefficient_ratio", "metric_value": 16, "instances_tested": 100}
```

TRIAL: {"seed": 547, "metric_name": "plethysm_coefficient_ratio", "metric_value": 8, "instances_test": 1000}

TRIAL: {"seed": 593, "metric_name": "plethysm_coefficient_ratio", "metric_value": 351, "instances_tested": 1000}

TRIAL: {"seed": 631, "metric_name": "plethysm_coefficient_ratio", "metric_value": 300, "instances_tested": 1000}

TRIAL: {"seed": 677, "metric_name": "plethysm_coefficient_ratio", "metric_value": 10, "instances_tested": 1000}

TRIAL: {"seed": 727, "metric_name": "plethysm_coefficient_ratio", "metric_value": 39, "instances_tested": 1000000, "metric_min": 39, "metric_max": 39, "metric_mean": 39, "metric_std": 0, "metric_min_max_mean_std": [39, 39, 39, 0]}

TRIAL: {"seed": 773, "metric_name": "plethysm_coefficient_ratio", "metric_value": 153, "instances_t

TRIAL: {"seed": 821, "metric_name": "plethysm_coefficient_ratio", "metric_value": 703, "instances_tested": 1000}

TRIAL: {"seed": 877, "metric_name": "plethysm_coefficient_ratio", "metric_value": 210, "instances_tested": 1000}

```
TRIAL: {"seed": 929, "metric_name": "plethysm coefficient ratio", "metric_value": 18, "instances tes
```

Judge reasoning

Safety rail: critic_challenge_falsified | original: Multiple trials show plethysm_coefficient $< n^{(k/2)}$ for specific k,m parameters, directly contradicting the conjecture | next: Analyze the k=7,m=1 case to understand why plethysm coefficients fail to meet the $\Omega(n^{\{k/2\}})$ bound

Noncommutative L^p Norm Lower Bound for Disjointness Matrices

- **Verdict:** INCONCLUSIVE
- **Bridge:** Noncommutative L^p Geometry $\times$ Communication Complexity of Disjointness
- **Recorded:** 2026-05-19 13:29 UTC
- **Entry ID:** 4a2bf93b3b28

Statement

The noncommutative L^p norm of the communication matrix M for DISJOINTNESS satisfies $\|M\|_p \geq \Omega(n)$ for $p = \log n$, and this norm lower bounds the randomized communication complexity of the function.

Rationale

Noncommutative L^p norms capture operator-theoretic structure of matrices, which may expose geometric barriers to efficient communication. For DISJOINTNESS, the matrix’s high norm could reflect the need for $\Omega(n)$ bits to resolve disjointness via adaptive protocols.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.04s

```
"conjecture holds": True, "counterexample": ""}
```

```
TRIAL: {"seed": 463, "metric name": "noncommutative Lp norm", "metric value": 0.2858705913056837, "i
```

```
TRIAL: {"seed": 503, "metric name": "noncommutative Lp norm", "metric value": 0.2918007642274359, "i
```

```

TRIAL: {"seed": 547, "metric_name": "noncommutative_Lp_norm", "metric_value": 0.5421441477680837, "i
TRIAL: {"seed": 593, "metric_name": "noncommutative_Lp_norm", "metric_value": 0.23203590952770686, "i
TRIAL: {"seed": 631, "metric_name": "noncommutative_Lp_norm", "metric_value": 0.24900934832637536, "i
TRIAL: {"seed": 677, "metric_name": "noncommutative_Lp_norm", "metric_value": 0.4523708904513098, "i
TRIAL: {"seed": 727, "metric_name": "noncommutative_Lp_norm", "metric_value": 0.1730611083822355, "i
TRIAL: {"seed": 773, "metric_name": "noncommutative_Lp_norm", "metric_value": 0.31022528349817463, "i
TRIAL: {"seed": 821, "metric_name": "noncommutative_Lp_norm", "metric_value": 0.18516347328321206, "i
TRIAL: {"seed": 877, "metric_name": "noncommutative_Lp_norm", "metric_value": 0.2670436577208097, "i
TRIAL: {"seed": 929, "metric_name": "noncommutative_Lp_norm", "metric_value": 0.3006046524354541, "i
RESULT: SUPPORTED mean=0.28556090944639073 std=0.09065176755208919 support_fraction=1.0

```

Judge reasoning

The test used $n=1$ instances, preventing observation of $\Omega(n)$ scaling. Metric values (~ 0.285) are below 1, making $\Omega(n)$ impossible for $n \geq 2$. | next: Test with $n \geq 1000$ and multiple instances to validate $\Omega(n)$ scaling

Matroid Rank Separation in Read-Twice Branching Programs

- **Verdict:** INCONCLUSIVE
- **Bridge:** Matroid Theory $\times$ Read-Twice Branching Programs
- **Recorded:** 2026-05-19 13:41 UTC
- **Entry ID:** 99773cf419c5

Statement

For any read-twice branching program P over n variables, the rank of the matroid M_P derived from its transition matrix satisfies $\text{rank}(M_P) \geq \Omega(n)$, whereas for read-once programs, $\text{rank}(M_P) = O(\log n)$.

Rationale

Matroid rank captures the combinatorial structure of dependencies in branching programs. Read-twice programs introduce persistent dependencies across paths, inflating matroid rank compared to read-once programs, which have acyclic dependency graphs.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3341c286.py", line 109, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3341c286.py", line 89, in run_trial
    rank = matroid_rank(transition_matrix(bp))
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3341c286.py", line 63, in transition_ma
    j = random.randint(i + 1, n - 1)
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

```



```
File "/usr/lib/python3.12/random.py", line 336, in randint
    return self.randrange(a, b+1)
           ^^^^^^^^^^^^^^^^^^^^^
File "/usr/lib/python3.12/random.py", line 319, in randrange
    raise ValueError(f"empty range in randrange({start}, {stop})")
ValueError: empty range in randrange(33, 33)
```

Judge reasoning

Test crashed due to invalid randint range, preventing data collection. The error suggests a code bug rather than a conjecture failure. | next: Fix the test code to handle edge cases where $i+1 \geq n-1$, then re-run with proper parameter ranges.

Additive Energy Inverse Proportionality to Karchmer-Wigderson Communication Complexity

- **Verdict:** INCONCLUSIVE
- **Bridge:** Additive combinatorics $\times$ Karchmer-Wigderson game
- **Recorded:** 2026-05-19 13:55 UTC
- **Entry ID:** 521a5bf7dd7d

Statement

For every boolean function $f: \{0,1\}^n \rightarrow \{0,1\}$, the additive energy $E(f)$ of its truth table satisfies $E(f) \leq 2^{\{n\}} / (C(f) \cdot \log n)$, where $C(f)$ is the communication complexity of the Karchmer-Wigderson game for f .

Rationale

Additive energy captures the number of additive quadruples in the truth table, reflecting algebraic structure that could enable efficient communication protocols. The inverse proportionality suggests that higher structure (lower energy) correlates with lower communication complexity.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 240.08s

TIMEOUT after 240s

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Run test with extended timeout and higher resource allocation

Polymatroid Rank Inverse Proportionality to SOS Refutation Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Polymatroid Optimization $\times$ SOS Refutation Size for CSPs
- **Recorded:** 2026-05-19 14:14 UTC
- **Entry ID:** 75387455e1b8

Statement

For a CSP instance with constraint matrix M , the polymatroid rank $r(M)$ satisfies $r(M) \geq \Omega(n/\rho)$, where ρ is the SOS refutation size. For k -CLIQUE instances, $r(M) = \Omega(n^{\{1/2\}})$ while $\rho = O(n^{\{1/2\}} \log n)$.

Rationale

Polymatroid rank captures combinatorial structure of constraints, while SOS refutation size measures algebraic complexity. Their inverse proportionality suggests hidden geometric regularity in CSP hardness.

Novelty

- Judge: ADJACENT_OK over 7 arXiv hits

Top hits consulted: - [2604.27336v1] Strongly Refuting Random CSP without Literals - [1701.04521v1] Sum of squares lower bounds for refuting any CSP - [2411.15363v3] The Polymatroid Representation of a Greedoid, and Associated Galois Connections - [2402.07064v6] Piecewise SOS-Convex Moment Optimization and Applications via Exact Semi-Definite Programs - [2307.07605v1] First-order Methods for Affinely Constrained Composite Non-convex Non-smooth Problems: Lower Complexity Bound and Near-o

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_aafdd548.py", line 66, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_aafdd548.py", line 24, in run_trial
    sos_refutation_size = sos_refutation_size(M)
                        ^^^^^^^^^^^^^^^^^^^^^
```

UnboundLocalError: cannot access local variable 'sos_refutation_size' where it is not associated with

Judge reasoning

Test crashed due to unbound local variable error, preventing data collection | next: Fix the sos_refutation_size function implementation and re-run experiments

Noncommutative Fourier Coefficient Inverse Proportionality to Communication Complexity Lower Bound

- **Verdict:** INCONCLUSIVE
- **Bridge:** Noncommutative Harmonic Analysis × Communication Complexity (two-party protocols for disjointness)
- **Recorded:** 2026-05-19 14:29 UTC
- **Entry ID:** 6623505195c8

Statement

For a two-party disjointness problem with matrix $M \in \{0,1\}^{\{n \times n\}}$, the absolute value of its non-commutative Fourier coefficient λ_k (computed via irreducible representations of S_n) satisfies $|\lambda_k|$

$\geq c / L$, where L is the communication complexity lower bound and $c > 0$ is a universal constant. Equality holds for all k with $|\lambda_k| > 0$.

Rationale

Noncommutative harmonic analysis decomposes communication matrices into group representations, exposing hidden symmetries. These symmetries may correlate with protocol efficiency, as Fourier coefficients capture correlation with group actions that constrain information flow.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': None, 'instances_tested': 0, 'con
```

```
--STDERR--
```

```
Traceback (most recent call last):
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_af447f9f.py", line 118, in <module>
    total_metric_value += trial_result["metric_value"]
```

```
TypeError: unsupported operand type(s) for +=: 'int' and 'NoneType'
```

Judge reasoning

Test crashed before producing valid data; no instances tested and metric values were None | next:
Fix the test to handle NoneType metric values and re-run with proper input instances

Jordan Rank Lower Bound for Tseitin Resolution Width

- **Verdict:** INCONCLUSIVE
- **Bridge:** Jordan Algebras $\times$ Resolution Width
- **Recorded:** 2026-05-19 14:38 UTC
- **Entry ID:** 6c4faf4f8bc5

Statement

For a Tseitin formula derived from a d -regular expander graph with n vertices, the minimal Jordan rank of its clause incidence matrix over $\text{GF}(2)$ is $\Omega(\sqrt{n})$. This implies that the resolution width of the formula is $\Omega(\sqrt{n})$ for all such instances.

Rationale

Jordan algebras provide a framework to analyze symmetric structures in clause incidence matrices. The Jordan rank captures non-linear dependencies in the clauses, which may correlate with the inherent width required for resolution proofs in expander-based Tseitin formulas.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5073fcf6.py", line 106, in <module>
    trial_result = run_trial(seed)
                    ^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5073fcf6.py", line 80, in run_trial
    jordan_rank = gaussian_elimination(matrix)
                  ^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5073fcf6.py", line 29, in gaussian_elim
    factor = Fraction(1, matrix[i][i])
              ^^^^^^^^^^^^^^^^^^^^^
File "/usr/lib/python3.12/fractions.py", line 281, in __new__
    raise ZeroDivisionError('Fraction(%s, 0)' % numerator)
ZeroDivisionError: Fraction(1, 0)
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing data collection. No counterexample found but support condition unmet. | next: Fix Gaussian elimination to handle singular matrices or verify matrix construction validity

Matroid Rank Inverse Proportionality to k-CLIQUE Communication Complexity

- **Verdict:** INCONCLUSIVE
- **Bridge:** Matroid Theory × Communication Complexity of k-CLIQUE
- **Recorded:** 2026-05-19 14:50 UTC
- **Entry ID:** 53abf37e69c1

Statement

For a graph G on n vertices, let $r(G)$ be the rank of its graphic matroid. The communication complexity of k -CLIQUE on G is $\Omega(n^{\{1 - r(G)/n\}})$ for $k \geq 2$. When $r(G) \leq n^{\{1/2\}}$, the complexity is $\Omega(n^{\{1/2\}})$

Rationale

Matroid rank captures structural dependencies in graphs; high rank implies sparse connectivity, requiring more communication to verify clique properties. This links matroid-theoretic sparsity to communication complexity barriers in monotone computation.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.10s

```
TRIAL: {"seed": 421, "metric_name": "communication_complexity", "metric_value": 1.156013, "instances": 1}
TRIAL: {"seed": 463, "metric_name": "communication_complexity", "metric_value": 1.174187, "instances": 1}
TRIAL: {"seed": 503, "metric_name": "communication_complexity", "metric_value": 1.197860, "instances": 1}
```

```

TRIAL: {"seed": 547, "metric_name": "communication_complexity", "metric_value": 1.320469, "instances": 1000000}
TRIAL: {"seed": 593, "metric_name": "communication_complexity", "metric_value": 1.137411, "instances": 1000000}
TRIAL: {"seed": 631, "metric_name": "communication_complexity", "metric_value": 1.146055, "instances": 1000000}
TRIAL: {"seed": 677, "metric_name": "communication_complexity", "metric_value": 1.276518, "instances": 1000000}
TRIAL: {"seed": 727, "metric_name": "communication_complexity", "metric_value": 1.100457, "instances": 1000000}
TRIAL: {"seed": 773, "metric_name": "communication_complexity", "metric_value": 1.189207, "instances": 1000000}
TRIAL: {"seed": 821, "metric_name": "communication_complexity", "metric_value": 1.104665, "instances": 1000000}
TRIAL: {"seed": 877, "metric_name": "communication_complexity", "metric_value": 1.167623, "instances": 1000000}
TRIAL: {"seed": 929, "metric_name": "communication_complexity", "metric_value": 1.181352, "instances": 1000000}
RESULT: SUPPORTED mean=1.178150 std=0.063764 support_fraction=1.00

```

Judge reasoning

Safety rail: critic_challenge | original judge: All 12 trials supported the conjecture with 100% confidence. The metric value remains consistent within the tested range. | next: Test larger graphs with varying $r(G)$ values to validate the $\Omega(n^{\{1/2\}})$ lower bound for $r(G) \leq n^{\{1/2\}}$

Real Rank Lower Bound for AC^0 Communication Matrices of PARITY

- **Verdict:** INCONCLUSIVE
- **Bridge:** Real Algebraic Geometry $\times$ Communication Complexity
- **Recorded:** 2026-05-19 15:23 UTC
- **Entry ID:** 50fc0919c645

Statement

For any AC^0 circuit C computing the parity function on n bits, the real rank of the communication matrix of C is $\Omega(n^{\{1/(d-1)\}})$, where d is the depth of C . This implies that the communication complexity of C is $\Omega(n^{\{1/(d-1)\}})$.

Rationale

The real rank of the communication matrix captures the geometric complexity of the function's representation. High real rank suggests that the function cannot be approximated by low-depth circuits, aligning with known lower bounds from switching lemmas.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.11s

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9e3ce890.py", line 73, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9e3ce890.py", line 53, in run_trial
    comm_matrix = communication_matrix(circuit)
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9e3ce890.py", line 27, in communication_matrix
    comm_matrix[i * n + k][j * n + k] = 1
                  ^

```

NameError: name 'k' is not defined

Judge reasoning

Test crashed due to NameError, preventing data collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Fix the NameError in the communication_matrix function and re-run tests with multiple seeds

Free Entropy Gap in Read-Twice Branching Programs for Inner Product Modulo 2

- **Verdict:** INCONCLUSIVE
- **Bridge:** Free Probability $\times$ Read-Twice Branching Programs
- **Recorded:** 2026-05-19 15:30 UTC
- **Entry ID:** 4283c19a6a01

Statement

Let P be a read-twice branching program computing IP_2 (inner product mod 2). Its free entropy $\rho(P) = \Theta(n)$. For all other read-twice BPs Q computing functions in $P/poly$, $\rho(Q) = O(\log \text{size}(Q))$.

Rationale

Free entropy quantifies non-commutative randomness; IP_2 's exponential lower bound for read-once BPs suggests its read-twice structure exhibits unique non-commutative correlations, while $P/poly$ BPs have structured, low-entropy transitions.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.05s

```
e_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'Free Entropy', 'metric_value': -8752.467003367005, 'instances_tested': 6, 'c
TRIAL: {'metric_name': 'Free Entropy', 'metric_value': -8759.306451418952, 'instances_tested': 6, 'c
TRIAL: {'metric_name': 'Free Entropy', 'metric_value': -8730.682094757096, 'instances_tested': 6, 'c
TRIAL: {'metric_name': 'Free Entropy', 'metric_value': -8739.801358826358, 'instances_tested': 6, 'c
TRIAL: {'metric_name': 'Free Entropy', 'metric_value': -8747.65405844156, 'instances_tested': 6, 'c
TRIAL: {'metric_name': 'Free Entropy', 'metric_value': -8755.76007094757, 'instances_tested': 6, 'c
TRIAL: {'metric_name': 'Free Entropy', 'metric_value': -8744.614303751803, 'instances_tested': 6, 'c
TRIAL: {'metric_name': 'Free Entropy', 'metric_value': -8750.187187349688, 'instances_tested': 6, 'c
TRIAL: {'metric_name': 'Free Entropy', 'metric_value': -8746.13418109668, 'instances_tested': 6, 'c
TRIAL: {'metric_name': 'Free Entropy', 'metric_value': -8753.98688071188, 'instances_tested': 6, 'c
TRIAL: {'metric_name': 'Free Entropy', 'metric_value': -8764.37270923521, 'instances_tested': 6, 'c
TRIAL: {'metric_name': 'Free Entropy', 'metric_value': -8730.935407647909, 'instances_tested': 6, 'c
TRIAL: {'metric_name': 'Free Entropy', 'metric_value': -8745.120929533428, 'instances_tested': 6, 'c
RESULT: SUPPORTED mean=-8745.543117684783 std=0 support_fraction=1.0
```

Judge reasoning

The test passed all 12 trials with 100% support, but $n \leq 15$ is insufficient to observe scaling behavior required for $\Omega(n)$ vs $O(\log \text{ size})$ claims. Free entropy metrics often saturate for fixed n . | next: Test with $n \geq 1000$ to observe scaling behavior and distinguish between $\Omega(n)$ and $O(\log \text{ size})$ regimes

Tensor Rank of Clause Incidence Matrix Bounds ACC⁰ Circuit Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Tensor Rank over Finite Fields $\times$ ACC⁰ Circuit Size
- **Recorded:** 2026-05-19 15:38 UTC
- **Entry ID:** 86fd76e5eb2d

Statement

For any CNF formula with n variables, the tensor rank of its incidence matrix over $\text{GF}(2)$ is $\Theta(\log n)$ if and only if the formula can be computed by an ACC⁰ circuit of size $O(n^k)$ for some k . Conversely, if the tensor rank exceeds $\Omega(n^\epsilon)$, then no ACC⁰ circuit of size $n^{o(1)}$ can compute the formula.

Rationale

Tensor rank captures the minimal number of rank-1 tensors needed to represent the incidence matrix, which reflects the algebraic complexity of the CNF's interaction structure. This invariant may expose inherent limitations in ACC⁰'s ability to compress combinatorial dependencies, aligning with Williams' framework for circuit lower bounds.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
also, 'counterexample': 'tensor_rank=15 > log2(15)'}
TRIAL: {'metric_name': 'tensor_rank', 'metric_value': 7, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'tensor_rank', 'metric_value': 25, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'tensor_rank', 'metric_value': 23, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'tensor_rank', 'metric_value': 9, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'tensor_rank', 'metric_value': 38, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'tensor_rank', 'metric_value': 16, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'tensor_rank', 'metric_value': 36, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'tensor_rank', 'metric_value': 19, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'tensor_rank', 'metric_value': 17, 'instances_tested': 1, 'conjecture_holds': 1}
```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e2b97f68.py", line 81, in <module>
    first_failing_seed = next(r["seed"] for r in results if r["counterexample"] != "")
                        ~~~~~^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e2b97f68.py", line 81, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if r["counterexample"] != "")
                        ~~~~~^
```

KeyError: 'seed'

Judge reasoning

Test crashed before producing valid data, rendering all trial results unreliable. | next: Re-run test with increased time limits and deterministic seed configuration

Schur-Weyl Component Count Separation in 3-CNF Incidence Tensors

- **Verdict:** BARRIER_HIT
- **Bridge:** Schur-Weyl Duality $\times$ 3-CNF SAT Instances
- **Recorded:** 2026-05-19 15:42 UTC
- **Entry ID:** 2165106dcdcd

Statement

For random 3-CNF formulas with n variables and m clauses, the number of distinct irreducible components in the decomposition of their incidence tensor under $GL(n) \times GL(m)$ action is $\Omega(n^2)$, while for formulas with determinant-like structure (e.g., rank-2 tensors), it is $O(n)$.

Rationale

Schur-Weyl duality decomposes tensor products into irreducible representations, revealing symmetry gaps between permutation-invariant and determinant-like structures. This could expose algebraic complexity differences between SAT instances and their tensor analogs.

Novelty

- Judge: “ over 0 arXiv hits

Empirical Test

- exit code: -1, elapsed: 0.00s

(no output)

Judge reasoning

Rejected by ALGEBRIZATION barrier

Real Rank of SOS Moment Matrix and Max-CUT Approximation Ratio

- **Verdict:** INCONCLUSIVE
- **Bridge:** Real Algebraic Geometry $\times$ SOS Degree for Max-CUT
- **Recorded:** 2026-05-19 15:51 UTC
- **Entry ID:** 95772d04a210

Statement

For any max-CUT instance with SOS approximation ratio ≥ 0.878 , the real rank of its degree- d SOS moment matrix satisfies $\text{rank}(M_d) \geq \Omega(d^2)$.

Rationale

The real rank of the moment matrix encodes geometric constraints on the SOS hierarchy's ability to approximate max-CUT. Higher rank implies stricter algebraic conditions on the feasible region, potentially limiting the hierarchy's effectiveness for certain instances.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f43a7346.py", line 106, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f43a7346.py", line 81, in run_trial
    if not sos_approximation_ratio(edges, d):
                                   ^
```

NameError: name 'd' is not defined. Did you mean: 'id'?

Judge reasoning

Test crashed due to undefined variable 'd', preventing data collection. Pre-registered support condition cannot be evaluated without valid results. | next: Fix the NameError in test_f43a7346.py by defining 'd' in the sos_approximation_ratio call

Persistent Homology Betti Numbers and Communication Complexity of 3-CNF Formulas

- **Verdict:** INCONCLUSIVE
- **Bridge:** Persistent Homology × Communication Complexity of 3-CNF Formulas
- **Recorded:** 2026-05-19 16:12 UTC
- **Entry ID:** a7266de96644

Statement

For a random 3-CNF formula with n variables, the sum of persistent homology Betti numbers (across all scales) is $\Theta(\log n)$, while the deterministic communication complexity is $\Omega(n)$. Thus, the sum of Betti numbers is inversely proportional to the communication complexity.

Rationale

Persistent homology captures the topological structure of the solution space, which may influence the communication complexity required to determine the function's value. Higher Betti numbers indicate more complex topological features, which could necessitate higher communication.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.03s

```
L: {'metric_name': 'Sum of Betti Numbers', 'metric_value': 52, 'instances_tested': 1, 'conjecture_hol
TRIAL: {'metric_name': 'Sum of Betti Numbers', 'metric_value': 53, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'Sum of Betti Numbers', 'metric_value': 40, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'Sum of Betti Numbers', 'metric_value': 35, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'Sum of Betti Numbers', 'metric_value': 29, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'Sum of Betti Numbers', 'metric_value': 13, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'Sum of Betti Numbers', 'metric_value': 48, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'Sum of Betti Numbers', 'metric_value': 45, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'Sum of Betti Numbers', 'metric_value': 16, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'Sum of Betti Numbers', 'metric_value': 74, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'Sum of Betti Numbers', 'metric_value': 26, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'Sum of Betti Numbers', 'metric_value': 70, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'Sum of Betti Numbers', 'metric_value': 38, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'Sum of Betti Numbers', 'metric_value': 33, 'instances_tested': 1, 'conjecture
RESULT: SUPPORTED mean=39.8 std=17.952529998118184 support_fraction=1.0
```

Judge reasoning

Single instance tested ($n \leq 15$) makes $\log n$ trivial compared to Betti sums ≥ 13 . Metric saturation and selection bias raise doubts about $\Theta(\log n)$ claim. | next: Test with $n \geq 1000$ and measure Betti sums across multiple random 3-CNF instances to validate $\Theta(\log n)$ scaling.

Gröbner Basis Monomial Count and SAT Complexity

- **Verdict:** INCONCLUSIVE
- **Bridge:** Computational Algebraic Geometry × SAT Problem
- **Recorded:** 2026-05-19 16:36 UTC
- **Entry ID:** 8315424c584f

Statement

For any unsatisfiable CNF formula with n variables, the number of monomials in the reduced Gröbner basis of its corresponding ideal over $\text{GF}(2)$ is at least $2^{\lfloor n/2 \rfloor}$.

Rationale

Gröbner bases capture algebraic dependencies of polynomial systems derived from CNF clauses. The monomial count reflects the complexity of resolving logical contradictions, potentially linking algebraic structure to circuit size lower bounds.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 1.45s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_26c9f5cf.py", line 78, in <module>
    result = run_trial(seed)
```

```

^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_26c9f5cf.py", line 62, in run_trial
    metric_value = buchberger(clauses)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_26c9f5cf.py", line 47, in buchberger
    G = [set([tuple(sorted(c))] for c in clauses)]
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
TypeError: unhashable type: 'list'

```

Judge reasoning

Test crashed due to unhashable type error, preventing metric evaluation | next: Fix the test code to handle clause lists properly and re-run experiments

Free Entropy Lower Bound for Disjointness Communication Complexity

- **Verdict:** INCONCLUSIVE
- **Bridge:** Free Probability $\times$ Communication Complexity of DISJOINTNESS
- **Recorded:** 2026-05-19 17:04 UTC
- **Entry ID:** b4aca3359472

Statement

For a DISJOINTNESS matrix $M \in \{0,1\}^{\{n \times n\}}$, define $\tau(M)$ as the free entropy of the random variable $X = (M_{\{i,j\}} + 1)/2$ under the semicircular distribution. Then $\tau(M) \geq \Omega(n)$ for all M with support size $\geq n^{\{1.5\}}$.

Rationale

Free entropy quantifies non-commutative complexity via Voiculescu's entropy formula. DISJOINTNESS requires $\Omega(n)$ bits due to its combinatorial structure, and free entropy captures this via matrix-valued distributions over non-commutative algebras.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.65s

```

n'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': 1.239300320182068, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'free_entropy', 'metric_value': 1.239544873717278, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'free_entropy', 'metric_value': 1.2798415306829964, 'instances_tested': 1, 'c
TRIAL: {'metric_name': 'free_entropy', 'metric_value': 1.2061111562331812, 'instances_tested': 1, 'c
TRIAL: {'metric_name': 'free_entropy', 'metric_value': 1.2485277782386066, 'instances_tested': 1, 'c
TRIAL: {'metric_name': 'free_entropy', 'metric_value': 1.2464538909911635, 'instances_tested': 1, 'c
TRIAL: {'metric_name': 'free_entropy', 'metric_value': 1.2873156682669213, 'instances_tested': 1, 'c
TRIAL: {'metric_name': 'free_entropy', 'metric_value': 1.2887115871569925, 'instances_tested': 1, 'c
TRIAL: {'metric_name': 'free_entropy', 'metric_value': 1.1916827655139577, 'instances_tested': 1, 'c

```

--STDERR--

Traceback (most recent call last):

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f90acabd.py", line 79, in <module>

Judge reasoning

Test crashed due to NameError before producing results; cannot evaluate conjecture validity | next: Fix the NameError in the is_satisfiable function by using 'vars' instead of 'var' in the generator expression

Plethysm Coefficient Exponential Gap in Symmetric Powers of Permanent Matrices

- **Verdict:** INCONCLUSIVE
- **Bridge:** Schur-Weyl Duality × Monotone Circuits for Permanent
- **Recorded:** 2026-05-19 17:34 UTC
- **Entry ID:** 5e26332ed673

Statement

For any $n \times n$ matrix M with entries in $\{0,1\}$, the plethysm coefficient of $\text{Sym}^{n(\text{Sym}^2(\mathbb{R}^n))}$ containing the irreducible representation $V_{\{n\}}$ is $\Omega(2^n)$, while for the determinant matrix D_n , this coefficient is $O(1)$.

Rationale

Schur-Weyl duality decomposes tensor powers into irreducible representations, and plethysm coefficients measure how these decompositions interact with symmetric powers. The permanent's higher plethysm coefficients suggest richer representation-theoretic structure, which could imply inherent complexity barriers for monotone circuits.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1713057d.py", line 88, in <module>
    trial_result = run_trial(seed)
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1713057d.py", line 62, in run_trial
    coeff_perm = plethysm_coefficient(M, 2)
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1713057d.py", line 52, in plethysm_coef
    return gaussian_elimination(M_k, b)[0]
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1713057d.py", line 31, in gaussian_elim
    factor = -A[j][i] / A[i][i]
```

ZeroDivisionError: float division by zero

Judge reasoning

Test crashed with division by zero error, preventing evaluation of plethysm coefficients. Support fraction cannot be assessed due to failed trials. | next: Debug Gaussian elimination implementation to handle singular matrices, then re-run tests with diversified seeds

A5 Cayley-Walk Equidistribution Defect Detects 3-CNF Density Threshold

- **Verdict:** INCONCLUSIVE
- **Bridge:** Geometric group theory: equidistribution and mixing of Cayley walks on the finite simple group $A_5 \times \text{Width-5 permutation branching programs (Barrington construction)}$ for random 3-CNF formulas
- **Recorded:** 2026-05-19 17:54 UTC
- **Entry ID:** dd47721574e1

Statement

Fix generators $a = (1\ 2\ 3\ 4\ 5)$ and $b = (1\ 3\ 5\ 2\ 4)$ of A_5 . For a 3-CNF F with n variables and m clauses, define the Barrington walk $W(F; x)$ in A_5 by, for each clause $C_i = (l_{i,1} \vee l_{i,2} \vee l_{i,3})$ and each literal position j in $\{0,1,2\}$, multiplying on the right by $g(l_{i,j}, x)$ where $g = a$ if $(i+j)$ is even and the literal is satisfied by x , $g = a^{-1}$ if $(i+j)$ is even and the literal is falsified, $g = b$ (resp. b^{-1}) if $(i+j)$ is odd and the literal is satisfied (resp. falsified). Let $\mu_F(g) = \Pr\{x \sim U(\{0,1\}^n) [W(F; x) = g]$ and define the equidistribution defect $\delta(F) = (1/2) * \sum_{g \in A_5} |\mu_F(g) - 1/60|$. Conjecture: there exists a universal α_* in $[4.0, 4.5]$ such that for uniformly random 3-CNF F with $m = \text{round}(\alpha_* n)$, (i) for $\alpha \leq 3.5$, $E[\delta(F)] = O(n^{-1/2})$; (ii) for $\alpha \geq 5.0$, $E[\delta(F)] \geq 0.10$ uniformly in n ; moreover for every individual F whose minimum resolution refutation has width $> n/3$, $\delta(F) \geq c_0 > 0$ for an absolute constant c_0 .

Rationale

A_5 is the smallest non-solvable group and is the algebraic anchor of Barrington's theorem; geometric-group invariants of walks on its Cayley graph (mixing, equidistribution) are genuinely non-commutative measures that do not pull back to polynomial extensions of the Boolean ring, sidestepping algebrization. Heuristically, a low-density (satisfiable) formula has many cancelling assignments, letting the walk equidistribute on the simple group; high-density (rigid, UNSAT) formulas force commutator obstructions that pin the walk distribution away from uniform. This bridges a geometric-group-theoretic mixing invariant with the combinatorial structure of random 3-SAT, a linkage with essentially no prior literature.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1510.05486v2] Two types of branching programs with bounded repetition that cannot efficiently compute monotone 3-CNFs - [1112.6285v3] Singularities of theta divisors and the geometry of A_5 - [2307.07683v2] Single and Multi-Speaker Cloned Voice Detection: From Perceptual to Learned Features - [1011.4255v3] Characterising planar Cayley graphs and Cayley complexes in terms of group presentations - [2604.09703v1] Cayley Graph Optimization for Scalable Multi-Agent Communication Topologies

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_19f16b59.py", line 126, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_19f16b59.py", line 82, in run_trial
    delta_F = equidistribution_defect(F)
```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_19f16b59.py", line 62, in equidistribut
    mu_F = {g: estimate_mu(F, g, n) for g in A5}
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_19f16b59.py", line 52, in estimate_mu
    count = sum(1 for x in product([0, 1], repeat=n) if barrington_walk(F, list(x)) == g)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_19f16b59.py", line 52, in <genexpr>
    count = sum(1 for x in product([0, 1], repeat=n) if barrington_walk(F, list(x)) == g)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_19f16b59.py", line 46, in barrington_wa
    g = g * g(literal, x[j])
              ^^^^^^^^^^^^^^^^^
TypeError: 'tuple' object is not callable

```

The test crashed with a `TypeError` (`'tuple' object is not callable`) due to a variable name collision where `g` is used both as the accumulator and the generator-selection function inside `barrington_walk`. No data was produced, so the pre-registered criterion cannot be evaluated. | next: Fix the name shadowing in `barrington_walk` (rename the local generator function, e.g., `gen = ...`; `g = g * gen(literal, x[j])`) and re-run the experiment.

- **Verdict:** INCONCLUSIVE
- **Bridge:** Sandpile groups / chip-firing on multigraphs (critical group theory via Smith normal form of the discrete Laplacian) $\times$ $\text{AC}^0[\text{MOD}_p]$ circuit structure approximating PARITY (Williams-style ACC^0 lower bounds)
- **Recorded:** 2026-05-19 18:01 UTC
- **Entry ID:** 496534d5bd8f

Let C be any $AC^0[MOD_p]$ circuit (p an odd prime ≥ 3) of size s and depth d on n inputs that ε -approximates $PARITY_n$, i.e. $\Pr_{\{x \in \{0,1\}^n\}}[C(x) = PARITY(x)] \geq 1/2 + \varepsilon$. Construct the underlying multigraph G_C by giving each MOD_p gate p parallel edges to each of its inputs, each AND/OR/NOT gate one edge per input, and each circuit input one edge to a common ‘sink’ node. Then the 2-rank of the sandpile (critical) group $K(G_C) = \mathbb{Z}^{|V|-1}/im\tilde{L}$ satisfies $rk_2(K(G_C)) \geq c \cdot \varepsilon \cdot n^{1/d} / \log(s+1)$ for an absolute constant $c > 0$; equivalently, any circuit with sandpile 2-rank below this threshold cannot ε -approximate $PARITY$.

Sandpile p-rank is a *syntactic, relational* invariant of the circuit DAG (computed from the Laplacian’s Smith normal form, not from the truth table), so it sidesteps Razborov–Rudich naturalness. Smith normal form over $\mathbb{Z}$ is fundamentally a torsion/integer-lattice operation, with no clean lift to a polynomial extension of the Boolean ring, so it dodges Aaronson–Wigderson algebrization. The conjecture leverages the known fact that $\text{AC}^0[\text{MOD}_p]$ is Smolensky-obstructed against PARITY by *modular-algebraic* reasons, and proposes that the obstruction reveals itself as torsion in the spanning-tree module of the circuit’s multigraph.

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [0610012v1] Tevatron-for-LHC Report of the QCD Working Group - [1908.06409v1] Schur multipliers of special p -groups of rank 2 - [0608534v2] On Automorphisms of Finite p -groups - [1412.3178v3] Some approximation problems in semi-algebraic geometry - [0906.0693v3] An improved lower bound on the counterfeit coins problem

Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_44e7f4da.py", line 176, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_44e7f4da.py", line 123, in run_trial
    s_values = [2 * n, 4 * n, 8 * n]
                ^
```

UnboundLocalError: cannot access local variable 'n' where it is not associated with a value

Judge reasoning

The test crashed with an UnboundLocalError before any trials ran, so no data on R was collected and neither the support nor falsification condition can be evaluated. | next: Fix the scoping bug so 'n' is defined before constructing s_values (e.g., move s_values inside the per-(n,d) loop), then rerun the pre-registered sweep.

Continued-Fraction Depth of Greedy-Restriction Bias Sequence Bounds ACC⁰ Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Diophantine approximation — continued fraction depth and Khinchin-Lévy statistics of rational approximants $\times$ ACC⁰ circuit size for explicit Boolean functions (Williams program; Sipser-style candidates)
- **Recorded:** 2026-05-19 18:13 UTC
- **Entry ID:** cf05799b6ef2

Statement

For a Boolean function $f: \{0,1\}^n \rightarrow \{0,1\}$, define the *greedy-bias trajectory* $\tau(f)$ as the sequence of rationals p_i/q_i obtained by iteratively (for $i=1, \dots, t=\lceil \log_2 n \rceil$) extending the current partial restriction ρ_{i-1} by the (variable, value) pair (v,b) that maximizes $|\Pr_{x \sim \rho}[f(x)=1] - 1/2|$, with $p_i/q_i := \Pr[f=1 \mid \rho_i]$ expressed in lowest terms over denominators that are powers of two divided by the current restriction count. Let $D(f)$ be the total number of partial quotients across the continued-fraction expansions of $(p_1/q_1, \dots, p_t/q_t)$. Conjecture: for every f computable by an ACC⁰ circuit of size s and depth d , $D(f) \leq 8 \cdot d \cdot (\log_2 s)^2 + d \cdot \log_2 n$; equivalently, $D(f) > 8 \cdot d \cdot (\log_2 s)^2 + d \cdot \log_2 n$ on a single instance refutes the existence of such a circuit.

Rationale

ACC⁰ gates (AND, OR, MOD _{m} for small m) admit Hastad-style random-restriction collapse: greedily biased restrictions concentrate the conditional acceptance probability onto a small number of dyadic rationals whose continued-fraction expansions are short by elementary Diophantine arguments. The trajectory is *relational* (depends on the choice sequence of restrictions, not on the truth table alone via a single statistic), which lets it detect the modular-collapse structure of ACC⁰ that ring-only invariants miss. Sipser-style functions, engineered to remain width-robust against

shallow restriction families, should force the bias trajectory through generically-irrational-looking approximants — long continued fractions are the obstruction signature.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1510.00068v2] On the complete solution of the general quintic using Rogers-Ramanujan continued fraction - [1111.6023v2] On a General Sextic Equation Solved by the Rogers Ramanujan Continued Fraction - [1908.00121v4] Continued fractions over non-Euclidean imaginary quadratic rings - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [2407.04826v1] Multi-strategy Based Quantum Cost Reduction of Quantum Boolean Circuits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_79724e62.py", line 132, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_79724e62.py", line 81, in run_trial
    circuit = build_random_circuit(n, size, depth)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_79724e62.py", line 40, in build_random_
    add_gate(gate_type)
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_79724e62.py", line 35, in add_gate
    input_indices = [random.randint(0, len(circuit) - 1) for _ in range(3)]
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/usr/lib/python3.12/random.py", line 336, in randint
    return self.randrange(a, b+1)
           ^^^^^^^^^^^^^^^^^^^
```

```
File "/usr/lib/python3.12/random.py", line 319, in randrange
    raise ValueError(f"empty range in randrange({start}, {stop})")
```

ValueError: empty range in randrange(0, 0)

Judge reasoning

The test crashed with a ValueError in build_random_circuit (empty range in randrange) before any trials produced data, so neither support nor falsification conditions could be evaluated. | next: Fix build_random_circuit to seed the circuit with input literals before sampling gate inputs (ensure len(circuit) > 0 prior to randint), then rerun the 30-seed sweep across $n \in \{12, 15, 18\}$.

Baker-Norine Rank Gap Bounds Tseitin Resolution Width

- **Verdict:** INCONCLUSIVE
- **Bridge:** Tropical Riemann-Roch: Baker-Norine divisor theory and chip-firing on graphs $\times$ Resolution width and DPLL tree size of Tseitin formulas on 3-regular expanders
- **Recorded:** 2026-05-19 18:30 UTC
- **Entry ID:** 753c91c0c440

Statement

Let G be a connected 3-regular graph on n vertices with cyclomatic genus $g(G) = |E| - |V| + 1$, and let $c: V(G) \rightarrow \{0,1\}$ be a 0/1 vertex charge with $\sum_v c(v)$ odd, defining the unsatisfiable Tseitin

CNF Tseitin(G, c). View c as an effective divisor of degree $|c|$ on the combinatorial graph G and compute its Baker-Norine rank $r_{\text{BN}}(c)$ via Dhar's burning / q -reduction algorithm; let $D(G, c) := g(G) - r_{\text{BN}}(c)$ be the rank gap. Conjecture: for every such (G, c) , the fixed-order DPLL backtrack tree of Tseitin(G, c) has at least $2^{\lceil D(G, c)/3 \rceil}$ leaves, and the maximum clause width touched in that DPLL run lies in the interval $[D(G, c)/3, 3 D(G, c) + 5]$.

Rationale

Tseitin's parity obstruction is exactly the divisor-class obstruction $[c]$ in the Picard group of G under chip-firing equivalence, and Baker-Norine rank quantifies how 'movable' c is across the graph — the same combinatorial degree of freedom that resolution must traverse to derive the global parity contradiction. Tropical Riemann-Roch on graphs has been entirely absent from proof complexity, yet it provides a sharp non-spectral, non-zeta handle: the genus-minus-rank gap is graph-theoretically invariant under chip-firing moves and is conceptually orthogonal to Ihara-zeta/Laplacian eigenvalue invariants. If true, the conjecture explains existing BSW Tseitin-on-expander width lower bounds via the fact that for 0/1 divisors of degree $\sim n/2$ the Baker-Norine rank generically saturates near -1 , forcing $D(G, c) \sim n/2$.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [0811.2800v2] Parallel Chip-Firing on the Complete Graph: Devil's Staircase and Poincare Rotation Number - [2103.09609v1] Characterizing Tseitin-formulas with short regular resolution refutations - [2407.15889v3] Periods and atomic firing sequences of parallel chip-firing games on directed graphs - [1806.02734v3] Spectral lower bounds for the orthogonal and projective ranks of a graph - [1906.08515v1] An overview on the bipartite divisor graph for the set of irreducible character degrees

Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_34a147e1.py", line 264, in <module>
    result = run_trial(seed)
              ~~~~~^~~~~~

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_34a147e1.py", line 182, in run_trial
    G = generate_3_regular_graph(n)
        ~~~~~^~~~~~

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_34a147e1.py", line 33, in generate_3_regular_graph
    if is_connected(G) and spectral_gap(G) >= 0.15:
        ~~~~~^~~~~~

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_34a147e1.py", line 54, in spectral_gap
    eigenvalues_A = eigenvalues(A)
                     ~~~~~^~~~~~

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_34a147e1.py", line 119, in eigenvalues
    A_copy[0][j] /= pivot
ZeroDivisionError: division by zero
```

Judge reasoning

The test crashed with a ZeroDivisionError in a custom eigenvalue routine during graph generation, before any trial data was produced. No support fraction could be computed, so the pre-registered criterion is unmet. | next: Replace the buggy hand-rolled eigenvalue computation with numpy.linalg.eigvalsh (or use networkx connectivity + a robust spectral gap) and rerun the 120-trial protocol.

Dual-VC Dimension of KW Conflict Family Bounds Formula Depth

- **Verdict:** INCONCLUSIVE
- **Bridge:** Vapnik-Chervonenkis dimension theory (Sauer-Shelah dual shatter function) $\times$ Karchmer-Wigderson game / de Morgan formula depth
- **Recorded:** 2026-05-19 18:40 UTC
- **Entry ID:** 4b8d8bea6f73

Statement

Let $f: \{0,1\}^n \rightarrow \{0,1\}$ be non-constant and define the Karchmer-Wigderson conflict family $L_f = \{D(x,y) : x \in f^{-1}(1), y \in f^{-1}(0)\}$ over ground set $[n]$, where $D(x,y) = \{i \in [n] : x_i \neq y_i\}$. Let $dvc(f)$ be the dual VC dimension of L_f : the largest k such that some k -subset T of $[n]$ is shattered by L_f , i.e., every $U \subseteq T$ equals $A \cap T$ for some A in L_f . CONJECTURE: for every non-constant f on $n \geq 4$, the minimum de Morgan formula depth satisfies $d(f) \geq \lceil \log_2(dvc(f)+1) \rceil$; equivalently, $dvc(f) \leq 2^{d(f)} - 1$. A single triple $(f, dvc(f), d(f))$ with $d(f) < \lceil \log_2(dvc(f)+1) \rceil$ falsifies it.

Rationale

A depth- d KW protocol partitions $f^{-1}(1) \times f^{-1}(0)$ into at most 2^d rectangles each pinned to a coordinate that must occur in every conflict set within the rectangle, so L_f 's projection patterns on any fixed coordinate set T are constrained by a coverage of at most 2^d 'leaf stars', bounding the achievable shattered cardinality. Dual VC dimension is a *relational* invariant of the bipartite KW structure (not a truth-table single number), and it fails to exclude ACC0 (XOR is in ACC0[2] yet has $dvc = n-1$), so it is not 'useful' in the Razborov-Rudich sense and dodges the natural-proofs trap while still giving a genuine depth lower bound. Sauer-Shelah duality is essentially unused in KW analyses — under five hits on arXiv linking VC-shatter to Karchmer-Wigderson protocols.

Novelty

- Judge: NOVEL over 8 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [1503.07648v2] Sign rank versus VC dimension - [2402.18250v1] String Dimension: VC Dimension for Infinite Shattering - [0111062v2] One-way communication complexity and the Neciporuk lower bound on formula size - [2203.12211v2] Tree Dimension and the Sauer-Shelah Dichotomy

Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_61a2a69e.py", line 141, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_61a2a69e.py", line 79, in run_trial
    if f(x) == 1 and f(y) == 0:
       ^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_61a2a69e.py", line 34, in MAJ_n
    return MAJ_k(n // 3, x)
           ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_61a2a69e.py", line 22, in MAJ_k
    return sum(x[i] for i in range(k)) > k // 2
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_61a2a69e.py", line 22, in <genexpr>
    return sum(x[i] for i in range(k)) > k // 2
           ~^^^
```

TypeError: 'int' object is not subscriptable

Judge reasoning

The test crashed with a `TypeError` ('int' object is not subscriptable) inside `MAJ_k` before any (f, dvc, d) data was collected, so the pre-registered support condition cannot be evaluated. | next: Fix the `MAJ_n` recursion bug (pass the sliced bit-vector `x[:k]` or proper sub-input to `MAJ_k` instead of recursing in a way that yields an int), then rerun the full grid `n` in `{4,6,8,10}` with 30 seeds per cell.

Max-Influence DPLL Branching Matches KKL Bound on Random 3-CNF

- **Verdict:** INCONCLUSIVE
- **Bridge:** KKL Fourier-influence theory (Kahn-Kalai-Linial; Friedgut influence/junta toolkit) $\times$ DPLL leaf count on random 3-CNF with influence-greedy branching
- **Recorded:** 2026-05-19 18:46 UTC
- **Entry ID:** 211ba56d901e

Statement

Let F be a 3-CNF on n variables at clause density $\alpha \in [3.0, 4.2]$, let $f_F: \{0,1\}^n \rightarrow \{0,1\}$ be its satisfying-assignment indicator, and let $I_{\max}(F) = \max_{i \in [n]} \Pr_{x \sim U}[f_F(x) \neq f_F(x \oplus e_i)]$ be its maximum single-variable influence. Let $\tau_{\text{KKL}}(F)$ be the leaf count of a DPLL run that uses unit-propagation and, at every internal node with residual formula $F|_\alpha$ and residual indicator $f\{F|_\alpha\}$, branches on the unassigned variable x_i maximizing $\text{Inf}_i(f\{F|_\alpha\})$ (recomputed per node). Conjecture: for every random 3-CNF F at the densities above and every $n \leq 40$, $\log_2(\tau_{\text{KKL}}(F)+1) \leq (1 - I_{\max}(F)) \cdot (n - 1) + 5\sqrt{n}$; a single instance violating this strict inequality falsifies the conjecture.

Rationale

KKL guarantees that any non-trivially-balanced Boolean function has a variable of influence $\Omega(\log n / n)$, and Friedgut's junta theorem says low-total-influence functions concentrate on few coordinates — together these suggest influence-greedy branching repeatedly cleaves the satisfying set along its 'pivot axis'. Testing whether the KKL-advantage compounds through restrictions probes whether influence is preserved (or amplified) under conditioning, a question untouched in the SAT-solver literature. The bound is an upper bound on an explicit deterministic algorithm's leaf count on a specific input distribution, so it is unrelated to lower bounds against P/poly and immune to the Razborov-Rudich natural-proofs trap.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1402.4455v1] Symbiosis of Search and Heuristics for Random 3-SAT - [0506069v1] A generating function method for the average-case analysis of DPLL - [1908.05361v2] Placing quantified variants of 3-SAT and Not-All-Equal 3-SAT in the polynomial hierarchy - [1103.5740v2] Generating and Searching Families of FFT Algorithms - [2003.09703v1] Variance function of boolean additive convolution

Empirical Test

- exit code: 1, elapsed: 0.69s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41947201.py", line 76, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41947201.py", line 58, in run_trial
    inf = influence(f_F, n)
          ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41947201.py", line 35, in influence
    flips = {tuple(assignment[:var] + [1-assignment[var]] + assignment[var+1:]) for assignment in truth}
              ~~~~~^~~~~~
```

TypeError: can only concatenate tuple (not "list") to tuple

Judge reasoning

The test crashed with a `TypeError` in the `influence()` function (tuple/list concatenation bug) before any trials completed, so no data was produced to evaluate the pre-registered $\Delta \geq 0$ criterion across the 240 trials. | next: Fix the `influence()` helper by converting the assignment tuple to a list (or building the flipped tuple via slicing on tuples) and rerun the full 240-trial sweep.

Restriction-Local Fourier Doubling Gap for ACC[2] vs Sipser

- **Verdict:** INCONCLUSIVE
- **Bridge:** Additive combinatorics: Plünnecke-Ruzsa doubling of heavy Fourier supports in $F_2^k \times \text{ACC}[2]$ circuits and the depth- d Sipser function (worst-case lower bound via average over restrictions)
- **Recorded:** 2026-05-19 18:51 UTC
- **Entry ID:** 51d7c2e9512a

Statement

For $f: \{0,1\}^n \rightarrow \{\pm 1\}$ and a random restriction ρ that zeros a uniformly random $(n-k)$ -subset of inputs and keeps $k = \lceil \log_2 n \rceil$ alive, let $f|_\rho$ be the resulting k -variable function with Walsh-Hadamard spectrum $\hat{g}$. Define $\text{HF}(\hat{g}) = \{S \subseteq [k]: |\hat{g}(S)| \geq 1/4\}$ and $\text{Doub}(f) = E_\rho[|\text{HF}(f|_\rho) \Delta \text{HF}(f|_\rho)| / \max(1, |\text{HF}(f|_\rho)|)]$. Conjecture: (i) every $\text{AC}^0[\oplus]$ (a.k.a. $\text{ACC}[2]$) circuit C of size s and depth d satisfies $\text{Doub}(f_C) \leq (1 + \log s)^{2d}$; (ii) the depth-3 Sipser function S_n with alternating fan-ins $\approx n^{1/3}$ satisfies $\text{Doub}(S_n) \geq \sqrt{(\log n)/6}$ for all $n \geq 16$. A single (n, seed) instance violating either bound refutes the conjecture.

Rationale

Razborov-Smolensky says $\text{ACC}[2]$ functions are approximated by low-degree F_2 -polynomials, whose Fourier mass is concentrated on a low-weight, additively-structured (Freiman-like) family of characters; restricting to a log-sized window preserves this structure, so the heavy support inherits small doubling. Sipser's recursive AND-OR-AND amplifier instead spreads heavy Fourier mass across characters with no F_2 -affine structure, so restriction averaging exposes a doubling gap. The construction escapes natural proofs by being non-large: random f has empty HF after restriction (so $\text{Doub}=0$), meaning random functions trivially satisfy the $\text{ACC}[2]$ bound — the property does not distinguish random from easy, only Sipser from easy.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [2511.07739v3] A Lower Bound for the Fourier Entropy of Boolean Functions on the Biased Hypercube - [1712.07615v2] A Plünnecke-Ruzsa inequality in compact abelian groups - [0906.0693v3] An improved lower bound on the counterfeit coins problem - [1508.06336v1] SPRIGHT: A Fast and Robust Framework for Sparse Walsh-Hadamard Transform - [0104014v1] Heavy-Ion Physics at the LHC

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_246f894e.py", line 112, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_246f894e.py", line 84, in run_trial
    f_acc = [evaluate_circuit(circuit, i) for i in k_window]
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_246f894e.py", line 37, in evaluate_circuit
    stack.append(gate(input))
                  ^^^^^^^^^^^
```

TypeError: 'int' object is not callable

Judge reasoning

The test crashed with a TypeError ('int' object is not callable) in evaluate_circuit before producing any data, so neither the support nor falsification condition can be evaluated. | next: Fix the circuit representation/evaluation bug (gate appears to be stored as an int rather than a callable) and re-run the 120-instance protocol.

Adjacency Spectrum Log-Discriminant Tracks Max-Cut SDP Slack

- **Verdict:** INCONCLUSIVE
- **Bridge:** Real algebraic geometry: Hermite-Vandermonde discriminant of a real-rooted characteristic polynomial and Fekete-Szegő logarithmic capacity on the real line $\times$ Degree-2 SOS (Goemans-Williamson / Delorme-Poljak) spectral upper bound for max-CUT, viewed as a moment-matrix invariant relevant to algebraic proof systems related to Extended Frege
- **Recorded:** 2026-05-19 19:02 UTC
- **Entry ID:** 2deb9c6539ac

Statement

For a graph G on n vertices with adjacency eigenvalues $\lambda_1 \leq \dots \leq \lambda_n$, define $\text{LogDisc}(G) := (1/n^2) \sum_{i < j} \log|\lambda_i - \lambda_j|$ (the normalized Hermite-Vandermonde log-discriminant of the real-rooted characteristic polynomial $\det(tI - A)$) and define the degree-2 SOS spectral slack $S(G) := (|E|/2 - (n/4)\lambda_1 - \text{MaxCut}(G))/n$. *Conjecture: over independent uniformly random 3-regular graphs at each fixed n in $\{12, 16, 20\}$, the Pearson correlation $\rho(S, \text{LogDisc})$ satisfies $\rho \geq 0.4$, and there exist universal constants a in $(0, 5)$, b in $\mathbb{R}$ such that $|E[S] - aE[\text{LogDisc}] - b| \leq 0.05$ for all such n . A single random instance with $S(G) > 0.2$ while $\text{LogDisc}(G) \leq -1$ falsifies the conjecture.*

Rationale

The Delorme-Poljak spectral bound for max-CUT is tight precisely when the adjacency spectrum is concentrated near its extreme eigenvalue, whereas the Hermite-Vandermonde log-discriminant

is the real-algebraic obstruction to root coalescence (Fekete-Szego logarithmic energy on the real line). Linking the SOS spectral slack to a classical RAG invariant of the characteristic polynomial sidesteps ring-extension arithmetization (so it should evade algebrization) and avoids any single-truth-table predicate (so it should evade natural proofs). The invariant is relational (compares pairs of eigenvalues), not a single-number truth-table statistic.

Novelty

- Judge: NOVEL over 10 arXiv hits

Top hits consulted: - [1411.1698v2] On the Max-Cut of Sparse Random Graphs - [2109.02238v1] Optimal solutions and ranks in the max-cut SDP - [1207.5696v1] Beyond Max-Cut: λ -Extendible Properties Parameterized Above the Poljak-Turzik Bound - [1801.09531v2] Fekete-Szego Inequality for Analytic and Bi-univalent Functions Subordinate to Chebyshev Polynomials - [1806.07358v1] No Threshold graphs are cospectral

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0ed02682.py", line 132, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0ed02682.py", line 93, in run_trial
    eigenvalues = adjacency_matrix_to_eigenvalues(A)
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0ed02682.py", line 37, in adjacency_mat
    v /= math.sqrt(sum(x * x for x in v))
```

TypeError: unsupported operand type(s) for /=: 'list' and 'float'

Judge reasoning

The test crashed with a `TypeError` in `adjacency_matrix_to_eigenvalues` (`list /= float`) before producing any data, so neither the support nor the falsification condition can be evaluated. | next: Fix the eigenvector normalization bug (convert `v` to a numpy array or use list comprehension division) and rerun the 90-graph trial to obtain valid Pearson correlations and OLS residuals.

Discrete Morse Betti Growth in Resolution-Frege Triangle Complexes for PHP

- **Verdict:** INCONCLUSIVE
- **Bridge:** Discrete Morse theory and cellular homology of 2-CW complexes (Forman 1998; Benedetti-Lutz) $\times$ Resolution refutations of the Pigeonhole Principle $\text{PHP}_{\{n+1 \rightarrow n\}}$ as a sub-system of Frege/Extended Frege
- **Recorded:** 2026-05-19 19:08 UTC
- **Entry ID:** 3b8307405b8f

Statement

Given any resolution refutation P of $\text{PHP}_{\{n+1 \rightarrow n\}}$, define its triangle 2-CW complex $K(P)$: 0-cells are the distinct clauses occurring in P , 1-cells are unordered pairs $\{C, D\}$ of clauses that co-appear as inputs/output of at least one resolution step, and 2-cells are triangular faces $\{A, B, R\}$ for each resolution inference $A, B \vdash R$. Let $\beta_1(K(P); F_2) = (|E| - |V| + \beta_0) - \text{rank}_{\{F_2\}}(\partial_2)$ be the first

mod-2 Betti number. Conjecture: for every $n \geq 3$ and every (tree-like or DAG) resolution refutation P of $\text{PHP}\{n+1 \rightarrow n\}$, $\beta_1(K(P); F_2) \geq \lfloor n \cdot \log_2(n+1) \rfloor$, and moreover the ratio $\beta_1(K(P))/|P|$ stabilizes to a strictly positive constant $c_{\text{PHP}} > 0$ as n grows; a single instance with $\beta_1 < \lfloor n \cdot \log_2(n+1) \rfloor$ refutes the conjecture.

Rationale

Resolution proofs are routinely measured by size/width but almost never by the cellular topology of the inference DAG itself; treating each resolution rule as a glued triangle yields a 2-complex whose homology is a functorial invariant of the proof object, not of the encoded truth-table — sidestepping the Razborov-Rudich natural-proofs trap that killed prior single-number truth-table attempts in this approach. The construction is purely combinatorial (no arithmetization, no oracle), so it does not algebrize or relativize. If β_1 genuinely scales with n it provides a Frege-relevant complexity lower bound complementing Haken's width method while exposing a previously unmeasured topological resource.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1608.07792v1] A Generalized Resolution Proof Schema and the Pigeonhole Principle - [0304041v1] $P \neq NP$, propositional proof complexity, and resolution lower bounds for the weak pigeonhole principle - [0909.5038v1] An Exponential Lower Bound on OBDD Refutations for Pigeonhole Formulas - [2504.15729v2] Strong discrete Morse theory - [2603.28382v1] Anick Resolution for Lawvere Theories from Algebraic Discrete Morse Theory

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ae02aa7.py", line 149, in <module>
    result = run_trial(seed)
    ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ae02aa7.py", line 88, in run_trial
    if dpll(clauses, assignment):
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ae02aa7.py", line 57, in dpll
    if dpll([c for c in clauses if not (set(c) & {pos_var, neg_var})], new_assignment):
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ae02aa7.py", line 51, in dpll
    pos_var = next(var for var in clause if var not in assignment)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

StopIteration

Judge reasoning

The test crashed with a StopIteration exception in the DPLL routine before producing any data; no RESULT line or measurements were emitted, so neither the support nor falsification conditions can be evaluated. | next: Fix the DPLL unit-propagation bug (guard the next(...) call against fully-assigned clauses) and re-run the 90-trial pre-registered protocol to obtain β_1 statistics.

Polymatroid Spectral Gap of Monotone DNFs Detects k-Cliques

- Verdict: INCONCLUSIVE

- **Bridge:** Polymatroid expansion via Lasserre level-1 SOS spectral gap of term-variable incidence (Barak-Steurer-style polymatroid rank lift) $\times$ Monotone DNF size lower bounds for the k -CLIQUE indicator (Razborov monotone program)
- **Recorded:** 2026-05-19 19:16 UTC
- **Entry ID:** 1ae8691b7cfd

Statement

For a monotone DNF $D = T_1 \vee \dots \vee T_s$ on n Boolean variables, let $M_D \in \{0,1\}^{\{s \times n\}}$ be its term-variable incidence matrix ($M[i,v]=1$ iff variable v occurs in term T_i) and let $G_D := D_r^{\{-1/2\}} M_D D_c^{\{-1/2\}}$ be the normalized bipartite incidence (D_r, D_c are the row/column degree diagonals). Define the polymatroid SOS-gap $\mu(D) := \sigma_1(G_D) - \sigma_2(G_D)$, the gap between the top two singular values of the level-1 Lasserre moment lift. Conjecture: (i) μ is submodular under conjunctive merge, $\mu(D \wedge D') \leq \mu(D) + \mu(D')$; (ii) for any monotone DNF D with $s \leq n^c$ terms representing a function with a $\text{poly}(n)$ -size monotone circuit, $\mu(D) \leq C \cdot c \cdot \log n$ for an absolute constant C ; (iii) for the canonical CLIQUE $_k$ DNF on K_N (terms = all k -cliques, variables = the $n = C(N,2)$ edges), $\mu \geq c_0 \cdot \sqrt{N}$ for every $k \geq 3$ and some absolute $c_0 > 0$.

Rationale

Razborov's approximation method replaces $\wedge/\vee$ with lattice operations whose distortion is exactly measured by polymatroid spread over positive/negative test sets; the singular spectrum of the term-variable incidence is the linear-algebraic shadow of this spread at SOS level 1, so a spectral gap predicts an unavoidable polymatroid obstruction to covering k -cliques compactly. The k -clique DNF's terms inherit Johnson-scheme structure $J(N,k)$, whose second eigenvalue is bounded away from the top by an $\sqrt{N}$ -scale gap that random poly-size monotone DNFs cannot reproduce. Because μ acts on the SYNTACTIC representation (not the truth table) it sidesteps natural-proofs largeness, and because it is purely real-spectral on $\{0,1\}$ -matrices it does not algebrize over polynomial extensions of $\text{GF}(2)$.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1801.08709v2] Adaptive Lower Bound for Testing Monotonicity on the Line - [2403.19550v2] Spectral gap for surfaces of infinite volume with negative curvature - [1806.02734v3] Spectral lower bounds for the orthogonal and projective ranks of a graph - [2501.09545v1] Hardness of clique approximation for monotone circuits - [2311.04204v3] Sharp Thresholds Imply Circuit Lower Bounds: from random 2-SAT to Planted Clique

Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_22556a5e.py", line 149, in <module>
    result = run_trial(seed)
    ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_22556a5e.py", line 106, in run_trial
    mu_clique = compute_polymatroid_spectral_gap(n)
    ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_22556a5e.py", line 95, in compute_polymatroid_spectral_gap
    eigenvalues = compute_spectral_gap(M_normalized)
    ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_22556a5e.py", line 61, in compute_spectral_gap
    AT_A = [[sum(A[i][k] * A[k][j] for k in range(n)) for j in range(m)] for i in range(m)]
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_22556a5e.py", line 61, in <genexpr>
    AT_A = [[sum(A[i][k] * A[k][j] for k in range(n)) for j in range(m)] for i in range(m)]
```

IndexError: list index out of range

Judge reasoning

The test crashed with an `IndexError` due to a matrix dimension bug in `compute_spectral_gap` (indexing `A[i][k]` with `k` in `range(n)` for an incompatible shape), so no data was produced for any of the three claims. | next: Fix the matrix multiplication in `compute_spectral_gap` (use `numpy.linalg.svd` on `G.D` directly) and rerun the pre-registered N sweep and 30-seed protocol.

Negative SoS-Fourier Hankel Spectrum Separates Sipser from ACC0[3]

- **Verdict:** INCONCLUSIVE
- **Bridge:** Sum-of-Squares hierarchy pseudo-moment matrices built from Walsh-Hadamard (Fourier-Hankel) signatures $\times$ ACC0[3] circuit complexity of the depth-2 Sipser function S_n (top OR of bottom ANDs of fan-in $\sim \sqrt{n}$)
- **Recorded:** 2026-05-19 19:25 UTC
- **Entry ID:** c1588ea452b8

Statement

For $f: \{0,1\}^n \rightarrow \{-1,+1\}$, let $B_d = \{S \subseteq [n] : |S| \leq d\}$ and let $Q_d(f)$ be the $|B_d| \times |B_d|$ symmetric matrix with entries $Q_d(f)[S,T] = \mathbb{E}(\mathbb{1}_S \cdot \mathbb{1}_T)$, where $\mathbb{1}_U = 2^{-n} \sum_x \mathbb{1}_U(x) \cdot (-1)^{\langle U, x \rangle}$. Define the SoS-Fourier negativity gap $NG_d(f) := -\lambda_{\min}(Q_d(f))$. The conjecture: (i) for every f computable by a depth-3 ACC0[3] circuit of size s , $NG_d(f) \leq s^{-1/2}$ for all $d \leq \log s$, and (ii) for the explicit Sipser function S_n with bottom fan-in $\lfloor \sqrt{n} \rfloor$ and top fan-in $\lceil \sqrt{n} \rceil$, $NG_2(S_n) \geq 1/(8n)$. A single ACC0[3] circuit (with size up to $n^* \log n$, depth 3) achieving $NG_2 \geq 1/(8n)$, or a single n in $\{6,8,10\}$ where Sipser yields $NG_2 < 1/(8n)$, refutes it.

Rationale

$Q_d(f)$ is the Gram matrix of the SoS pseudo-expectation $E[\chi_i \cdot \chi_j]$; its negative-spectrum component encodes obstructions to writing f as a sum-of-squares of degree- d Fourier polynomials. Random and ACC0[3] truth tables have Fourier mass concentrated at low $|U|$, forcing Q_d to be near-PSD (small NG). Sipser’s tight DNF/CNF gap creates a sign-cancellation pattern on medium-sized Fourier coefficients that, viewed through the Hankel skew, should produce a robust negative eigenvalue at degree 2 - and crucially the property ‘high negativity’ is SMALL (random f gives $NG = O(2^{-n/2})$), so the property avoids the Razborov-Rudich largeness condition while remaining useful and subexp-computable.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [2511.07739v3] A Lower Bound for the Fourier Entropy of Boolean Functions on the Biased Hypercube - [2003.09703v1] Variance function of boolean additive convolution - [2406.18700v4] Structure of sparse Boolean functions over Abelian groups, and its application to testing - [2504.19966v3] Quantum circuit lower bounds in the magic hierarchy - [1508.06336v1] SPRIGHT: A Fast and Robust Framework for Sparse Walsh-Hadamard Transform

Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3f4e32f9.py", line 89, in <module>
    print(f"TRIAL: {run_trial(seed)}")
    ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3f4e32f9.py", line 57, in run_trial
    Q_2_S_n = gram_matrix(f, 2)
    ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3f4e32f9.py", line 22, in gram_matrix
    B_d = list(combinations(range(len(f)), d))
    ~~~~~
```

TypeError: object of type 'function' has no len()

Judge reasoning

The test crashed with a `TypeError` (`len()` called on a function) before any `NG_2` values were computed, so the pre-registered support condition could not be evaluated. | next: Fix the `gram_matrix` call to use `n` (the input length) instead of `len(f)`, then rerun the trials for `n` in `{6,8,10}`.

Sturm Real-Root Count of GW Moment Matrix Bounds Max-CUT Slack

- **Verdict:** INCONCLUSIVE
- **Bridge:** Real algebraic geometry — Hermite-Sturm real root counting on intervals via signed Hankel/subresultant signatures × Degree-2 Sum-of-Squares (Goemans-Williamson) pseudo-expectation moment matrix for max-CUT on 3-regular graphs
- **Recorded:** 2026-05-19 19:45 UTC
- **Entry ID:** 6afd248521a0

Statement

Let G be a 3-regular graph on n vertices with Laplacian L_G ; obtain a degree-2 GW Gram matrix $M_G \in \mathbb{R}^{\{n \times n\}}$ by 80 steps of projected gradient ascent of $\langle L_G, V V^T \rangle / 4$ over rank- r unit-row matrices V with $r = \lceil \sqrt{2n} \rceil$ from a fixed deterministic init ($M_G = \bar{V} V^T$ at convergence, all $\text{diag}(M_G) = 1$). Let $R(G) := (\# \text{ eigenvalues of } M_G \text{ lying in the open interval } (-1 + 1/\sqrt{n}, 1 - 1/\sqrt{n}))$, equivalently the positive index minus the negative index of the Hermite-Hankel form built from the first $2n$ traces $\text{tr}(M_G^k)$ restricted to that interval. Conjecture: for $n \in \{8, 10, 12, 14, 16, 18, 20, 25, 30, 35, 40\}$ and 30 i.i.d. uniform 3-regular graphs per n , $R(G)/n$ is concentrated within ± 0.07 of a universal constant $c \in [0.62, 0.80]$, and every instance with GW objective $\langle L_G, M_G \rangle / 4 \geq 0.878 \cdot \text{UB}(G)$ (where $\text{UB}(G) = |E|/2 + n \cdot \lambda_{\max}(L_G)/4$ is the spectral upper bound on MaxCut) satisfies $R(G)/n \geq 0.55$ — a single counterexample (high SDP ratio with $R/n < 0.55$, or low SDP ratio with $R/n \geq 0.80$) refutes the conjecture.

Rationale

The Hermite-Sturm count of moment-matrix eigenvalues in $(-1, 1)$ is a real-semialgebraic invariant of the SOS pseudo-distribution that is preserved under monomial relabelings and odd-degree real-affine reparametrizations, hence under simple polynomial reductions of CSPs. Tightness of the GW relaxation forces correlations to saturate to ± 1 along the cut direction, pushing M_G 's spectrum away from the open middle of $[-1, 1]$; conversely, slack leaves eigenvalues stranded in the interior, so the Hermite signature provides a structural witness rather than a truth-table single-number invariant. Because the invariant is built from real-spectral data of an SDP solution (not from polynomial-ring extensions of the Boolean cube), the construction sidesteps Razborov-Rudich,

algebrization, and Karp-Lipton barriers while staying inside the Williams-style spectral lower-bound toolkit.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [2004.09833v2] Expressing the largest eigenvalue of a singular beta F-matrix with heterogeneous hypergeometric functions - [2604.08367v2] Per-Shot Evaluation of QAOA on Max-Cut: A Black-Box Implementation Comparison with Goemans-Williamson - [2112.11354v4] Warm-Started QAOA with Custom Mixers Provably Converges and Computationally Beats Goemans-Williamson's Max-Cut at Low C_i - [0812.4567v1] On degenerate Hamburger moment problem and extensions of positive semidefinite Hankel block matrices - [2108.02009v3] Isolation Intervals of the Real Roots of the Parametric Cubic Equation and Improved Complete Root Classification

Empirical Test

- exit code: 1, elapsed: 11.55s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0a8a5c97.py", line 158, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0a8a5c97.py", line 127, in run_trial
    G = generate_random_3_regular(n)
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0a8a5c97.py", line 45, in generate_random_3_regular
    if is_3_regular(G):
       ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0a8a5c97.py", line 23, in is_3_regular
    for u, v in G:
       ^^^^
```

ValueError: not enough values to unpack (expected 2, got 1)

Judge reasoning

The test crashed in `generate_random_3_regular` with a `ValueError` before producing any data, so no instances were evaluated against the pre-registered criterion. | next: Fix the graph generator's edge unpacking bug (iterate edges as tuples, e.g., `G.edges()`) and rerun the full 330-instance sweep.

Hermite Spectral-Diversity Lower Bound on Tseitin GW Slack

- **Verdict:** INCONCLUSIVE
- **Bridge:** Real algebraic geometry — Hermite-Sylvester signature theorem for counting distinct real roots via the rank/signature of moment Hankel matrices $\times$ Degree-2 SOS / Delorme-Poljak-Goemans-Williamson integrality slack for max-CUT on 3-regular Tseitin-type graphs (a degree-2 pseudo-moment matrix property tied to expander Tseitin proof complexity)
- **Recorded:** 2026-05-19 20:01 UTC
- **Entry ID:** 78c9742281c4

Statement

For every connected 3-regular graph G on n vertices, let $A(G)$ be its adjacency matrix and let H_G be the $n \times n$ Hankel matrix $H_G[i,j] = (1/n) \cdot \text{trace}(A^{\{i+j\}})$ for $0 \leq i,j \leq n-1$. By the Hermite-Sylvester theorem the rank $\nu(G)$ of H_G equals the number of distinct eigenvalues of $A(G)$, an invariant of

the real-algebraic moment problem. Let $S(G) := \text{UB_DP}(G) - \text{MaxCut}(G)$ where $\text{UB_DP}(G) = |E|/2 - (n/4) \cdot \lambda_{\min}(A(G))$ is the Delorme-Poljak degree-2 SOS upper bound. Conjecture: there is an absolute constant $c \in (0.02, 0.15)$ such that $S(G)/n \geq c \cdot \nu(G)/n$ for all 3-regular G with $n \geq 8$, with the bound saturating up to a factor of 3 on random 3-regular graphs; equivalently, every graph whose normalized GW slack is $o(\nu(G)/n)$ is a counterexample.

Rationale

The Hermite-Sylvester signature of the trace-moment Hankel matrix is the canonical real-algebraic-geometry encoding of spectral diversity, and spectral diversity is exactly what forces the Delorme-Poljak eigenvalue relaxation to be slack: a graph whose A has many distinct eigenvalues cannot align its top eigenspace with any single ± 1 cut indicator, so the SOS Lagrangian leaves a residual that scales linearly in distinct-eigenvalue count. This links a Positivstellensatz-style moment-matrix invariant (preserved under the polynomial reductions that act linearly on adjacency moments) to integrality slack on the Tseitin-style 3-regular expanders where SOS lower bounds are known to be hardest. It sidesteps natural proofs (the invariant is a graph property, not a truth-table predicate), algebrization (no polynomial extension of the Boolean ring is used), and relativization (no oracle TM is invoked).

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1107.1630v3] On the Integrality Gap of the Subtour LP for the 1,2-TSP - [1207.5696v1] Beyond Max-Cut: λ -Extendible Properties Parameterized Above the Poljak-Turzík Bound - [2605.14994v1] Sharp Bounds on the Eigenvalues of Kikuchi Graphs and Applications to Quantum Max Cut - [0812.4567v1] On degenerate Hamburger moment problem and extensions of positive semidefinite Hankel block matrices - [1508.00617v2] Hankel determinants of random moment sequences

Empirical Test

- exit code: 1, elapsed: 0.15s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df09d2e2.py", line 133, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df09d2e2.py", line 23, in run_trial
    UB_DP = ub_dp(G)
            ^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df09d2e2.py", line 90, in ub_dp
    lambda_min_A = min(eigenvalues(A)[0], eigenvalues(A)[-1])
                    ^
```

NameError: name 'A' is not defined

Judge reasoning

The test crashed with a NameError ('A' is not defined) inside `ub_dp` before any trials produced data, so the pre-registered support condition cannot be evaluated. | next: Fix the `ub_dp` function (pass/define A correctly, e.g., $A = \text{nx.adjacency_matrix}(G).toarray()$) and rerun the full 180-graph protocol.

Edge-Expansion Rank Bound on Read-Twice BPs for Tseitin

- **Verdict:** INCONCLUSIVE
- **Bridge:** Discrepancy / sign-rank theory of communication matrices (real-rank as a discrepancy proxy, in the spirit of Razborov-Sherstov) $\times$ Syntactic read-twice branching program size for Tseitin formulas on 3-regular expanders
- **Recorded:** 2026-05-19 20:26 UTC
- **Entry ID:** 22abf1f27d7f

Statement

Let G be a connected 3-regular graph on m vertices with edge expansion $h(G) := \min_{1 \leq |S| \leq m/2} |\partial S|/|S|$, and let $\sigma: V(G) \rightarrow \{0,1\}$ be any odd charging. Let π be the median edge bipartition (first $\lfloor |E|/2 \rfloor$ edges by canonical lex order vs. the rest), and let $M_{\{G,\sigma,\pi\}} \in \{0,1\}^{\lfloor |E|/2 \rfloor \times 2^{\lfloor |E|/2 \rfloor}}$ be the bipartite truth-table of the Tseitin function $f_{\{G,\sigma\}}$. Conjecture (BP-READTWICE-Tseitin Rank): $\log_2 \text{rank}_{\mathbb{R}}(M_{\{G,\sigma,\pi\}}) \geq (1/4) \cdot h(G) \cdot m$ for every such (G,σ) ; equivalently, since every syntactic read-twice BP P satisfies $\text{rank}_{\mathbb{R}}(M_{\{f,\pi\}}) \leq \text{size}(P)^2$, every read-twice BP for $f_{\{G,\sigma\}}$ has size $\geq 2^{\{h(G) \cdot m/8\}}$.

Rationale

Read-once BPs admit $\text{rank}(M_f) \leq \text{size}(P)$ under matching orderings, and the only known size lower bounds for Tseitin on expanders go through this rank/width route; pushing the exponent from 1 to 2 captures exactly the ‘one extra read’ of read-twice and lets edge-expansion (the same combinatorial gadget Ben-Sasson-Wigderson used for resolution width) drive a BP size bound. Real-rank of the communication matrix is a clean, purely combinatorial discrepancy proxy, sidestepping algebraization (the bound uses a non-algebraic graph invariant $h(G)$) and natural-proofs (the property is not P-time on truth tables, only on BPs). If true it would yield the first quantitative read-twice lower bound for an explicit family beyond IP_2 .

Novelty

- Judge: NOVEL over 11 arXiv hits

Top hits consulted: - [2209.05839v3] On bounded depth proofs for Tseitin formulas on the grid; revisited - [1709.08890v2] Partial matching width and its application to lower bounds for branching programs - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [1806.02734v3] Spectral lower bounds for the orthogonal and projective ranks of a graph - [0906.0693v3] An improved lower bound on the counterfeit coins problem

Empirical Test

- exit code: 124, elapsed: 240.09s

TIMEOUT after 240s

Judge reasoning

The test timed out after 240s (returncode 124) without producing any RESULT line or instance data, so neither the support nor falsification conditions can be evaluated. | next: Reduce the instance set (e.g., $m \in \{4,6,8\}$ with 10 graphs each) or optimize the rank/expansion computation (sparse linear algebra, cached BFS) to fit within the time budget, then rerun.

Lie Stabilizer Codimension Bounds ABP Width

- **Verdict:** INCONCLUSIVE

- **Bridge:** Geometric Complexity Theory: Lie algebra stabilizer of a homogeneous polynomial under the natural GL_N action on $Sym^d C^N$ (Mulmuley-Sohoni, Burgisser-Ikenmeyer style symmetry invariants) $\times$ Algebraic Branching Program (ABP) width for homogeneous multilinear polynomials
- **Recorded:** 2026-05-19 20:38 UTC
- **Entry ID:** 82460edbaea0

Statement

For a homogeneous polynomial f of degree d in N variables, define $\sigma(f) = \dim\{A \in gl_N : \sum_{i,j} A_{ij} x_j \partial f / \partial x_i \equiv 0 \text{ as a polynomial}\}$, the dimension of the Lie-algebra stabilizer of f under the standard GL_N derivation action. Conjecture: if f is computed by an algebraic branching program of width w (with affine-in-variables edge labels), then $\sigma(f) \geq N - 2w^2$. Equivalently, the GL_N -orbit codimension obeys $\text{codim}(GL_N \cdot f) \geq N - 2w^2$, so any polynomial whose Lie-stabilizer codimension exceeds $N - 2w^2$ requires ABP width strictly greater than w .

Rationale

GCT's central thesis is that the symmetry (Lie stabilizer) of an explicit polynomial is the obstruction that separates orbit closures; $\det$ has a huge Lie stabilizer while permanent's is tiny, mirroring their conjectured complexity gap. Bounded-width ABPs build polynomials by 2×2 -type recursive products and partial sums, which we conjecture force a large 'inherited' continuous symmetry algebra at least linearly in N once w is bounded, whereas random polynomials of degree ≥ 3 generically have $\sigma = 0$. The mapping is purely differential-linear-algebraic (kernel of the polynomial-coefficient map on gl_N), so it avoids ring-only constructions that algebrize and is not a truth-table predicate, sidestepping Razborov-Rudich.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_10694d13.py", line 149, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_10694d13.py", line 118, in run_trial
    f, T = build_ABP(N, w, d, 2**30 - 1)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_10694d13.py", line 73, in build_ABP
    f = [[M[i][j][k] * M[(i + j) % w][k][l] for k in range(d + 1)] for j in range(w) for l in range(2)]
           ^
```

UnboundLocalError: cannot access local variable 'i' where it is not associated with a value

Judge reasoning

Test crashed due to unbound local variable 'i' in ABP construction, preventing data collection | next: Fix variable scoping in build_ABP function and rerun tests with validated code

Real Rank of Clause-Indicator Polynomial Bounds ACC⁰ Circuit Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Real Algebraic Geometry × ACC⁰ Circuit Size
- **Recorded:** 2026-05-19 20:52 UTC
- **Entry ID:** f849016d205f

Statement

For any CNF formula with n variables, the real rank of its clause-indicator polynomial over $\mathbb{R}$ is $\Theta(\log n)$ if and only if the formula can be computed by an ACC⁰ circuit of size $\Theta(n)$.

Rationale

Real algebraic geometry provides invariants like real rank that capture the structure of polynomial representations, which may mirror the layered complexity of ACC⁰ circuits. The real rank of the clause-indicator polynomial could reflect the minimal number of real-valued polynomial components needed to compute the CNF, directly tied to circuit size.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1932a445.py", line 68, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1932a445.py", line 49, in run_trial
    rank = sum(1 for row in gaussian_elimination(A, b) if any(row))
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1932a445.py", line 25, in gaussian_elim
    factor = A[j][i] / A[i][i]
             ^^^^^^^^^
```

ZeroDivisionError: float division by zero

Judge reasoning

Test crashed with division by zero error, preventing evaluation of the conjecture's validity | next:
Debug gaussian_elimination implementation to handle singular matrices

Singular Value Gap in Read-Twice BPs for IP₂

- **Verdict:** INCONCLUSIVE
- **Bridge:** Random Matrix Theory × Read-Twice Branching Programs for IP₂
- **Recorded:** 2026-05-19 21:00 UTC
- **Entry ID:** f3a21c12c399

Statement

The sum of singular values of the transition matrix of a read-twice BP for IP₂ is $\Omega(n)$, whereas for read-once BPs, it is $O(\log n)$.

Rationale

Random matrix theory provides tools to analyze the spectral structure of BP transition matrices. IP_2's inherent symmetry forces large singular values in read-twice BPs, contrasting with the logarithmic bounds for read-once BPs, which could reflect structural differences in their computation.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6fdb702f.py", line 85, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6fdb702f.py", line 69, in run_trial
    sv_sum_twice = sum_of_singular_values(n)
                  ^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6fdb702f.py", line 31, in sum_of_singular_values
    n = len(T)
        ^^^^^
```

TypeError: object of type 'int' has no len()

Judge reasoning

Test crashed with TypeError due to invalid len() call on integer | next: Debug the sum_of_singular_values function to fix type handling

Real Stable Polynomial Root Count Separation for Monotone DNF

- **Verdict:** INCONCLUSIVE
- **Bridge:** Real Stable Polynomials $\times$ Monotone DNF Formulas for k-CLIQUE
- **Recorded:** 2026-05-19 21:12 UTC
- **Entry ID:** 63ae2b41cb6d

Statement

Let f be a monotone DNF formula for the k-CLIQUE problem on n -vertex graphs. Let $P_f(x)$ be the real stable polynomial whose coefficients encode the inclusion-exclusion of f 's clauses. Then, the number of real roots of $P_f(x)$ is $\Omega(n)$, whereas for general monotone DNF formulas of size $\text{poly}(n)$, it is $O(\log n)$.

Rationale

Real stable polynomials encode positivity conditions on their roots, which may mirror the combinatorial rigidity of k-CLIQUE's clause structure. The root count acts as a proxy for the formula's 'algebraic complexity', potentially exposing inherent limitations in representing k-CLIQUE with small DNFs.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_11971f3e.py", line 108, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_11971f3e.py", line 69, in run_trial
    clique_poly = real_stable_polynomial(generate_k_clique(n, k))
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_11971f3e.py", line 28, in generate_k_clique
    if (clique[i], clique[j]) not in edges and (clique[j], clique[i]) not in edges:
        ^^^^^
```

NameError: name 'edges' is not defined

Judge reasoning

Test crashed due to NameError before producing results, preventing evaluation of conjecture support/falsification. | next: Fix the NameError in generate_k_clique by properly initializing edges, then rerun tests with multiple seeds

Automorphism Group Conjugacy Classes and Resolution Width of Tseitin Formulas

- **Verdict:** INCONCLUSIVE
- **Bridge:** Representation Theory of Finite Groups × Resolution Width of Tseitin Formulas on Expanders
- **Recorded:** 2026-05-19 21:23 UTC
- **Entry ID:** b1190ecf760d

Statement

For a Tseitin formula derived from an expander graph G , the resolution width is at least the number of conjugacy classes in the automorphism group of G . Formally, if $C(G)$ denotes the number of conjugacy classes of $\text{Aut}(G)$, then $\text{res_width}(G) \geq C(G)$.

Rationale

The automorphism group's conjugacy classes encode symmetry properties of the graph. These symmetries may impose structural constraints on the resolution proof complexity, suggesting a lower bound on the width related to the group's representation-theoretic structure.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f9a344b0.py", line 76, in <module>
    result = run_trial(seed)
```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f9a344b0.py", line 52, in run_trial
    backtrack({}, 0)
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f9a344b0.py", line 46, in backtrack
    if all(current_mapping[u] != i for u, v in edges if v == next_node and u not in current_mapping):
    ~~~~~^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f9a344b0.py", line 46, in <genexpr>
    if all(current_mapping[u] != i for u, v in edges if v == next_node and u not in current_mapping):
    ~~~~~^
KeyError: 10

```

Test crashed with `KeyError` before producing results, preventing validation of the conjecture. | next: Debug and fix the test code to handle node indexing properly, then re-run with multiple seeds

- **Verdict:** INCONCLUSIVE
- **Bridge:** Representation Theory of Algebraic Groups $\times$ Tensor Rank of Permanent vs Determinant
- **Recorded:** 2026-05-19 21:42 UTC
- **Entry ID:** cf52d7a98cd5

For all $n \geq 2$, the dimension of the space of GL_n -invariant polynomials on the n -th tensor power of the permanent polynomial exceeds that of the determinant polynomial by a factor of $\Omega(2^n)$.

Invariant spaces capture symmetry under linear transformations. If permanent’s invariant dimension grows exponentially faster than determinant’s, it suggests a structural complexity barrier aligning with GCT’s orbit closure separation goals.

- Judge: NOVEL over 0 arXiv hits

- exit code: 124, elapsed: 240.05s

Judge reasoning

Real Rank of Communication Matrix Bounds AC^0 Circuit Size for PARITY

- **Verdict:** INCONCLUSIVE
- **Bridge:** Real Algebraic Geometry $\times$ AC^0 Circuit Size for PARITY
- **Recorded:** 2026-05-19 22:18 UTC
- **Entry ID:** 9bf36053ab4b

Statement

For any AC^0 circuit C computing PARITY on n bits, the real rank of its communication matrix is $\Omega(\log n)$.

Rationale

Real rank captures the minimal number of rank-1 matrices needed to express the communication matrix, reflecting the function's geometric complexity. High real rank implies the function cannot be compressed into small AC^0 circuits, aligning with known lower bounds for PARITY.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 1.17s

```
se, 'counterexample': 'singular_matrix'}
TRIAL: {'metric_name': 'real_rank', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
TRIAL: {'metric_name': 'real_rank', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
TRIAL: {'metric_name': 'real_rank', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
TRIAL: {'metric_name': 'real_rank', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
TRIAL: {'metric_name': 'real_rank', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
TRIAL: {'metric_name': 'real_rank', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
TRIAL: {'metric_name': 'real_rank', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
TRIAL: {'metric_name': 'real_rank', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
TRIAL: {'metric_name': 'real_rank', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
TRIAL: {'metric_name': 'real_rank', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
TRIAL: {'metric_name': 'real_rank', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
TRIAL: {'metric_name': 'real_rank', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
TRIAL: {'metric_name': 'real_rank', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
TRIAL: {'metric_name': 'real_rank', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
TRIAL: {'metric_name': 'real_rank', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
RESULT: FALSIFIED counterexample="singular_matrix" first_failing_seed=929
```

Judge reasoning

The FALSIFIED result rests on a buggy metric: 'singular_matrix' is not a valid counterexample to an $\Omega(\log n)$ rank lower bound (singular matrices can have rank $\leq \log n$), and metric_value was None on every trial, so the actual real rank was never measured. The critic's CHALLENGE is well-founded and the pre-registered criterion cannot be evaluated. | next: Fix the test to compute `numpy.linalg.matrix_rank` of a communication matrix actually derived from an AC^0 PARITY circuit, then compare it against

Restriction-Stable Fourier Tail Mass Excludes ACC0[2] Functions

- **Verdict:** BARRIER_HIT

- **Bridge:** Fourier analysis of Boolean functions (Bourgain-Kindler-Mossel tail decay under random restrictions) × ACC0[2] circuit lower bounds for explicit functions in P (Williams/Sipser program)
- **Recorded:** 2026-05-19 22:20 UTC
- **Entry ID:** 085bd128ea20

Statement

Define for a Boolean function $f: \{-1,1\}^n \rightarrow \{-1,1\}$ the restriction-stable Fourier tail $RT_k(f) := E_{\{\rho \sim R_p\}}[\sum_{|S| \geq k} \hat{f}(\rho)(S)^2]$ where ρ is a p -random restriction with $p = 1/\log(n+2)$ and $k = \text{floor}(\sqrt{n})$. Conjecture: every f computable by a depth- d ACC0[2] circuit of size s satisfies $RT_k(f) \leq (s * \log(n))^{\{O(d)\}} / 2^{\{\sqrt{n}\}}$, while the explicit P-time function Sip_n (Sipser's depth-3 read-once monotone AND-OR-AND with fan-in $\sim n^{\{1/3\}}$) satisfies $RT_k(Sip_n) \geq 1/\text{poly}(n)$. Hence RT_k is a natural distinguisher that is constructive and large on Sip_n but small on all ACC0[2] (with -1 polynomial-method exponent), and any explicit P-time f with $RT_k(f) \geq 1/\text{poly}(n)$ lies outside ACC0[2].

Rationale

Switching-lemma-type arguments give exponentially decaying Fourier tail under random restriction for AC0; the Razborov-Smolensky polynomial method forces similar low-degree concentration after restriction even with MOD₂ gates, because a restricted ACC0[2] circuit collapses to a low-degree GF(2) polynomial whose Walsh-Hadamard mass concentrates below degree polylog. Sipser-type read-once formulas, by contrast, retain a constant fraction of high-degree mass after a $1/\log n$ restriction because their decision-tree depth is preserved by sparse restrictions. The 'restriction-stable' averaging is the minimal violation of natural-proofs constructivity: it is computable in time $2^{\{O(n)\}}$ but plausibly not in P, sidestepping the Razborov-Rudich constructivity clause while remaining algorithmically testable for $n \leq 40$.

Novelty

- Judge: “ over 0 arXiv hits

Empirical Test

- exit code: -1, elapsed: 0.00s

(no output)

Judge reasoning

Rejected by NATURAL_PROOFS barrier

Loomis-Whitney Marginal Defect of 3-XOR Solution Lattice Tracks SoS Refutation Threshold

- **Verdict:** INCONCLUSIVE
- **Bridge:** Loomis-Whitney / Bollobás-Thomason coordinate-projection inequalities on the discrete cube (combinatorial measure theory; Loomis-Whitney 1949, Bollobás-Thomason 1995, uniform cover inequalities) × Sum-of-Squares degree-2 pseudo-feasibility / refutation threshold for random 3-XOR CSPs
- **Recorded:** 2026-05-19 22:35 UTC
- **Entry ID:** 17460b6d540e

Statement

For a 3-XOR instance φ with $m = \alpha n$ clauses on n Boolean variables, let $S_\tau(\varphi) := \{x \in \{0,1\}^n : x \text{ satisfies } \geq \tau m \text{ clauses of } \varphi\}$ and let $\pi_i: \{0,1\}^n \rightarrow \{0,1\}^{\{n-1\}}$ drop coordinate i . Define the Loomis-Whitney marginal defect $LW(\varphi, \tau) := (1/(n-1)) \cdot \sum_{i=1}^n \log_2 |\pi_i(S_\tau(\varphi))| - \log_2 |S_\tau(\varphi)|$, which is ≥ 0 by Loomis-Whitney. The conjecture: at $\tau = 1 - 1/(8\alpha)$, the rescaled defect $LW(\varphi, \tau)/n$ converges in probability to a function $\psi(\alpha)$ that is monotone non-decreasing, equals 0 for $\alpha \leq \alpha_{\text{sat}} \approx 0.918$ (the random-3-XOR satisfiability/degree-2 SoS feasibility threshold), and is strictly positive for $\alpha > \alpha_{\text{sat}}$; equivalently, for every random 3-XOR instance with $\alpha \leq 0.85$ we have $LW(\varphi, \tau)/n < 0.02$, while at $\alpha \geq 1.10$ we have $LW(\varphi, \tau)/n > 0.05$ with probability ≥ 0.9 .

Rationale

The degree-2 SoS pseudo-distribution is a relaxation of the *uniform* distribution on near-satisfying assignments, so its existence is constrained by how ‘product-like’ the true satisfying set is across coordinate projections — exactly what Loomis-Whitney quantifies. Below the SAT threshold the solution lattice is essentially a full-dimensional union of orbits (Loomis-Whitney is tight on Hamming-thick sets), while above it the surviving near-satisfying assignments are forced to lie in a low-codimension affine variety whose marginal projections shrink faster than the set itself — a quantitative analogue of the freezing/condensation transition. This bridge avoids Natural Proofs because LW is a relational marginal quantity, not a single-number truth-table invariant, and it is not poly-time computable in n .

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_34e7de56.py", line 66, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_34e7de56.py", line 42, in run_trial
    instance = [[random.randint(0, 1) for _ in range(n)] for _ in range(alpha * n)]
                                   ^^^^^^^^^^^^^^^^^
```

TypeError: 'float' object cannot be interpreted as an integer

Judge reasoning

Test crashed due to type error before producing data; cannot assess support/falsification | next: Fix the integer conversion error in instance generation (replace *alphan* with *int(alphan)* or similar)

Noncommutative Fourier Coefficient Count Lower Bound for AC⁰ PARITY Circuits

- **Verdict:** INCONCLUSIVE
- **Bridge:** Noncommutative Harmonic Analysis × AC⁰ Circuit Size for PARITY
- **Recorded:** 2026-05-19 22:59 UTC
- **Entry ID:** 1c23316cce55

Statement

For any AC^0 circuit C of depth d computing the PARITY function on n variables, the number of non-zero coefficients in the noncommutative Fourier transform of PARITY is $\Omega(n^{\frac{1}{d-1}})$.

Rationale

The noncommutative Fourier transform captures the algebraic structure of PARITY in a noncommutative setting. Its coefficient count reflects the depth required to compute PARITY, aligning with the switching lemma's lower bounds on AC^0 circuit size.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6602b959.py", line 58, in <module>
    result = run_trial(seed)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6602b959.py", line 40, in run_trial
    transform = noncommutative_fourier_transform(n)
                ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6602b959.py", line 24, in noncommutative_fourier_transform
    transform = [0] * (1 << n)
                ~~~~~
```

MemoryError

Judge reasoning

Test crashed with MemoryError before producing results, preventing evaluation of conjecture. |
next: Optimize noncommutative Fourier transform implementation to handle larger n values

Invariant Ring Dimension and Circuit Complexity for Permanent vs Determinant

- **Verdict:** INCONCLUSIVE
- **Bridge:** Invariant Theory $\times$ Circuit Complexity
- **Recorded:** 2026-05-19 23:23 UTC
- **Entry ID:** 56f24c51fad0

Statement

Let P_n be the permanent polynomial and D_n the determinant polynomial on $n \times n$ matrices. For each $n \geq 2$, the dimension of the invariant ring of $GL(n)$ acting on the space of $n \times n$ matrices is exponentially larger for P_n than for D_n , implying that the minimal circuit size for P_n is exponentially greater than that for D_n .

Rationale

Invariant theory under $GL(n)$ provides tools to distinguish algebraic varieties like the permanent and determinant. The dimension of the invariant ring reflects the complexity of the orbit closure, which is hypothesized to correlate with circuit complexity. This conjecture leverages GCT's focus on orbit closures and representation-theoretic multiplicities to separate VP and VNP.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 240.10s

TIMEOUT after 240s

Judge reasoning

Test timed out before producing results, preventing definitive validation or falsification. | next: Increase timeout duration and re-run test with extended computation limits

Schur Positivity of Clause-Indicator Polynomials Bounds ACC^0 Circuit Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Symmetric Functions Theory $\times$ ACC^0 Circuit Size
- **Recorded:** 2026-05-19 23:38 UTC
- **Entry ID:** 413870e28527

Statement

For any CNF formula with n variables, the number of distinct Schur functions that vanish on its clause-indicator polynomial is $\Theta(\log n)$ if and only if the ACC^0 circuit size is bounded by $O(\text{polylog } n)$.

Rationale

Schur functions encode combinatorial symmetries of polynomials, and their vanishing conditions may reveal structural constraints on clause interactions. By linking Schur positivity to circuit complexity, we explore algebraic invariants that could separate ACC^0 from NEXP.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 240.10s

TIMEOUT after 240s

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run test with detailed logging to diagnose crash cause